

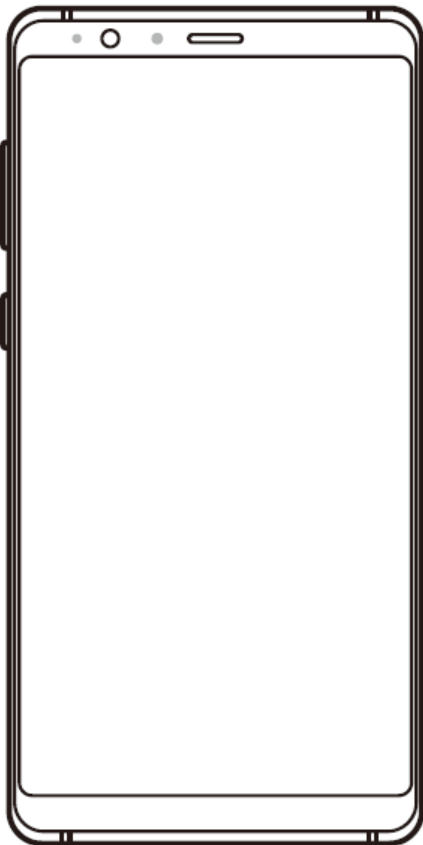
SAMSUNG

Mobile Device SM-G8850 Common

SERVICE Manual

Mobile Device

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1. Safety Precautions

1-1. Repair Precaution

Before attempting any repair or detailed tuning, shield the device from RF noise or static electricity discharges.

Use only demagnetized tools that are specifically designed for small electronic repairs, as most electronic parts are sensitive to electromagnetic forces.

Use only high quality screwdrivers when servicing products. Low quality screwdrivers can easily damage the heads of screws.

Use only conductor wire of the properly gauge and insulation for low resistance, because of the low margin of error of most testing equipment.

We recommend 22-gauge twisted copper wire.

Hand-soldering is not recommended, because printed circuit boards (PCBs) can be easily damaged, even with relatively low heat. Never use a soldering iron with a power rating of more than 100 watts and use only lead-free solder with a melting point below 250°C (482°F).

Prior to disassembling the battery charger for repair, ensure that the AC power is disconnected. Always use the replacement parts that are registered in the SEC system. Third-party replacement parts may not function properly.

1. Safety Precautions

1-2. ESD(Electrostatically Sensitive Devices) Precaution

Many semiconductors and ESDs in electronic devices are particularly sensitive to static discharge and can be easily damaged by it. We recommend protecting these components with conductive anti-static bags when you store or transport them.

Always use an anti-static strap or wristband and remove electrostatic buildup or dissipate static electricity from your body before repairing ESDs.

Ensure that soldering irons have AC adapter with ground wires and that the ground wires are properly connected.

Use only desoldering tools with plastic tips to prevent static discharge.

Properly shield the work environment from accidental electrostatic discharge before opening packages containing ESDs.

The potential for static electricity discharge may be increased in low humidity environments, such as air-conditioned rooms. Increase the airflow to the working area to decrease the chance of accidental static electricity discharges.

2. Specification

2-1. GSM General Specification

Item		GSM 850	EGSM 900	DCS1800	PCS1900
Freq. Band[MHz]		824~849	880~915	1710~1785	1850~1910
Uplink/Downlink		869~894	925~960	1805~1880	1930~1990
ARFCN range		128~251	0~124 & 975~1023	512~885	512~810
Tx/Rx spacing		45MHz	45MHz	95MHz	80MHz
Mod. Bit rate/ Bit Period		270.833kbps 3.692us	270.833kbps 3.692us	270.833kbps 3.692us	270.833kbps 3.692us
Time Slot Period/ Frame Period		576.9us 4.615ms	576.9us 4.615ms	576.9us 4.615ms	576.9us 4.615ms
Modulation	GSM/ EGPRS	GMSK/ 8PSK	GMSK/ 8PSK	GMSK/ 8PSK	GMSK/ 8PSK
MS Power		32.8dBm~5dBm	33dBm~5dBm	29.5dBm~0dBm	29.8dBm~0dBm
Power Class		4(GMSK) E2(8PSK)	4(GMSK) E2(8PSK)	1(GMSK) E2(8PSK)	1(GMSK) E2(8PSK)
Sensitivity		-103.5dBm	-109.5dBm	-108.5dBm	-108dBm
TDMA Mux		8	8	8	8

2. Specification

2-2. GSM Tx Power Class

TX Power control level	GSM850	TX Power control level	EGSM900	TX Power control level	DCS1800	TX Power control level	PCS1900
5	33±2 dBm	5	33±2 dBm	0	30±3 dBm	0	30±3 dBm
6	31±2 dBm	6	31±2 dBm	1	28±3 dBm	1	28±3 dBm
7	29±2 dBm	7	29±2 dBm	2	26±3 dBm	2	26±3 dBm
8	27±2 dBm	8	27±2 dBm	3	24±3 dBm	3	24±3 dBm
9	25±2 dBm	9	25±2 dBm	4	22±3 dBm	4	22±3 dBm
10	23±2 dBm	10	23±2 dBm	5	20±3 dBm	5	20±3 dBm
11	21±2 dBm	11	21±2 dBm	6	18±3 dBm	6	18±3 dBm
12	19±2 dBm	12	19±2 dBm	7	16±3 dBm	7	16±3 dBm
13	17±2 dBm	13	17±2 dBm	8	14±3 dBm	8	14±3 dBm
14	15±2 dBm	14	15±2 dBm	9	12±4 dBm	9	12±4 dBm
15	13±2 dBm	15	13±2 dBm	10	10±4 dBm	10	10±4 dBm
16	11±3 dBm	16	11±3 dBm	11	8±4 dBm	11	8±4 dBm
17	9±3 dBm	17	9±3 dBm	12	6±4 dBm	12	6±4 dBm
18	7±3 dBm	18	7±3 dBm	13	4±4 dBm	13	4±4 dBm
19	5±3 dBm	19	5±3 dBm	14	2±5 dBm	14	2±5 dBm
-	-	-	-	15	0±5 dBm	15	0±5 dBm

2. Specification

2-3. WCDMA General Specification

Item	WCDMA2100(B1)	WCDMA1900(B2)	WCDMA850(B5)	WCDMA900(B8)
Freq. Band[MHz] Uplink/Downlink	1920~1980 2110~2170	1850~1910 1930~1990	824~849 869~894	880~915 925~960
ARFCN range	UL: 9612~9888 DL: 10562~10838	UL: 9262~9538 DL: 9662~9938	UL: 4132~4233 DL: 4357~4458	UL: 2712~2868 DL: 2937~3088
Tx/Rx spacing	190MHz	80MHz	45MHz	45MHz
Mod. Bit rate/ Bit Period	42.2Mbps(DL) 5.42Mbps(UL)	42.2Mbps(DL) 5.42Mbps(UL)	42.2Mbps(DL) 5.42Mbps(UL)	42.2Mbps(DL) 5.42Mbps(UL)
Time Slot Period/ Frame Period	WCDMA 10ms/0.667ms HSPA 2ms/0.667ms	WCDMA 10ms/0.667ms HSPA 2ms/0.667ms	WCDMA 10ms/0.667ms HSPA 2ms/0.667ms	WCDMA 10ms/0.667ms HSPA 2ms/0.667ms
Modulation	QPSK 16QAM 64QAM	QPSK 16QAM 64QAM	QPSK 16QAM 64QAM	QPSK 16QAM 64QAM
MS Power (dBm)	23.5 ~ -49(↓)	22.5 ~ -49(↓)	23 ~ -49(↓)	22.5 ~ -49(↓)
Power Class	3(max+24dBm)	3(max+24dBm)	3(max+24dBm)	3(max+24dBm)
Sensitivity	-112.0dBm	-111dBm	-112dBm	-112dBm

2. Specification

2-4. LTE General Specification

Item	LTE Band1	LTE Band3	LTE Band4	LTE Band5	LTE Band7	LTE Band8
Freq. Band[MHz] Uplink/Downlink	1920~1980 2110~2170	1710~1785 1805~1880	1710~1755 2110~2155	824~849 869~894	2500~2570 2620~2690	880~915 925~960
ARFCN range	UL:18000~18599 DL:0~599	UL:19200~19949 DL:1200~1949	UL:19950~20399 DL:1950~2399	UL:20400~20649 DL:2400~2649	UL:20750~21449 DL:2750~3449	UL:21450~21799 DL:3450~3799
Tx/Rx spacing (MHz)	190	95	400	45	120	45
Channel Bandwidth (MHz)	5/10/15/20	1.4/3/5/10/15/20	1.4/3/5/10/15/20	1.4/3/5/10	5/10/15/20	1.4/3/5/10
Modulation	QPSK,16/64QAM 256QAM(DL only)	QPSK,16/64QAM 256QAM(DL only)	QPSK,16/64QAM 256QAM(DL only)	QPSK,16/64QAM 256QAM(DL only)	QPSK,16/64QAM 256QAM(DL only)	QPSK,16/64QAM 256QAM(DL only)
MS Power (dBm)	24~-39(↓)	24~-39(↓)	24~-39(↓)	24.5~-39(↓)	22.5~-39(↓)	24~-39(↓)
Sensitivity (QPSK, BW 10MHz) (dBm)	-101.5	-101	-102	-103	-101.5	-103.5

2. Specification

Item	LTE Band34	LTE Band38	LTE Band39	LTE Band40	LTE Band41
Freq. Band[MHz] Uplink/Downlink	2010~2025	2570~2620	1880~1920	2300~2400	2496~2690
ARFCN range	UL/DL:36200 ~ 36349	UL/DL:37750 ~ 38249	UL/DL:38250 ~ 38649	UL/DL:38650 ~ 39649	UL/DL:39650 ~ 41589
Tx/Rx spacing (MHz)	0	0	0	0	0
Channel Bandwidth (MHz)	5/10/15	5/10/15/20	5/10/15/20	5/10/15/20	5/10/15/20
Modulation	QPSK,16/64QAM 256QAM(DL only)	QPSK,16/64QAM 256QAM(DL only)	QPSK,16/64QAM 256QAM(DL only)	QPSK,16/64QAM 256QAM(DL only)	QPSK,16/64QAM 256QAM(DL only)
MS Power (dBm)	24~-39(↓)	23.5~-39(↓)	24~-39(↓)	24~-39(↓)	23.5~-39(↓)
Sensitivity (QPSK, BW 10MHz) (dBm)	-102	-101	-102	-102.5	-101

2. Specification

2-5. TDSCDMA General Specification

Item	TDSCDMA2010(A)	TDSCDMA1880(F)
Chip rate	1.28 Mcps	1.28 Mcps
OBW	1.6 MHz	1.6 MHz
Freq. Band[MHz] Uplink/Downlink	2010~2025	1880~1920
ARFCN range	10054~10121	9404~9596
Tx/Rx spacing (MHz)	0	0
MS Power (dBm)	24 ~ -48(↓)	24 ~ -48(↓)
Power Class	2(max+24dBm)	2(max+24dBm)
Sensitivity (dBm /1.28 MHz)	-110	-110

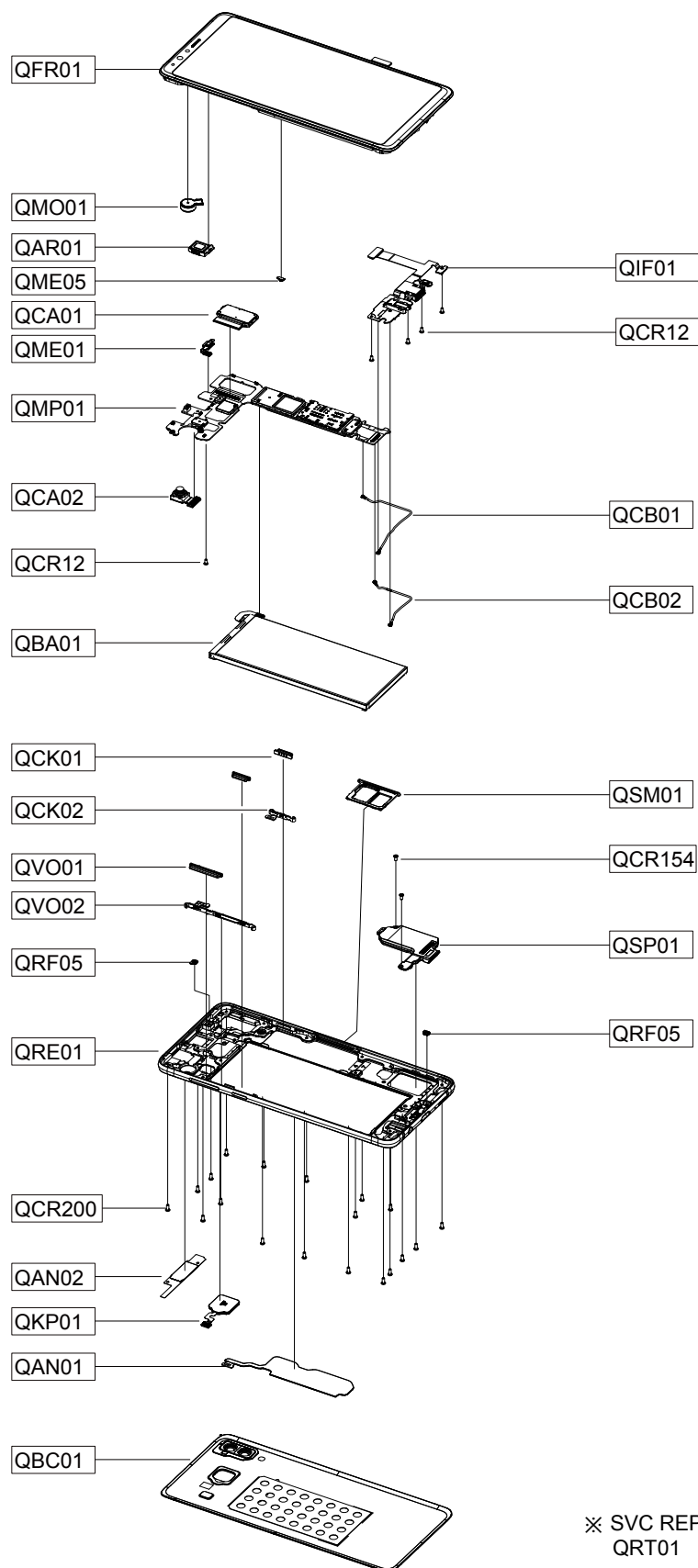
3. Product Function

Main Function

Item	Description
OS	Android V8.0
RF	CDMA800 GSM850 / GSM900 / DCS1800 / PCS1900 WCDMA: B1/ B2/ B5/ B8 TDSCDMA : B34/ B39 LTE: B1/ B3/ B4/ B5/ B7/ B8/ B34/ B38/ B39/ B40/ B41/
Battery	3700mAh
Base Band	2.2Ghz Quad + 1.8GHz Quad
Other RF	A-GPS, Glonass, BEIDOU, Galileo, BT5.0, USB 2.0 Type-C, WIFI 802.11 a/b/g/n/ac SISO,NFC,
Camera	Dual Camera (24MP+16MP F1.7 RGB) with LED Flash, 24MP F2.0 (Front)
LCD	6.2", Super AMOLED FHD+, 2220x1080
RAM	4GB
Sensor	Accelerometer, Fingerprint Sensor, Gyro Sensor, Geomagnetic Sensor, Proximity Sensor, RGB Light Sensor
Accessory	Charger: 5V/2A (AFC: 9V/1.67A) Data cable: 2.8pi, 1.2m(Type C/ USB-A) Ear phone: 3.5pi, 4pin

4. Exploded View and Parts List

4-1. Cellular phone Exploded View



5. MAIN Electrical Parts List

Parts Code	Design LOC	Description
0404-001250	D5000,D6001	DIODE-SCHOTTKY
0406-001506	ZD1001,ZD1002	DIODE-TVS
0406-001561	ZD6009,ZD6010,ZD6011	DIODE-TVS
0406-001561	ZD6012,ZD6013,ZD6022	DIODE-TVS
0406-001561	ZD8002,ZD8004	DIODE-TVS
0406-001592	ZD3002,ZD3003,ZD3004	DIODE-TVS
0406-001592	ZD3005,ZD3006,ZD3009	DIODE-TVS
0406-001592	ZD4001,ZD6002,ZD6008	DIODE-TVS
0406-001592	ZD7000,ZD7001,ZD7002	DIODE-TVS
0406-001592	ZD7003,ZD7004,ZD7005	DIODE-TVS
0406-001592	ZD7007,ZD7008,ZD7009	DIODE-TVS
0406-001592	ZD7010,ZD7011,ZD7012	DIODE-TVS
0406-001592	ZD7017	DIODE-TVS
0406-001659	ZD3000,ZD3001	DIODE-TVS
0406-001728	ZD1000,ZD4000,ZD5000	DIODE-TVS
0406-001728	ZD5001	DIODE-TVS
0406-001729	ZD8000,ZD8003	DIODE-TVS
0406-001743	D6000,ZD6003,ZD6004	DIODE-TVS
0406-001743	ZD6005,ZD6006,ZD6007	DIODE-TVS
0406-001762	ZD6027,ZD6028	DIODE-TVS
0406-001763	ZD6001	DIODE-TVS
0406-001780	ZD6025,ZD6026	DIODE-TVS
0406-001781	ZD6014,ZD6015	DIODE-TVS
0406-001789	ZD6000,ZD6024	DIODE-TVS
0406-001797	ZD6016,ZD6017	DIODE-TVS
0406-001808	ZD6023	DIODE-TVS
0505-002088	TR5001,TR5002	FET-SILICON
0505-002652	TR5000	FET-SILICON
0601-001905	LED3000	LED
0601-003629	LED5000	LED
0801-003437	U4001	IC
1001-001835	U3021	IC
1001-001911	U2027	IC
1001-001928	U8003	IC
1001-001974	U1001	IC
1001-001999	U1005	IC
1003-002911	U5006	IC
1108-000648	UME6000	MEMORY

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1201-003869	U3009	IC
1201-003974	U1010	IC
1201-003978	U1011	IC
1201-004029	U2032,U2033	IC
1201-004109	PAM2000	IC
1201-004135	U1004	IC
1201-004156	U8014	IC
1202-001120	U7010,U8001	IC
1203-007365	U8016	IC
1203-008249	U3000	IC
1203-008251	U5004	IC
1203-008379	C2054	IC
1203-008603	U3017	IC
1203-008696	U7009	IC
1203-008797	U6002	IC
1203-008819	U5002	IC
1203-008864	U6003	IC
1203-008865	U6001,U7002,U7004	IC
1203-008865	U7006	IC
1203-008867	U3001,U3015,U8008	IC
1203-008872	U7007	IC
1203-008924	U7005	IC
1203-008948	U8005	IC
1203-008955	U7000,U7001,U7003	IC
1203-008998	U6000	IC
1203-009012	U5003	IC
1203-009014	U5005	IC
1204-003685	U8010	IC
1205-005106	U3006	IC
1205-005731	U4002	IC
1205-005738	U4000	IC
1205-005830	UCP4000	IC
1205-005833	U2029	IC
1205-005881	U3011	IC
1209-002450	U3010	IC
1209-002474	U3003	IC
1209-002513	U3020	IC
1404-001724	TH2000,TH4000,TH5000	THERMISTOR
1404-001724	THM3000	THERMISTOR

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1405-001395	VR2001,VR6011,VR6012	VARISTOR
1405-001395	VR6013,VR6014	VARISTOR
1405-001403	VR2000,VR6015	VARISTOR
1405-001404	VR6010	VARISTOR
1405-001412	VR1007	VARISTOR
1405-001415	V2000	VARISTOR
1405-001418	VR3000	VARISTOR
1405-001434	VR2002	VARISTOR
2007-001292	R8026	R-CHIP
2007-003015	R6033,R6034	R-CHIP
2007-007014	R3011	R-CHIP
2007-007190	R6038,R6041	R-CHIP
2007-007308	R5029	R-CHIP
2007-007741	R3000,R4032,R4034	R-CHIP
2007-007741	R4035,R4049,R4050	R-CHIP
2007-007741	R4051,R5040,R5047	R-CHIP
2007-007741	R5054,R6000,R6017	R-CHIP
2007-007741	R6022,R6023,R6026	R-CHIP
2007-007741	R6027,R6035,R6036	R-CHIP
2007-007741	R6051	R-CHIP
2007-008056	R1008,R1009	R-CHIP
2007-008210	R6005	R-CHIP
2007-008211	R7012	R-CHIP
2007-008263	R4007	R-CHIP
2007-008312	R8011	R-CHIP
2007-008516	R6013	R-CHIP
2007-008531	R5005,R7010,R7011	R-CHIP
2007-008588	C3084,C3085,R4010	R-CHIP
2007-008588	R4011,R4013,R4014	R-CHIP
2007-008588	R4015,R4016,R4017	R-CHIP
2007-008588	R4021,R4022,R4023	R-CHIP
2007-008588	R4024,R4025,R4026	R-CHIP
2007-008588	R4028,R4029,R4030	R-CHIP
2007-008588	R4031,R8029	R-CHIP
2007-008766	R4005,R4006	R-CHIP
2007-008774	R3007,R3008,R3009	R-CHIP
2007-008785	R1007	R-CHIP
2007-009111	R3001,R8031	R-CHIP
2007-009155	R5009,R5020,R8006	R-CHIP

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2007-009157	R5019,R5022,R5026	R-CHIP
2007-009157	R5038,R6044	R-CHIP
2007-009158	R3020,R5003	R-CHIP
2007-009182	R8007	R-CHIP
2007-009201	R4019,R4020,R4052	R-CHIP
2007-009201	R8021,R8022	R-CHIP
2007-009212	R4045,R4047,R5014	R-CHIP
2007-009212	R8000	R-CHIP
2007-009223	R6037	R-CHIP
2007-009233	R6052	R-CHIP
2007-009314	R7009	R-CHIP
2007-009315	R5011,R6002,R6003	R-CHIP
2007-009315	R6004,R6006,R6007	R-CHIP
2007-009315	R6008,R6011,R6012	R-CHIP
2007-009315	R6014,R6015,R6016	R-CHIP
2007-009352	R4033,R7006	R-CHIP
2007-009408	R4037,R4038,R5036	R-CHIP
2007-009408	R5037,R6032	R-CHIP
2007-009410	R1010,R1011	R-CHIP
2007-009766	R8002	R-CHIP
2007-009805	R1000,R1001,R1002	R-CHIP
2007-009805	R1003,R1004,R1005	R-CHIP
2007-009805	R1006,R2001	R-CHIP
2007-009866	R5016,R7013,R7014	R-CHIP
2007-009920	R5043,R5044,R6009	R-CHIP
2007-009920	R6010,R8030	R-CHIP
2007-009954	R5027,R5028,R5030	R-CHIP
2007-009954	R6042	R-CHIP
2007-009969	R4009,R4012,R4018	R-CHIP
2007-009969	R8001	R-CHIP
2007-010202	R3015,R5041	R-CHIP
2007-010233	R8032	R-CHIP
2007-010685	R1012,R1013,R4002	R-CHIP
2007-010685	R6001,R8005,U8032	R-CHIP
2007-011043	R4027	R-CHIP
2007-011648	R2002,R2003,R2004	R-CHIP
2007-011648	R3010,R3016,R3017	R-CHIP
2007-011648	R3018,R3019,R6047	R-CHIP
2007-011648	R6048	R-CHIP

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2007-012033	R8028	R-CHIP
2007-012068	R5049	R-CHIP
2007-012202	R8010	R-CHIP
2203-000254	C5068,C5074	C-CERAMIC,CHIP
2203-000995	C5070	C-CERAMIC,CHIP
2203-001153	C8073,C8074	C-CERAMIC,CHIP
2203-001239	C4025	C-CERAMIC,CHIP
2203-005053	C3007	C-CERAMIC,CHIP
2203-005344	C5014	C-CERAMIC,CHIP
2203-005682	C1023,C2051,C2056	C-CERAMIC,CHIP
2203-005682	C2058,C2059,C2060	C-CERAMIC,CHIP
2203-005682	C2068,C2071,C2078	C-CERAMIC,CHIP
2203-005682	C2080,C2083,C2121	C-CERAMIC,CHIP
2203-005682	C2127,C2128,C3033	C-CERAMIC,CHIP
2203-005682	C6034,C6063,C6064	C-CERAMIC,CHIP
2203-005682	C6069,L1001,L1020	C-CERAMIC,CHIP
2203-005682	L1029,L1040,L2067	C-CERAMIC,CHIP
2203-005682	L2079	C-CERAMIC,CHIP
2203-005717	C1017,C1030,C1050	C-CERAMIC,CHIP
2203-005717	C1054,C1122,C2011	C-CERAMIC,CHIP
2203-005717	C2022,C2029,C2055	C-CERAMIC,CHIP
2203-005717	C2062,C2076,C2077	C-CERAMIC,CHIP
2203-005717	C2096	C-CERAMIC,CHIP
2203-005719	C3045,C3092	C-CERAMIC,CHIP
2203-005725	C1007,C3044,C7012	C-CERAMIC,CHIP
2203-005725	C7014	C-CERAMIC,CHIP
2203-005726	C8039	C-CERAMIC,CHIP
2203-005729	C8000,C8008,C8009	C-CERAMIC,CHIP
2203-005729	C8014,C8017	C-CERAMIC,CHIP
2203-005731	C3006,C3012,C3915	C-CERAMIC,CHIP
2203-005731	C6045,C6047	C-CERAMIC,CHIP
2203-005732	C2092,C5072,C8002	C-CERAMIC,CHIP
2203-005734	C3022,C3023,C3024	C-CERAMIC,CHIP
2203-005734	C6046,C6048	C-CERAMIC,CHIP
2203-005736	C1003,C1004,C1032	C-CERAMIC,CHIP
2203-005736	C1033,C1036,C1037	C-CERAMIC,CHIP
2203-005736	C1044,C1045,C1046	C-CERAMIC,CHIP
2203-005736	C1095,C1096,C1104	C-CERAMIC,CHIP
2203-005736	C1113,C1125,C1128	C-CERAMIC,CHIP

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2203-005736	C1129,C1132,C2002	C-CERAMIC,CHIP
2203-005736	C2005,C2006,C2012	C-CERAMIC,CHIP
2203-005736	C2015,C2016,C2017	C-CERAMIC,CHIP
2203-005736	C2018,C2125,C2126	C-CERAMIC,CHIP
2203-005736	C3000,C3028,C3046	C-CERAMIC,CHIP
2203-005736	C3047,C3049,C3051	C-CERAMIC,CHIP
2203-005736	C3053,C3093,C3906	C-CERAMIC,CHIP
2203-005736	C3914,C3919,C3920	C-CERAMIC,CHIP
2203-005736	C8063,C8064,L1048	C-CERAMIC,CHIP
2203-005736	L1078,L2004	C-CERAMIC,CHIP
2203-005777	C1006,C1080,C1090	C-CERAMIC,CHIP
2203-005777	C1092,C1126,C1127	C-CERAMIC,CHIP
2203-005777	C2036,C2063,C2093	C-CERAMIC,CHIP
2203-005777	C2095	C-CERAMIC,CHIP
2203-005789	C1031,C1069,C1102	C-CERAMIC,CHIP
2203-005789	C2038,C2042,C2053	C-CERAMIC,CHIP
2203-005789	C2082,C2091,C2094	C-CERAMIC,CHIP
2203-005789	L2075	C-CERAMIC,CHIP
2203-005792	C1094,C1106,C1107	C-CERAMIC,CHIP
2203-005792	C2075	C-CERAMIC,CHIP
2203-005806	C2067,C2074,C2081	C-CERAMIC,CHIP
2203-005806	C5017	C-CERAMIC,CHIP
2203-006121	C6044	C-CERAMIC,CHIP
2203-006123	C2066	C-CERAMIC,CHIP
2203-006187	C2028,C2033	C-CERAMIC,CHIP
2203-006305	C1077,C3008,C3009	C-CERAMIC,CHIP
2203-006305	C3095,C3096,C3921	C-CERAMIC,CHIP
2203-006305	C3925,C6008,C7024	C-CERAMIC,CHIP
2203-006305	C7025,C7031,C7069	C-CERAMIC,CHIP
2203-006305	C8056	C-CERAMIC,CHIP
2203-006318	C2088	C-CERAMIC,CHIP
2203-006379	C8052	C-CERAMIC,CHIP
2203-006400	C2024,C2052,C4021	C-CERAMIC,CHIP
2203-006400	C4024,C5054,C8004	C-CERAMIC,CHIP
2203-006423	C1134,C1135,C4082	C-CERAMIC,CHIP
2203-006423	R6019	C-CERAMIC,CHIP
2203-006556	C1065,C1071,C1074	C-CERAMIC,CHIP
2203-006556	C2070,C5006	C-CERAMIC,CHIP
2203-006562	C2072,C3003,C3013	C-CERAMIC,CHIP

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5. MAIN Electrical Parts List

2203-006562	C3086,C3912,C3913	C-CERAMIC,CHIP
2203-006562	C5022,C5061,C5064	C-CERAMIC,CHIP
2203-006562	C5083,C5085,C5086	C-CERAMIC,CHIP
2203-006562	C5088,C5089,C5091	C-CERAMIC,CHIP
2203-006562	C6041,C7058,C7059	C-CERAMIC,CHIP
2203-006562	C7060	C-CERAMIC,CHIP
2203-006604	C3926,C6033	C-CERAMIC,CHIP
2203-006648	C4077	C-CERAMIC,CHIP
2203-006665	C1093	C-CERAMIC,CHIP
2203-006674	C6053	C-CERAMIC,CHIP
2203-006707	C2064	C-CERAMIC,CHIP
2203-006839	C4003,C4011,C4083	C-CERAMIC,CHIP
2203-006839	C5011,C5019,C5037	C-CERAMIC,CHIP
2203-006839	C5071,C5084,C5087	C-CERAMIC,CHIP
2203-006839	C6067,C8057,C8060	C-CERAMIC,CHIP
2203-006844	C3043	C-CERAMIC,CHIP
2203-006846	L1066	C-CERAMIC,CHIP
2203-006872	C4004,C8072	C-CERAMIC,CHIP
2203-006979	C2050,C3021,C3027	C-CERAMIC,CHIP
2203-006979	C3029,C3030,C3031	C-CERAMIC,CHIP
2203-006979	C3038,C3039	C-CERAMIC,CHIP
2203-007143	C5059	C-CERAMIC,CHIP
2203-007210	C4006,C4028,C4029	C-CERAMIC,CHIP
2203-007210	U8019,U8028	C-CERAMIC,CHIP
2203-007271	C1067,C2073,C3040	C-CERAMIC,CHIP
2203-007271	C3090,C5045,C6051	C-CERAMIC,CHIP
2203-007391	C3041,C4005	C-CERAMIC,CHIP
2203-007392	C5076,C5078,C5079	C-CERAMIC,CHIP
2203-007392	C5080	C-CERAMIC,CHIP
2203-007393	C2099,C2102,C2111	C-CERAMIC,CHIP
2203-007393	C2115,C2120,C2123	C-CERAMIC,CHIP
2203-007393	C2124,C3032,C3036	C-CERAMIC,CHIP
2203-007393	C3042,C3908,C5033	C-CERAMIC,CHIP
2203-007393	C5034,C5035,C5043	C-CERAMIC,CHIP
2203-007393	C5052,C5069,C5075	C-CERAMIC,CHIP
2203-007393	C5077,C5081,C5082	C-CERAMIC,CHIP
2203-007393	C5093,C6040,C6061	C-CERAMIC,CHIP
2203-007393	C6065,C6066,C8062	C-CERAMIC,CHIP
2203-007425	C5095	C-CERAMIC,CHIP

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5. MAIN Electrical Parts List

2203-007544	C3014,C3016,C3017	C-CERAMIC,CHIP
2203-007544	C3018,C3055,C8065	C-CERAMIC,CHIP
2203-007634	C6036	C-CERAMIC,CHIP
2203-007775	C8024,C8071	C-CERAMIC,CHIP
2203-007796	C3005,C3025,C3026	C-CERAMIC,CHIP
2203-007796	C3035,C3037,C3057	C-CERAMIC,CHIP
2203-007796	C3060,C4000,C4001	C-CERAMIC,CHIP
2203-007796	C4002,C4007,C4008	C-CERAMIC,CHIP
2203-007796	C4009,C4010,C4012	C-CERAMIC,CHIP
2203-007796	C4016,C4017,C4039	C-CERAMIC,CHIP
2203-007796	C4040,C4041,C4042	C-CERAMIC,CHIP
2203-007796	C4043,C4044,C4046	C-CERAMIC,CHIP
2203-007796	C4047,C4050,C4051	C-CERAMIC,CHIP
2203-007796	C4052,C4053,C4055	C-CERAMIC,CHIP
2203-007796	C4069,C4070,C4071	C-CERAMIC,CHIP
2203-007796	C4072,C4073,C4074	C-CERAMIC,CHIP
2203-007796	C4075,C4076,C4080	C-CERAMIC,CHIP
2203-007796	C4081,C4084,C4086	C-CERAMIC,CHIP
2203-007796	C4087,C4088,C4089	C-CERAMIC,CHIP
2203-007796	C4091,C4092,C4093	C-CERAMIC,CHIP
2203-007796	C4094,C4096,C4097	C-CERAMIC,CHIP
2203-007796	C4099,C4100,C4103	C-CERAMIC,CHIP
2203-007796	C5010,C5031,C5032	C-CERAMIC,CHIP
2203-007796	C5036,C5040,C5048	C-CERAMIC,CHIP
2203-007796	C5050,C5060,C5062	C-CERAMIC,CHIP
2203-007796	C5065,C5090,C6004	C-CERAMIC,CHIP
2203-007796	C6005,C6006,C6007	C-CERAMIC,CHIP
2203-007796	C6012,C6016,C6017	C-CERAMIC,CHIP
2203-007796	C6018,C6019,C6021	C-CERAMIC,CHIP
2203-007796	C6022,C6023,C6024	C-CERAMIC,CHIP
2203-007796	C6025,C6026,C6027	C-CERAMIC,CHIP
2203-007796	C6028,C6029,C6030	C-CERAMIC,CHIP
2203-007796	C6042,C6049,C6050	C-CERAMIC,CHIP
2203-007796	C7007,C7009,C7017	C-CERAMIC,CHIP
2203-007796	C7020,C7021,C7022	C-CERAMIC,CHIP
2203-007796	C7023,C7026,C7027	C-CERAMIC,CHIP
2203-007796	C7030,C7032,C7033	C-CERAMIC,CHIP
2203-007796	C7035,C7042,C7055	C-CERAMIC,CHIP
2203-007796	C7056,C7070,C8005	C-CERAMIC,CHIP

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5. MAIN Electrical Parts List

2203-007796	C8019,C8020,C8027	C-CERAMIC,CHIP
2203-007796	C8030,C8075	C-CERAMIC,CHIP
2203-008095	C4037,C4038	C-CERAMIC,CHIP
2203-008097	C1066,C1072,C1073	C-CERAMIC,CHIP
2203-008097	C1075,C1117,C1131	C-CERAMIC,CHIP
2203-008097	C2027,C2100,C2101	C-CERAMIC,CHIP
2203-008097	C2103,C2104,C2105	C-CERAMIC,CHIP
2203-008097	C2106,C2107,C2108	C-CERAMIC,CHIP
2203-008097	C2109,C2110,C2112	C-CERAMIC,CHIP
2203-008097	C2113,C2114,C2116	C-CERAMIC,CHIP
2203-008097	C2117,C2118,C2119	C-CERAMIC,CHIP
2203-008097	C3052,C3910,C3911	C-CERAMIC,CHIP
2203-008097	C4048,C4049,C4090	C-CERAMIC,CHIP
2203-008097	C5049,C6000,C6001	C-CERAMIC,CHIP
2203-008097	C6002,C6003,C6015	C-CERAMIC,CHIP
2203-008097	C6059,C6062,C8010	C-CERAMIC,CHIP
2203-008097	C8021,C8025,C8059	C-CERAMIC,CHIP
2203-008097	R8012	C-CERAMIC,CHIP
2203-008177	C3034	C-CERAMIC,CHIP
2203-008242	C3091	C-CERAMIC,CHIP
2203-008394	C1049,L2047	C-CERAMIC,CHIP
2203-008529	C8069	C-CERAMIC,CHIP
2203-008572	C5096,C5097	C-CERAMIC,CHIP
2203-008749	C6035	C-CERAMIC,CHIP
2203-008795	C6039,C8012	C-CERAMIC,CHIP
2203-008860	C3004,C4068,C4078	C-CERAMIC,CHIP
2203-008860	C4079,C5012,C5025	C-CERAMIC,CHIP
2203-008860	C5044,C5066,C5067	C-CERAMIC,CHIP
2203-008860	C6043,C7028,C7029	C-CERAMIC,CHIP
2203-008860	C7034,C7036,C7041	C-CERAMIC,CHIP
2203-008860	C8007,C8026,C8028	C-CERAMIC,CHIP
2203-008860	C8034	C-CERAMIC,CHIP
2203-008876	C5000,C5007,C5009	C-CERAMIC,CHIP
2203-008876	C5013,C5020,C5021	C-CERAMIC,CHIP
2203-008876	C5039,C5041,C5042	C-CERAMIC,CHIP
2203-008876	C5046,C5047,C5051	C-CERAMIC,CHIP
2203-008876	C5053,C5056,C5058	C-CERAMIC,CHIP
2203-008876	C6010,C6011,C6013	C-CERAMIC,CHIP
2203-008876	C6014,C7057,C7062	C-CERAMIC,CHIP

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5. MAIN Electrical Parts List

2203-008876	C7072,C8067,C8068	C-CERAMIC,CHIP
2203-009064	C4018,C4019,C4020	C-CERAMIC,CHIP
2203-009064	C4030,C4031,C8076	C-CERAMIC,CHIP
2203-009167	C4022,C4026	C-CERAMIC,CHIP
2203-009444	C5018	C-CERAMIC,CHIP
2203-009618	C5028,C5029,C7040	C-CERAMIC,CHIP
2203-009618	C7051,C7052	C-CERAMIC,CHIP
2203-009732	C6009,C6057	C-CERAMIC,CHIP
2203-009733	C1076,C2049,C3907	C-CERAMIC,CHIP
2203-009733	C3916,C3917,C3918	C-CERAMIC,CHIP
2203-009733	C5001,C5002,C5003	C-CERAMIC,CHIP
2203-009733	C5004,C5005,C5015	C-CERAMIC,CHIP
2203-009733	C5016,C5023,C5030	C-CERAMIC,CHIP
2203-009733	C5073,C5094,C5100	C-CERAMIC,CHIP
2203-009733	C5101,C5102,C5103	C-CERAMIC,CHIP
2203-009733	C5104,C5105,C5106	C-CERAMIC,CHIP
2203-009733	C7000,C7002,C7003	C-CERAMIC,CHIP
2203-009733	C7005,C7011,C7018	C-CERAMIC,CHIP
2203-009733	C7037,C7043,C7047	C-CERAMIC,CHIP
2203-009733	C7048,C7067,C7068	C-CERAMIC,CHIP
2203-009733	C8053,C8054,C8055	C-CERAMIC,CHIP
2203-009734	C5055	C-CERAMIC,CHIP
2203-009735	C4057,C4058	C-CERAMIC,CHIP
2203-009738	C5008,C5057,C6020	C-CERAMIC,CHIP
2203-009738	U8018	C-CERAMIC,CHIP
2203-009858	C7046,C7064,C7065	C-CERAMIC,CHIP
2203-009858	C8006	C-CERAMIC,CHIP
2203-009969	C8011	C-CERAMIC,CHIP
2703-002649	C2122	INDUCTOR-SMD
2703-002900	L2014	INDUCTOR-SMD
2703-002901	C1024,L1021,L2018	INDUCTOR-SMD
2703-002903	L6003	INDUCTOR-SMD
2703-002953	L1033,L3911	INDUCTOR-SMD
2703-002958	C1034	INDUCTOR-SMD
2703-002959	L3916	INDUCTOR-SMD
2703-003004	L3001	INDUCTOR-SMD
2703-003915	C1059	INDUCTOR-SMD
2703-003970	C1114	INDUCTOR-SMD
2703-004000	C1098	INDUCTOR-SMD

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5. MAIN Electrical Parts List

2703-004001	L2016	INDUCTOR-SMD
2703-004012	C1018,C3002,L1008	INDUCTOR-SMD
2703-004012	L2005,L2020,L2036	INDUCTOR-SMD
2703-004012	L2044,L2050,L2052	INDUCTOR-SMD
2703-004012	L2055,L2056,L2063	INDUCTOR-SMD
2703-004012	L2076,L2081,L3910	INDUCTOR-SMD
2703-004013	L1009,L1023,L1036	INDUCTOR-SMD
2703-004014	C3001,L1006,L1024	INDUCTOR-SMD
2703-004014	L1035,L1046,L1074	INDUCTOR-SMD
2703-004014	L3913	INDUCTOR-SMD
2703-004018	L1022,L1044,L1057	INDUCTOR-SMD
2703-004030	L1032,L2025	INDUCTOR-SMD
2703-004032	L1077,L2046,L2057	INDUCTOR-SMD
2703-004032	L2059	INDUCTOR-SMD
2703-004034	L2048	INDUCTOR-SMD
2703-004035	L1063,L2029,L2043	INDUCTOR-SMD
2703-004035	L2078	INDUCTOR-SMD
2703-004036	L2008,L3912	INDUCTOR-SMD
2703-004038	C3922,L1028,L1038	INDUCTOR-SMD
2703-004038	L2006,L2021,L2045	INDUCTOR-SMD
2703-004038	L2058	INDUCTOR-SMD
2703-004259	L2035	INDUCTOR-SMD
2703-004286	L2051,L2068,L2077	INDUCTOR-SMD
2703-004287	C1040,L1058,L1072	INDUCTOR-SMD
2703-004287	L2042	INDUCTOR-SMD
2703-004288	L2026,L2038	INDUCTOR-SMD
2703-004302	L1049,L1060,L1069	INDUCTOR-SMD
2703-004302	L3012	INDUCTOR-SMD
2703-004328	L1053,L1056,L2073	INDUCTOR-SMD
2703-004328	L2084	INDUCTOR-SMD
2703-004366	L1002,L1070	INDUCTOR-SMD
2703-004367	L2007	INDUCTOR-SMD
2703-004703	L2033	INDUCTOR-SMD
2703-004763	L2082	INDUCTOR-SMD
2703-004764	L1030,L1031,L1041	INDUCTOR-SMD
2703-004764	L1064	INDUCTOR-SMD
2703-004853	L1079,L1080,L3915	INDUCTOR-SMD
2703-004862	L1052,L2049,L2080	INDUCTOR-SMD
2703-004862	L6013	INDUCTOR-SMD

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5. MAIN Electrical Parts List

2703-005066	L5012,L5013,L5014	INDUCTOR-SMD
2703-005085	C2065,L1061	INDUCTOR-SMD
2703-005087	L1059	INDUCTOR-SMD
2703-005101	L2039	INDUCTOR-SMD
2703-005117	L7008,L7011	INDUCTOR-SMD
2703-005118	L7007	INDUCTOR-SMD
2703-005136	L5004	INDUCTOR-SMD
2703-005201	L5001,L5003,L5005	INDUCTOR-SMD
2703-005201	L5007,L5009,L5011	INDUCTOR-SMD
2703-005225	L3909	INDUCTOR-SMD
2703-005295	C1105,C1108,L2011	INDUCTOR-SMD
2703-005295	L2017,L3905	INDUCTOR-SMD
2703-005296	L1067,L2072,L3011	INDUCTOR-SMD
2703-005366	L3000	INDUCTOR-SMD
2703-005505	L8001	INDUCTOR-SMD
2703-005555	L4002,L4003	INDUCTOR-SMD
2703-005581	L5002,L5006,L5008	INDUCTOR-SMD
2703-005581	L6002	INDUCTOR-SMD
2801-005481	OSC5000	CRYSTAL-UNIT
2901-001690	C4056	FILTER-EMI SMD
2901-001786	F7000,F7001,F7002	FILTER-EMI SMD
2901-001786	F7003,F7004,F7005	FILTER-EMI SMD
2901-001786	F7006,F7007,F7008	FILTER-EMI SMD
2901-001786	F7009	FILTER-EMI SMD
2901-001798	F7010,F7011,F7012	FILTER-EMI SMD
2901-001798	F7013,F7014	FILTER-EMI SMD
2903-001576	F3902	FILTER-CERAMIC
2904-002143	F2008	FILTER-SAW
2904-002253	F1006,F2002	FILTER-SAW
2904-002256	F1008	FILTER-SAW
2904-002257	F2005	FILTER-SAW
2904-002314	F3901	FILTER-SAW
2904-002349	F2003	FILTER-SAW
2904-002350	F2010	FILTER-SAW
2904-002356	F3003	FILTER-SAW
2904-002361	F2001	FILTER-SAW
2904-002365	F2004	FILTER-SAW
2910-000390	F2007	FILTER
2911-000430	U2031	FILTER

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5. MAIN Electrical Parts List

2911-000452	F1000	FILTER
3301-001885	L8000,L8006	CORE-FERRITE BEAD
3301-002085	L8003,L8004	CORE-FERRITE BEAD
3301-002122	L8002,L8005	CORE-FERRITE BEAD
3301-002223	L7001,L7004,L7005	CORE-FERRITE BEAD
3301-002237	L1000	CORE-FERRITE BEAD
3301-002238	L3010	CORE-FERRITE BEAD
3301-002244	L3908	CORE-FERRITE BEAD
3301-002312	L5010,L7006,L7009	CORE-FERRITE BEAD
3301-002312	L7010	CORE-FERRITE BEAD
3301-002331	L5000,L6007	CORE-FERRITE BEAD
3705-001937	RFS1000,RFS1001	CONNECTOR-COAXIAL
3709-001891	SIM6000	CONNECTOR-CARD EDGE
3711-007107	HEA7000	CONNECTOR-HEADER
3711-008593	HEA7001	CONNECTOR-HEADER
3711-008847	HDC5000	CONNECTOR-HEADER
3711-008997	HEA3000,HEA8000	CONNECTOR-HEADER
3711-009073	HEA6000	CONNECTOR-HEADER
3711-009302	HEA7002	CONNECTOR-HEADER
3712-001626	ANT2001,ANT2002	CONNECTOR
3712-001626	ANT2005,ANT2007	CONNECTOR
3712-001626	ANT2009,ANT3001	CONNECTOR
3712-001626	ANT6009	CONNECTOR
3712-001633	ANT1007,ANT2003	CONNECTOR
3712-001633	ANT3000	CONNECTOR
3712-001634	ANT6006,ANT6007	CONNECTOR
3712-001694	ANT3003,ANT3004	CONNECTOR
3712-001694	ANT6000,ANT6001	CONNECTOR
3712-001694	ANT6002,ANT6003	CONNECTOR
3712-001694	ANT6004,ANT6005	CONNECTOR
3712-001694	ANT6010,ANT8000	CONNECTOR
3712-001694	ANT8001,ANT8002	CONNECTOR
3712-001694	ANT8003	CONNECTOR
4709-002226	F1007	RF-MODULE
4709-002285	F3900	RF-MODULE
4709-002484	F2000	RF-MODULE
4709-002589	F1001,F1003	RF-MODULE

Please consult the GSPN website (Samsung Portal) for the most recent version of the product's part list.

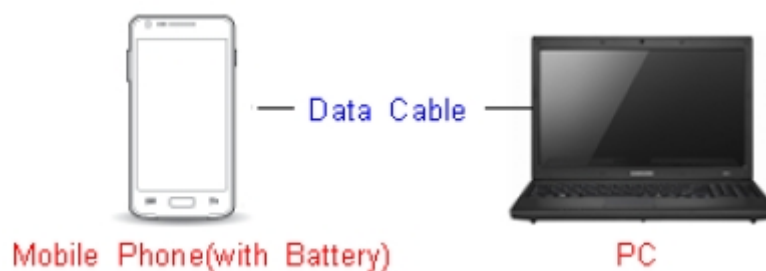
6. Level 1 Repair

6-1. S/W Update

6-1-1. Preparation

- S/W Update program : [Fenrir 5.17.xxxx](#)
- Mobile Phone
- Data Cable

※ Settings

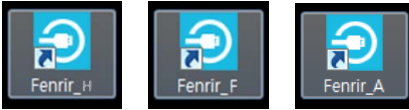


Data Cable : [GH39-01886A](#)

6. Level 1 Repair

6-1-2. How to use 'Fenrir' S/W update program.

1) Launch Fenrir by clicking on the icon on the desktop



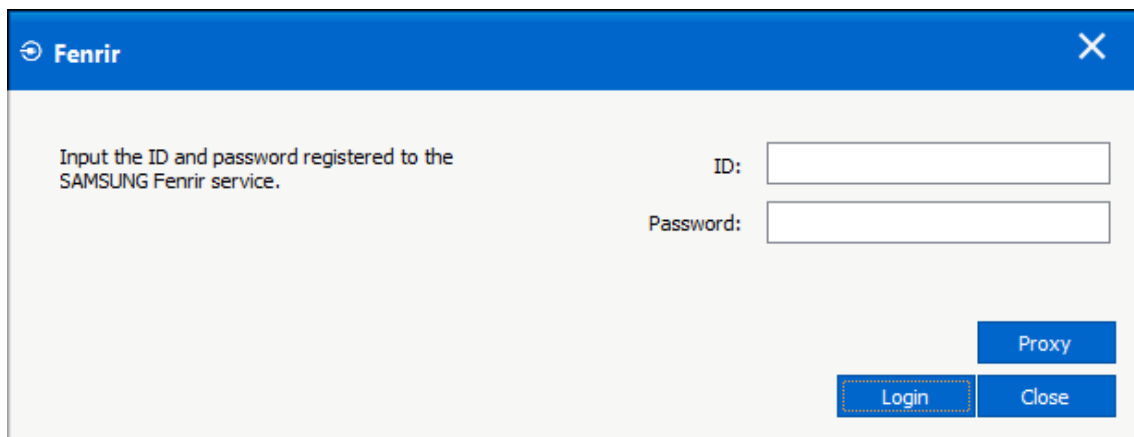
- SVH (Fenrir_Home) : It uses Home binary which does not have user data area in the memory when flashed to a device. (Keep user data)

- SVC (Fenrir_Factory) : It uses Factory binary which erases all user data in the memory when flashed to a device. (Clear user data)

- SVA (Fenrir_All) : It uses Factory and Home binaries. you can download Home and Factory binary in a PC(but requires double HDD storage and NW traffic)

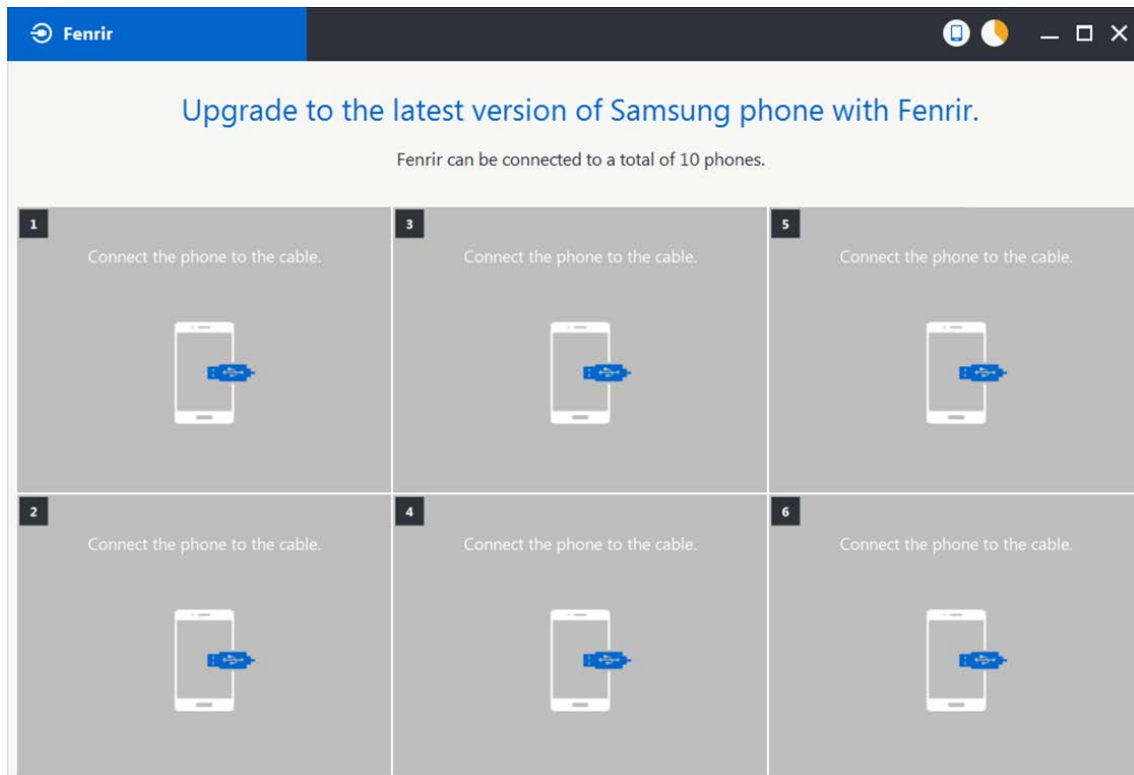
2) Input ID & password

※ You need to reset the ID information in case of PC change and format and repair, hard disk change

A screenshot of the Fenrir software login window. The window has a blue title bar with the 'Fenrir' logo and a close button. The main area is light gray. On the left, it says 'Input the ID and password registered to the SAMSUNG Fenrir service.' On the right, there are two input fields: 'ID:' and 'Password:'. Below the 'Password:' field, there are three buttons: 'Proxy' (blue), 'Login' (blue with a dashed border), and 'Close' (blue).

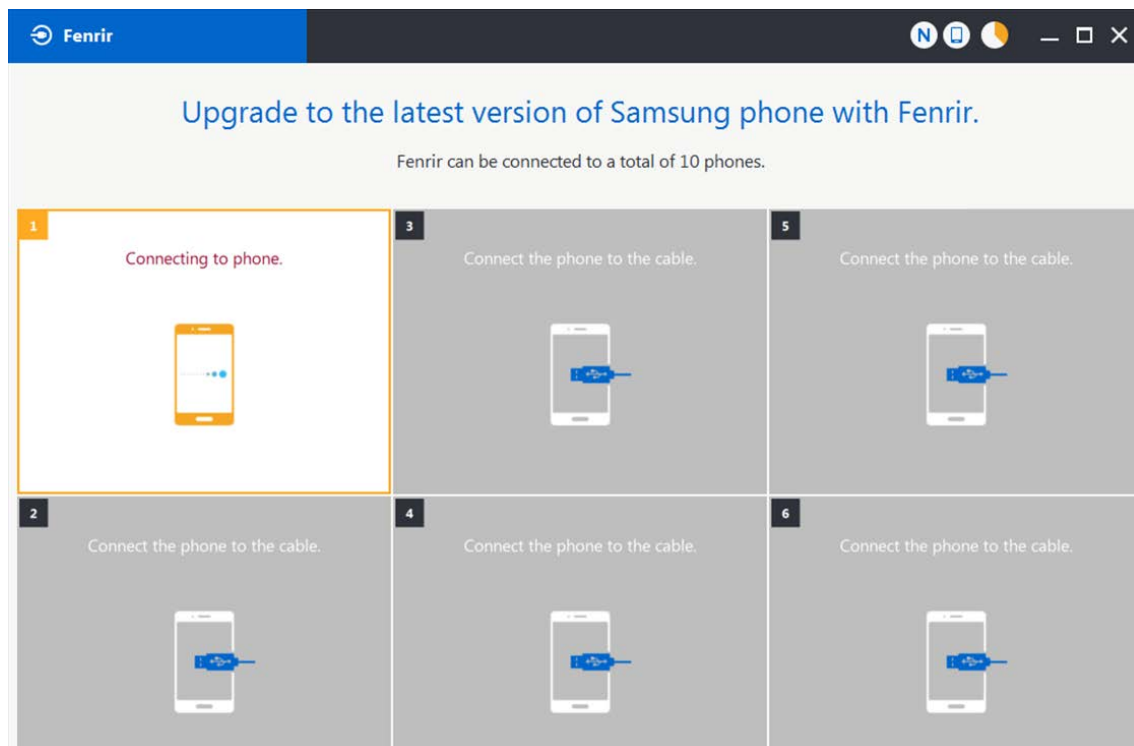
6. Level 1 Repair

3) Ensure device has sufficient charge (at least 20%) to start firmware update.



4) Connect the device to PC via data cable.

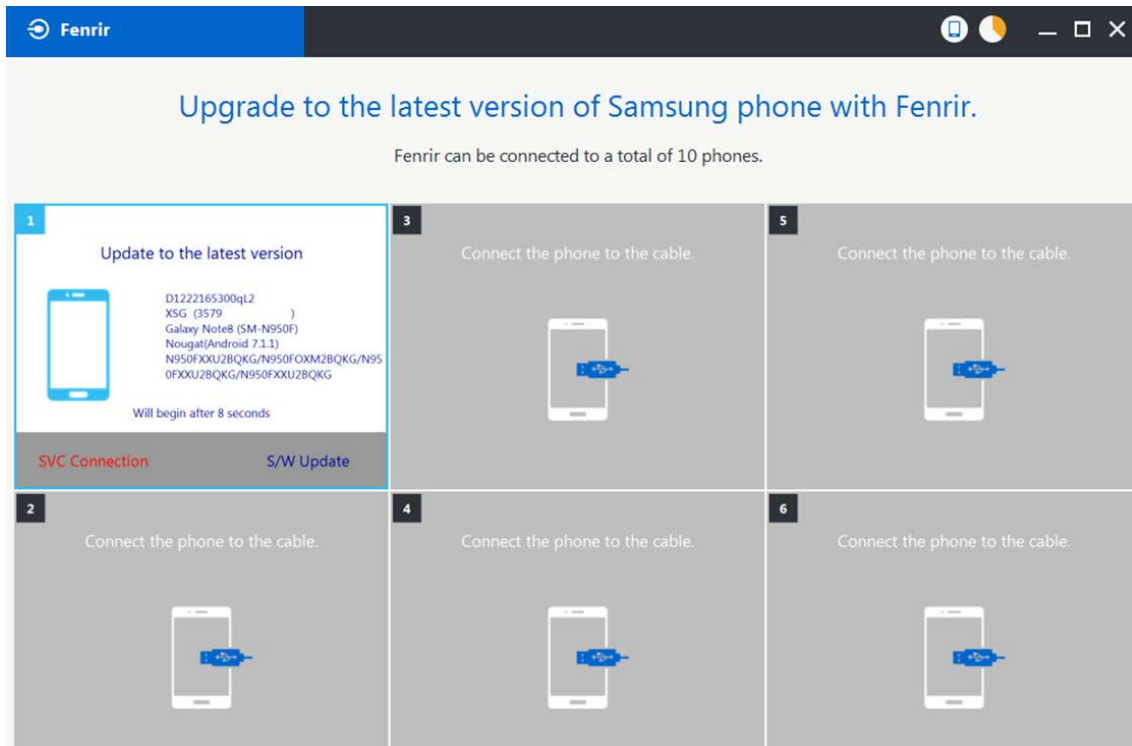
5) Upon USB connection, you will be presented with below screen.



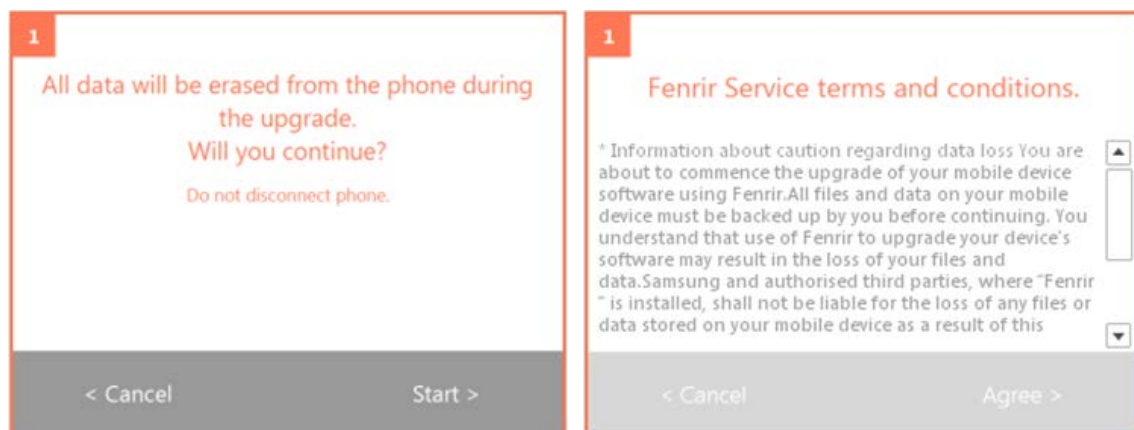
6. Level 1 Repair

=C

6) Once device is detected, you will be presented with below screen. To update S/W, select “S/W Update” or to exit select “SVC Connection”. If you select “SVC Connection”, only Fenrir connection history (record) will be stored in the FUS server to support warranty validation. (This is known as “Service Connection” history)

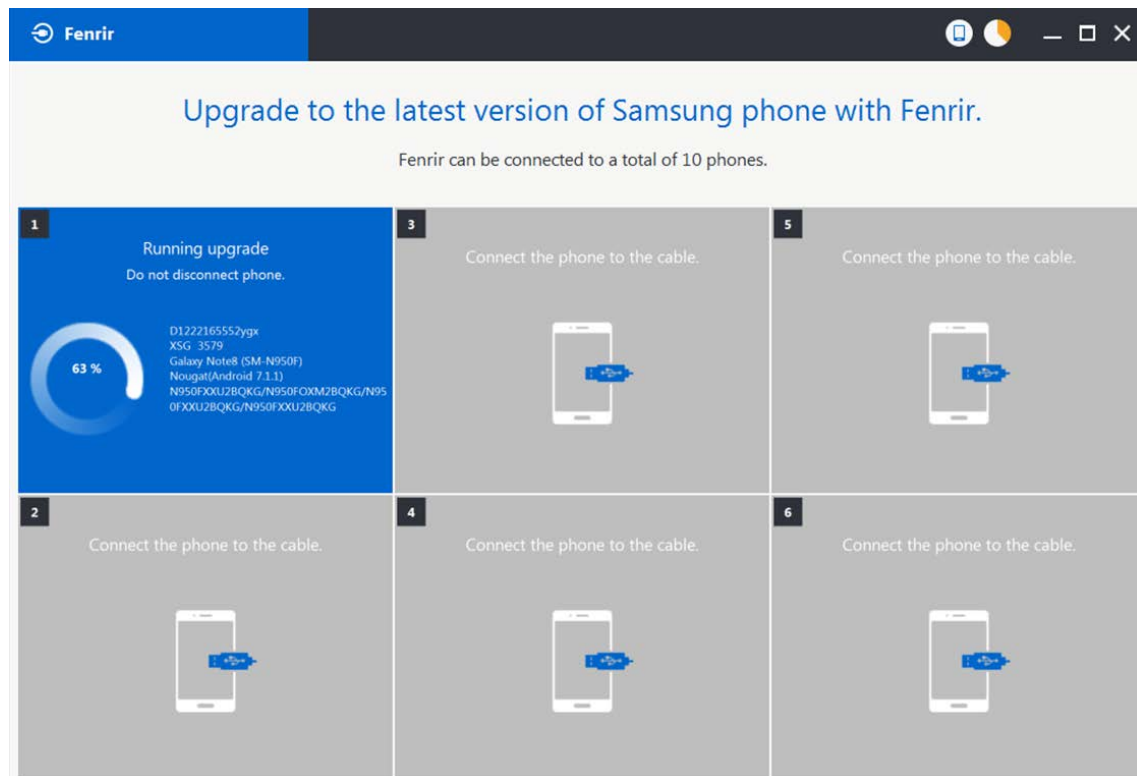


7) Once Fenrir starts, application will display the below screen. And select the Start button & Agree button.

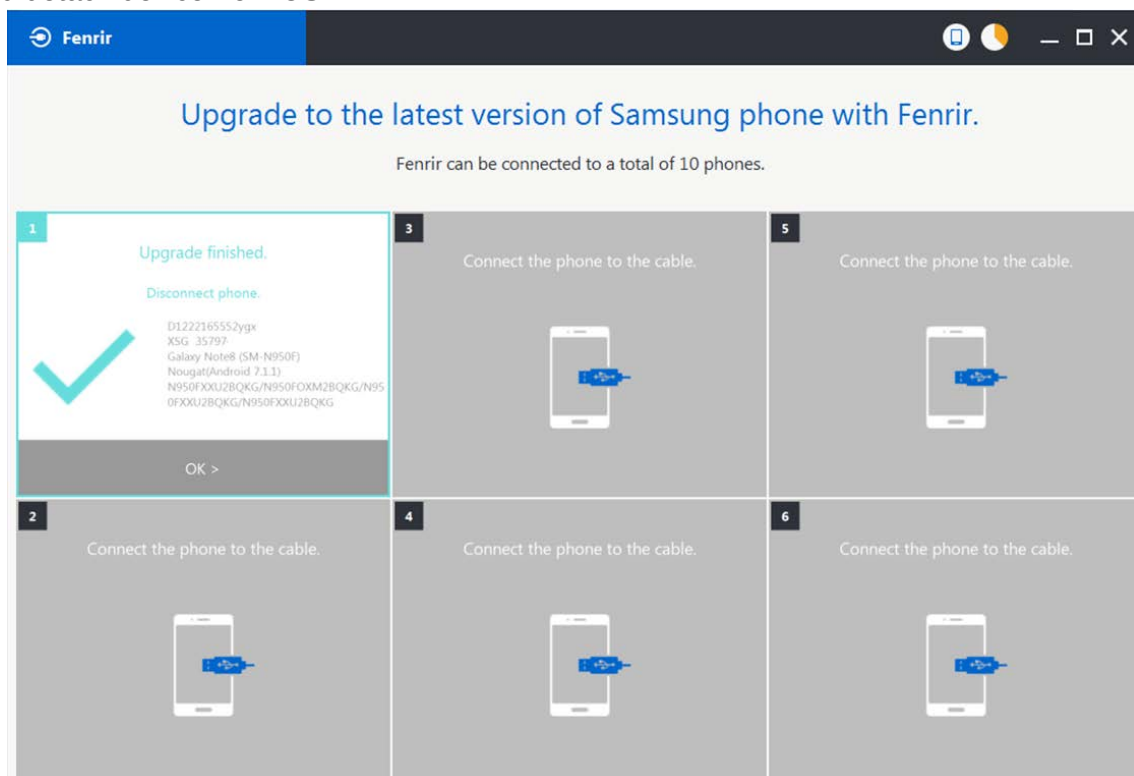


6. Level 1 Repair

8) The status circle increases as the update installs. The update process takes approximately 5-10 minutes to complete. Do not disconnect the device from USB during processing.



9) Once complete, application will present the below screen indicating update complete. Click Ok and detach device from USB.



6. Level 1 Repair

6-2. How to use 'Odin' program

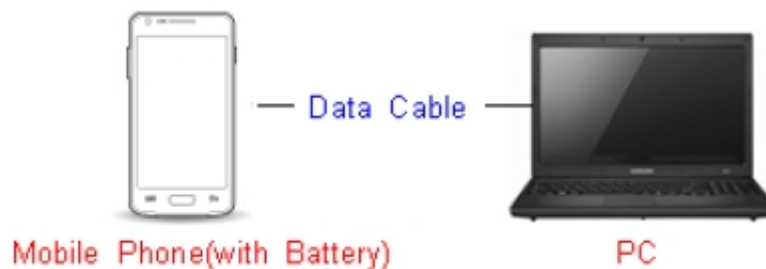
※ S/W Update via Fenrir is mandatory.

Below is the method to use 'Odin' program in any specific case.

6-2-1. Preparation

- Installation program : [Odin3 v3.13.2.exe or above](#)
- Mobile Phone
- Data Cable
- S/W Binary files (downloaded from GSPN)

※ Settings

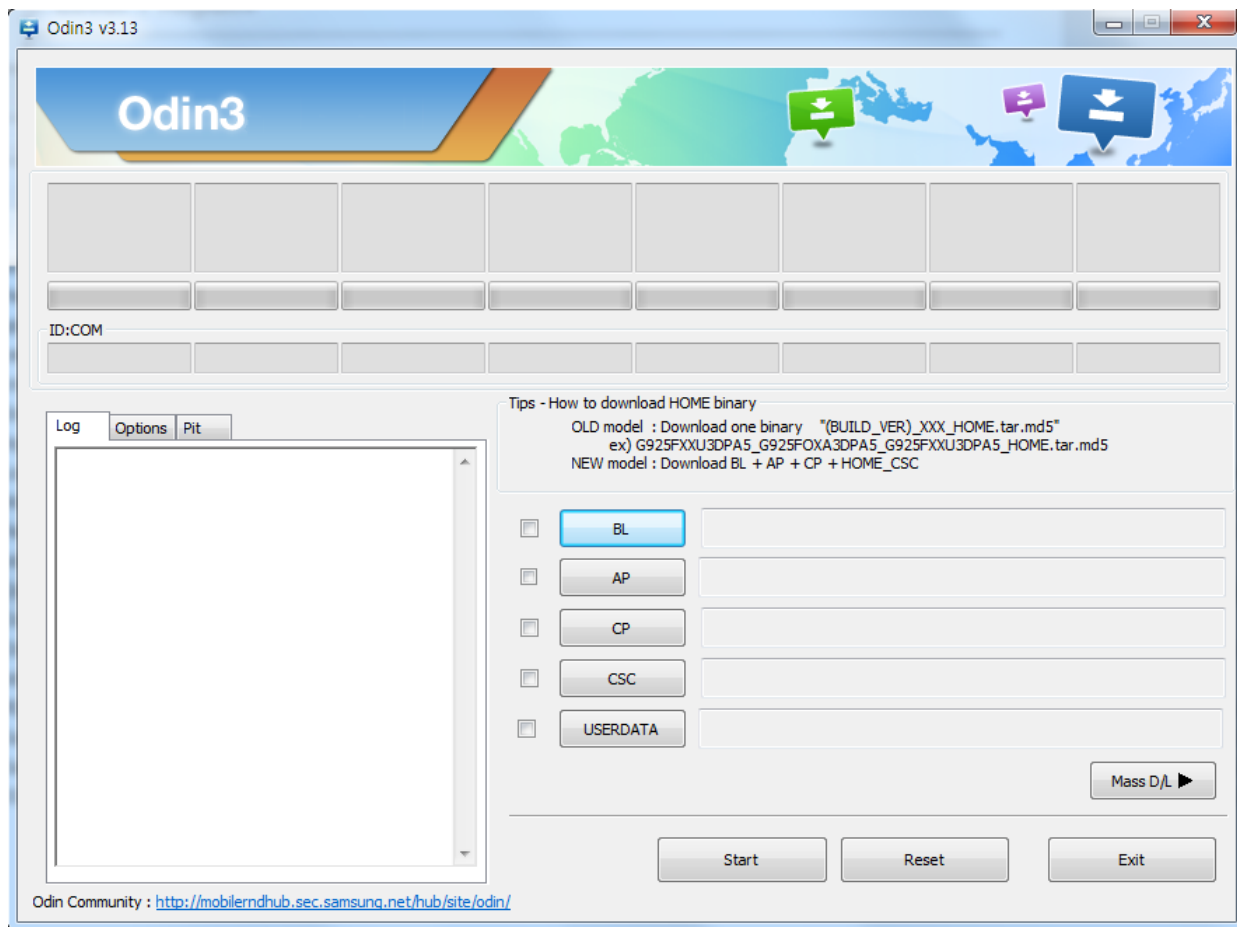


Data Cable : [GH39-01886A](#)

6. Level 1 Repair

6-2-2. S/W Installation Program (Downloader program)

Open up the S/W Installation Program by executing the "**Odin3 v3.13.2.exe**"

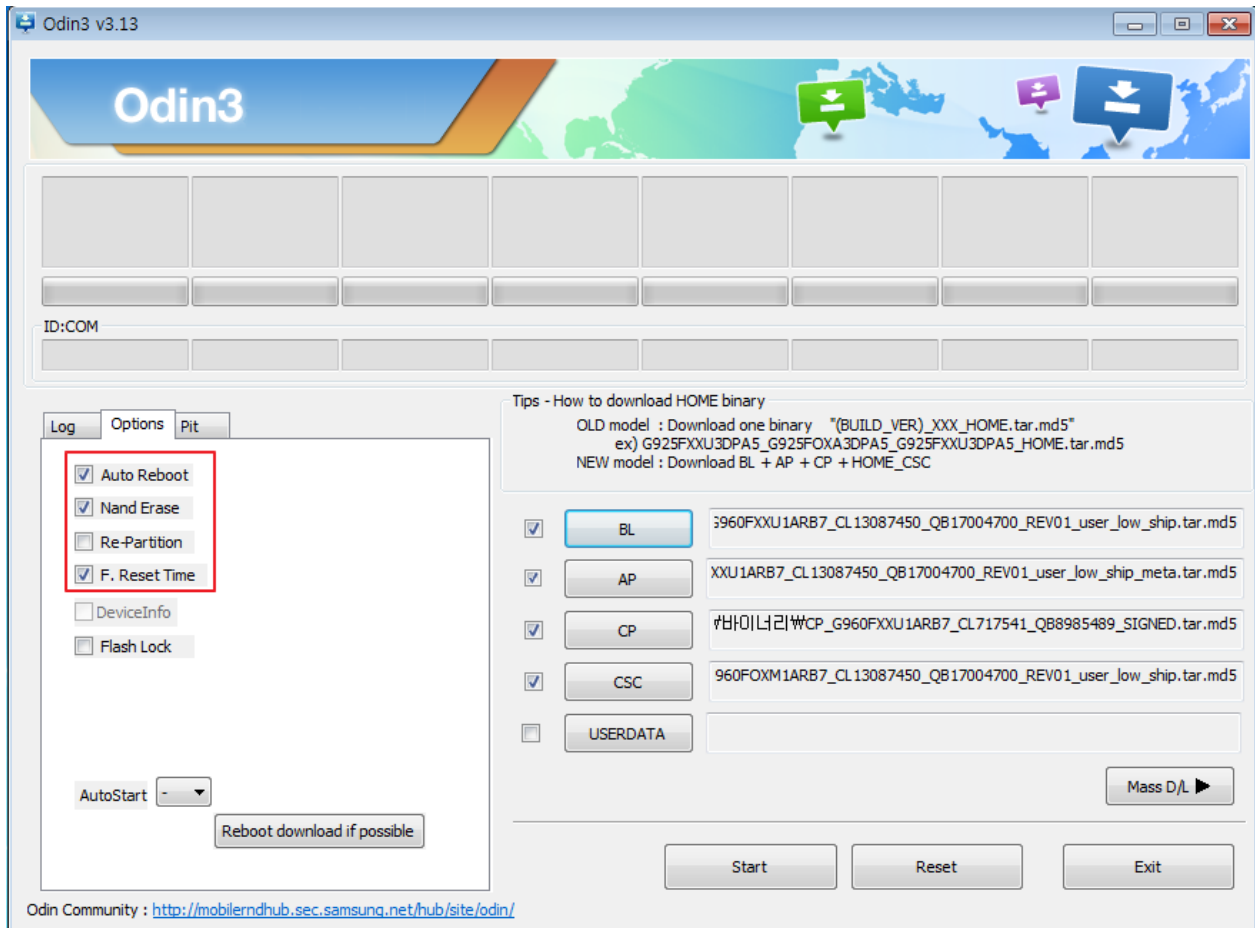


6. Level 1 Repair

1. Enable the check mark by click on the following options

- Check Auto Reboot, F. Reset Time, Nand Erase
- Check BL, AP, CP, CSC Files

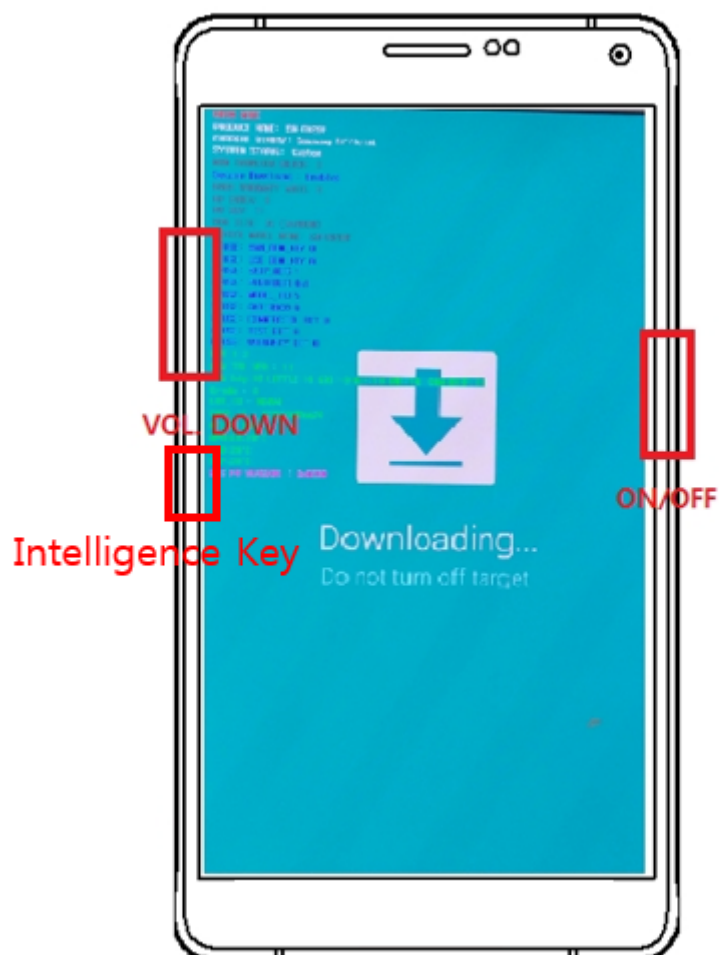
* Note : "Odin v3.13.2 or above" checks MD5 checksum just after file selection.



6. Level 1 Repair

2. Enter into Download Mode

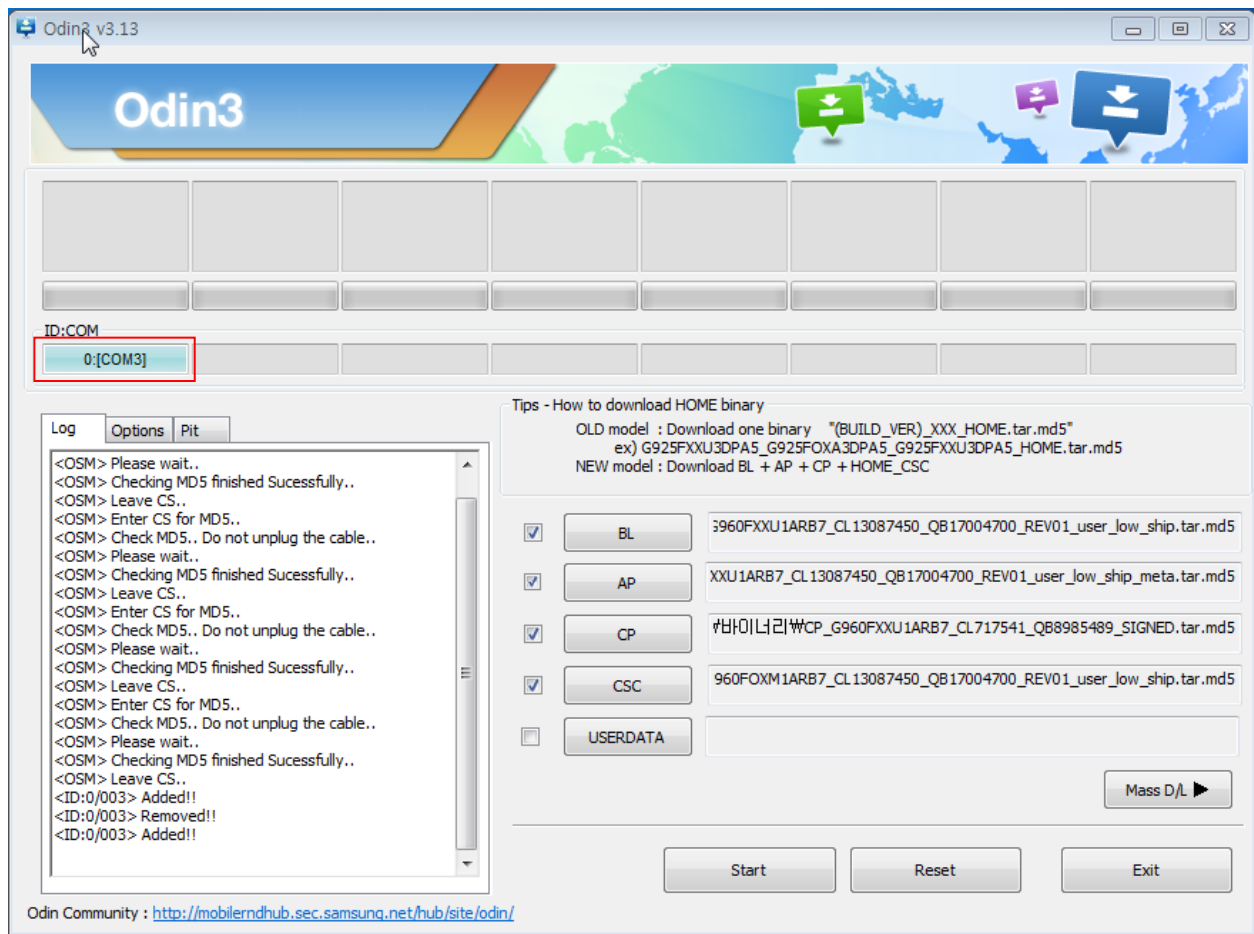
- Enter into Download Mode by pressing Volume Down button, Intelligence button and connecting USB cable



6. Level 1 Repair

3. Connect the device to PC via Data Cable.

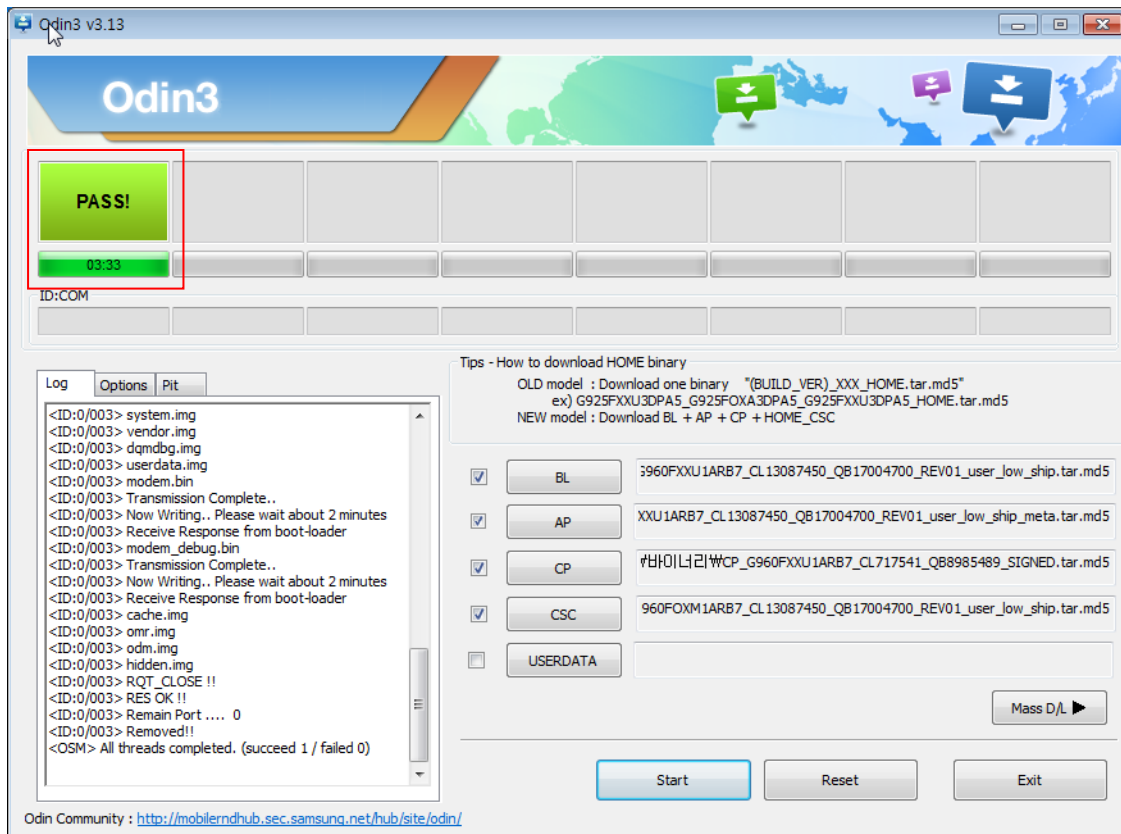
Make sure that the one of communication ports [ID:COM] box is highlighted in sky blue.
The device is now connected with the PC and ready to download the binary files in it.



6. Level 1 Repair

4. Start downloading the binary files into the device by clicking Start button on the screen.

The green colored "PASS!" sign will appear on the upper-left box if the binary files have been successfully downloaded into the device.



5. Disconnect the device from the Data cable.

6. Once the device boots up, you can check the version of the binary file or name by pressing the following code in sequence; ***#1234#**

You can perform Factory data Reset by Settings → General Management → Reset

※ Caution. Never disconnect during the S/W downloading.

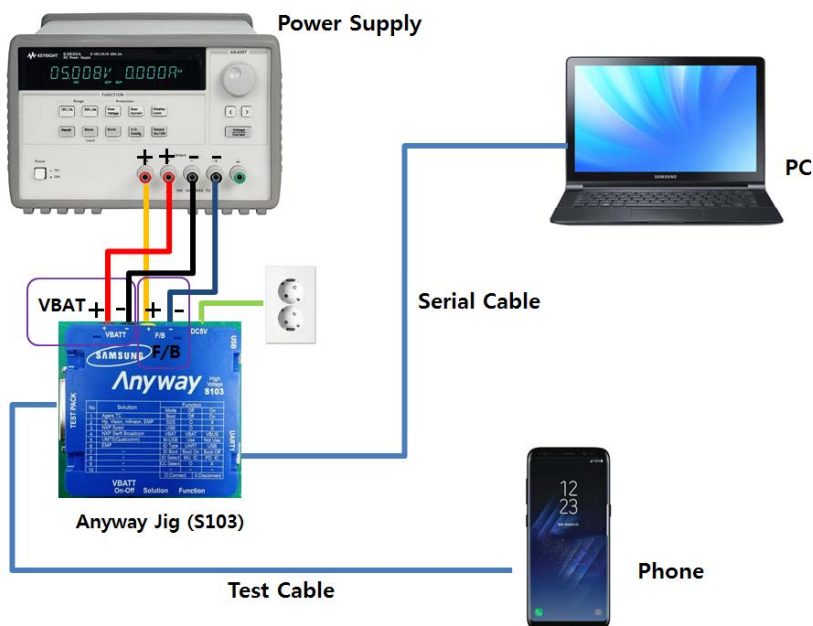
6. Level 1 Repair

6-3. IMEI writing

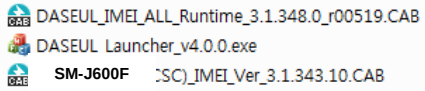
6-3-1. Preparation

- New IMEI writing Program has been released.
- Supported Model : Models which CAB files are uploaded on HHPsvc INI File category, instead of ini file.
- Refer to below IMEI writing procedure.

- H/W



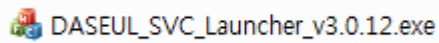
- S/W

① Library Install	To use Daseul, library files should be installed. Refer to SVC Bulletin “(11-82) Daseul (New IMEI writing Program) Library Install guide_rev1.0”
② Launcher	DASEUL_SVC_Launcher_v3.0.12 or higher -Uploaded on HHPsvc Notice
③ Runtime File	1. DASEUL_IMEI_ALL_Runtime_3.1.348.0_r00519.CAB or higher -Uploaded on HHPsvc Notice 2. Make 'ModelName' folder at the same position with launcher & Runtime file. 
④ Model File	Copy Model File under the 'SM-J600F' folder

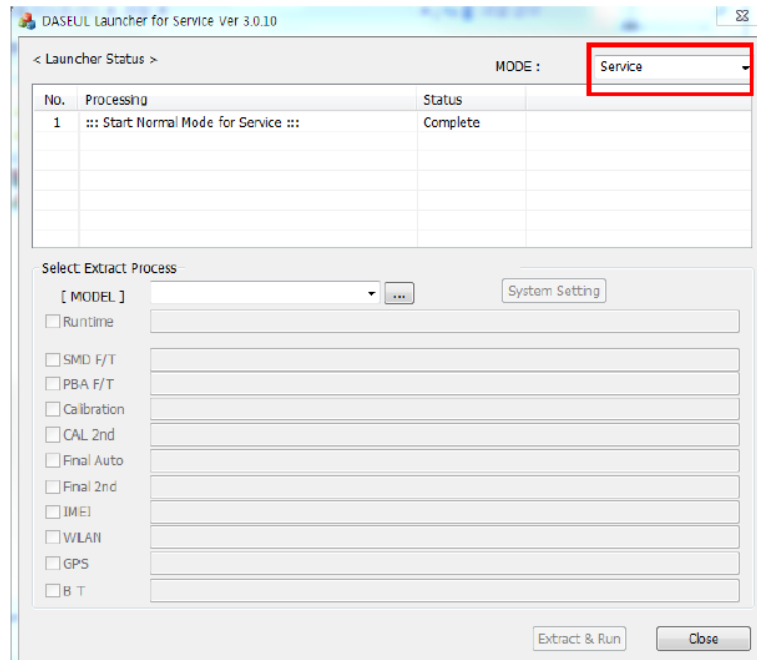
6. Level 1 Repair

6-3-2. IMEI writing Process

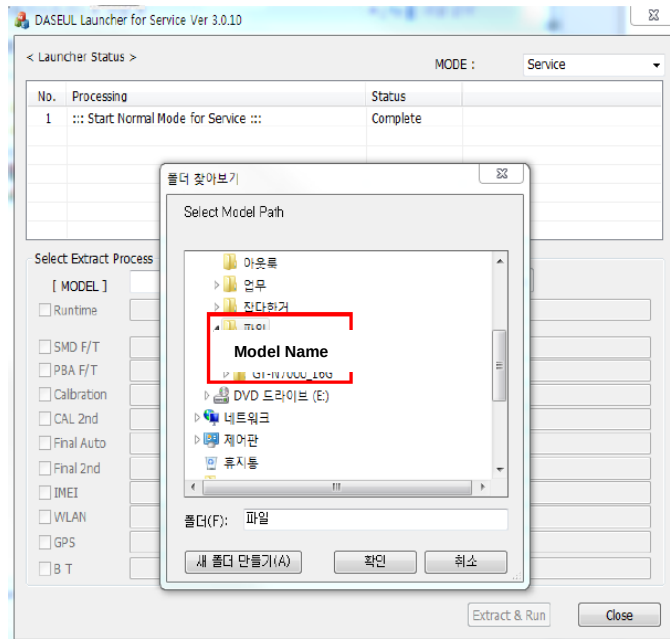
1. Run DASEUL_SVC_Launcher_v3.0.12.exe



2. Select Service Mode

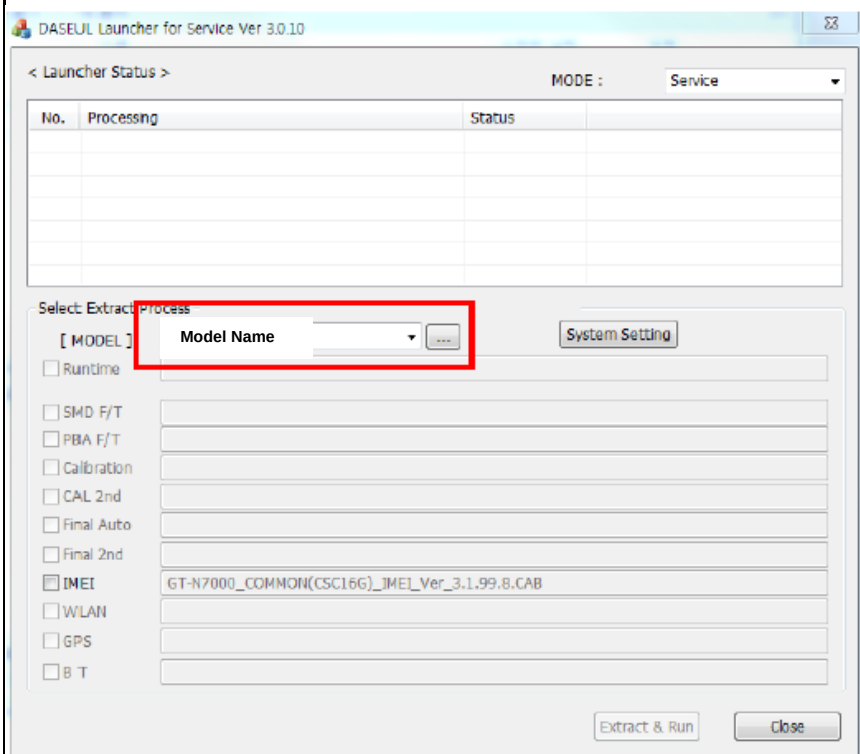


3. Click  and Select folder where the Launcher exists



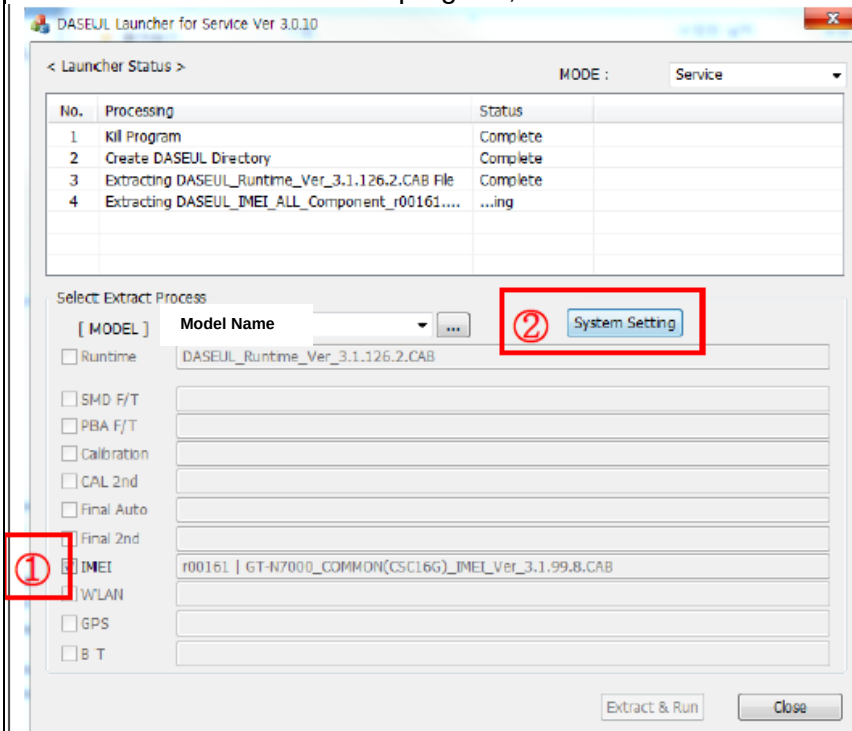
6. Level 1 Repair

4. Select Model



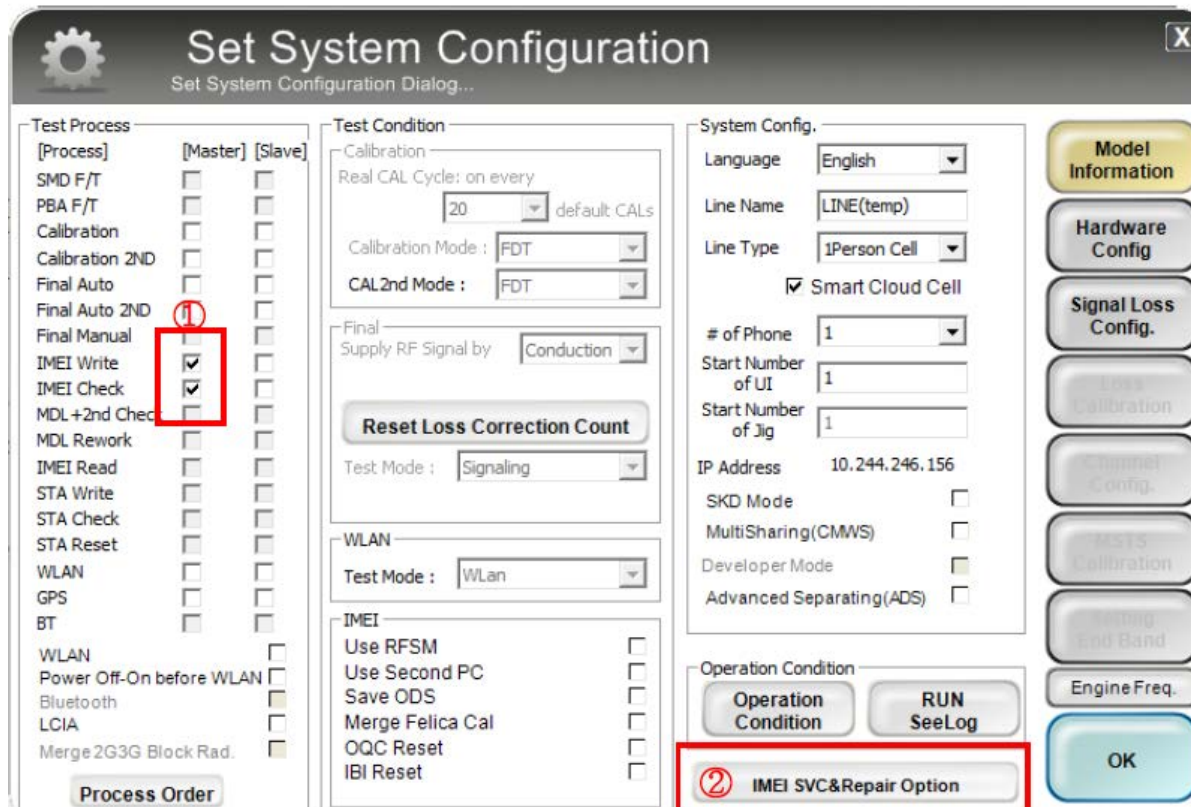
5. Check IMEI and click System Setting

※ Once you setup the setting, you don't have to do it again, unless there is change.
From second run of the IMEI program, check IMEI and click Extract & Run.



6. Level 1 Repair

6. Check IMEI Write / IMEI Check and click IMEI SVC & Repair Option.



The 'Set System Configuration' dialog box is shown. It has a title bar with a gear icon and a close button. The main area is divided into several sections: 'Test Process' (a list of checkboxes for various tests), 'Test Condition' (fields for calibration and RF signal), 'System Config.' (fields for language, line name, line type, etc.), and 'Operation Condition' (buttons for 'Operation Condition' and 'RUN SeeLog'). A red box highlights the 'IMEI Write' and 'IMEI Check' checkboxes in the 'Test Process' section, with a circled '1' next to them. Another red box highlights the 'IMEI SVC&Repair Option' button in the 'Operation Condition' section, with a circled '2' next to it.

Test Process	[Master]	[Slave]
SMD F/T	☐	☐
PBA F/T	☐	☐
Calibration	☐	☐
Calibration 2ND	☐	☐
Final Auto	☐	☐
Final Auto 2ND	☐	☐
Final Manual	☐	☐
IMEI Write	☒	☐
IMEI Check	☒	☐
MDL+2nd Check	☐	☐
MDL Rework	☐	☐
IMEI Read	☐	☐
STA Write	☐	☐
STA Check	☐	☐
STA Reset	☐	☐
WLAN	☐	☐
GPS	☐	☐
BT	☐	☐
WLAN	☐	☐
Power Off-On before WLAN	☐	☐
Bluetooth	☐	☐
LCIA	☐	☐
Merge 2G3G Block Rad.	☐	☐

Test Condition

Calibration

Real CAL Cycle: on every 20 default CALs

Calibration Mode: FDT

CAL2nd Mode: FDT

Final

Supply RF Signal by: Conduction

Reset Loss Correction Count

Test Mode: Signaling

WLAN

Test Mode: WLAN

IMEI

Use RFSM ☐

Use Second PC ☐

Save ODS ☐

Merge Felica Cal ☐

OQC Reset ☐

IBI Reset ☐

System Config.

Language: English

Line Name: LINE(temp)

Line Type: 1Person Cell

☒ Smart Cloud Cell

of Phone: 1

Start Number of UI: 1

Start Number of Jig: 1

IP Address: 10.244.246.156

SKD Mode ☐

MultiSharing(CMWS) ☐

Developer Mode ☐

Advanced Separating(ADS) ☐

Operation Condition

Operation Condition

IMEI SVC&Repair Option

Model Information

Hardware Config

Signal Loss Config.

Loss Calibration

Channel Config.

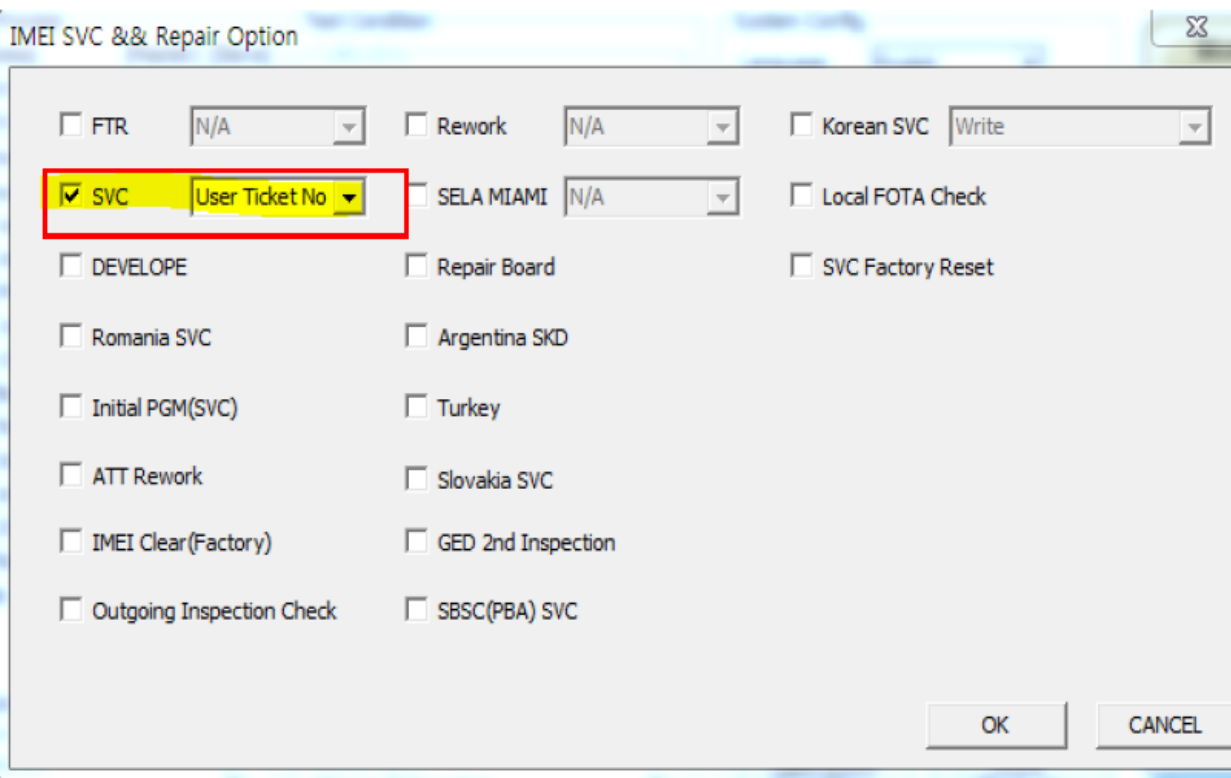
MSIS Calibration

Setting End Band

Engine Freq.

OK

7. Check 'SVC , User Ticket No' and click OK



The 'IMEI SVC & Repair Option' dialog box is shown. It has a title bar with a close button. The main area contains several checkboxes and dropdown menus. A red box highlights the 'SVC' checkbox and the 'User Ticket No' dropdown menu. The 'SVC' checkbox is checked. The 'User Ticket No' dropdown menu is set to 'User Ticket No'. Other options include 'FTR', 'Rework', 'Korean SVC', 'SELA MIAMI', 'Local FOTA Check', 'DEVELOPE', 'Repair Board', 'SVC Factory Reset', 'Romania SVC', 'Argentina SKD', 'Initial PGM(SVC)', 'Turkey', 'ATT Rework', 'Slovakia SVC', 'IMEI Clear(Factory)', 'GED 2nd Inspection', 'Outgoing Inspection Check', and 'SBSC(PBA) SVC'. The 'OK' and 'CANCEL' buttons are at the bottom right.

IMEI SVC & Repair Option

☐ FTR N/A ☐ Rework N/A ☐ Korean SVC Write

☒ SVC User Ticket No ☐ SELA MIAMI N/A ☐ Local FOTA Check

☐ DEVELOPE ☐ Repair Board ☐ SVC Factory Reset

☐ Romania SVC ☐ Argentina SKD

☐ Initial PGM(SVC) ☐ Turkey

☐ ATT Rework ☐ Slovakia SVC

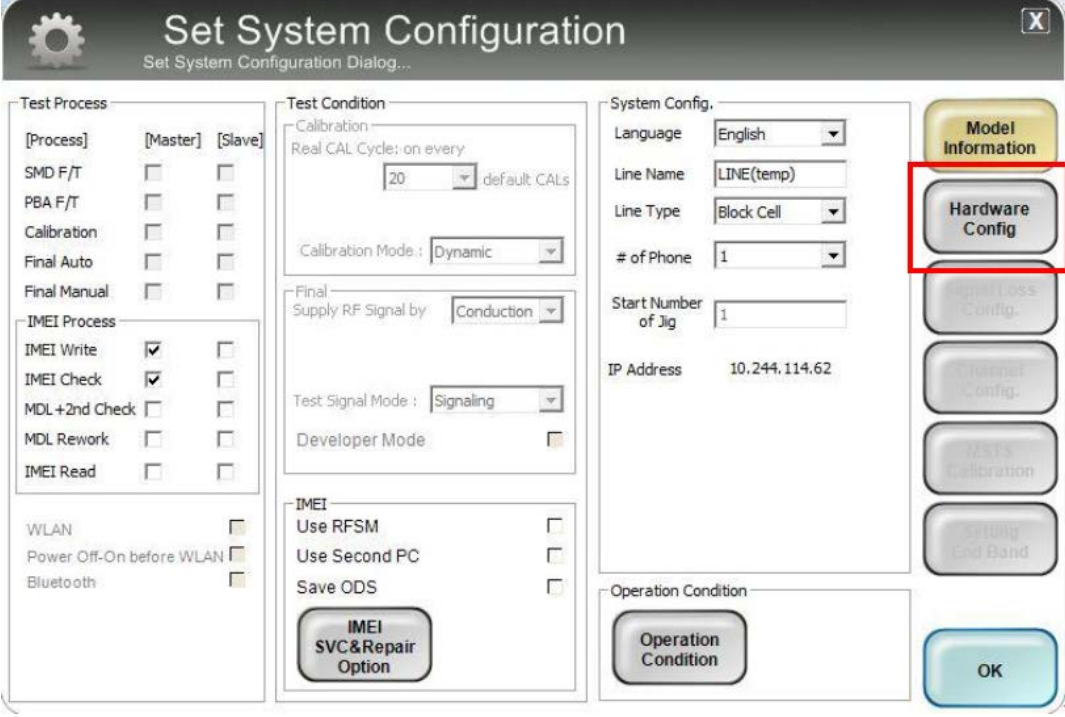
☐ IMEI Clear(Factory) ☐ GED 2nd Inspection

☐ Outgoing Inspection Check ☐ SBSC(PBA) SVC

OK CANCEL

6. Level 1 Repair

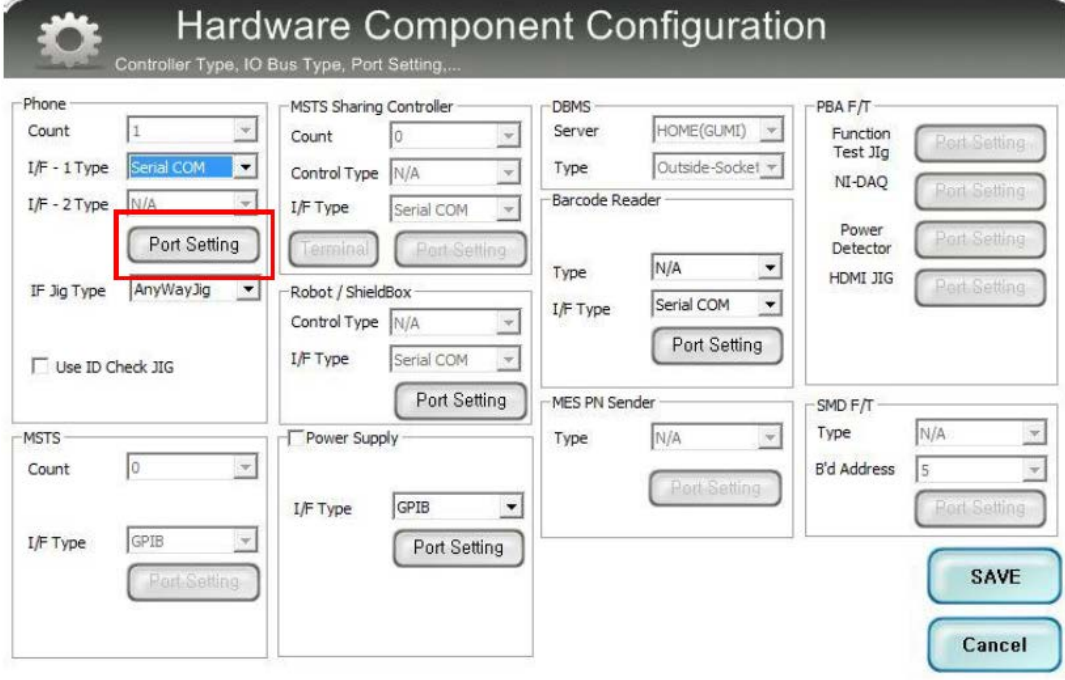
8. Click 'Hardware Config'



The 'Set System Configuration' dialog box is shown. It has a title bar with a gear icon and the text 'Set System Configuration Dialog...'. The dialog is divided into several sections:

- Test Process:** Includes checkboxes for [Process], [Master], [Slave], SMD F/T, PBA F/T, Calibration, Final Auto, Final Manual, IMEI Process, IMEI Write, IMEI Check, MDL+2nd Check, MDL Rework, IMEI Read, WLAN, Power Off-On before WLAN, and Bluetooth.
- Test Condition:** Includes a 'Calibration' section with 'Real CAL Cycle: on every' and a dropdown for '20' (default CALs), and a 'Calibration Mode' dropdown set to 'Dynamic'. It also has a 'Final' section with 'Supply RF Signal by' set to 'Conduction', a 'Test Signal Mode' dropdown set to 'Signaling', and a 'Developer Mode' checkbox.
- System Config.:** Includes 'Language' (English), 'Line Name' (LINE(temp)), 'Line Type' (Block Cell), '# of Phone' (1), 'Start Number of Jig' (1), and 'IP Address' (10.244.114.62).
- Operation Condition:** Includes a button labeled 'Operation Condition'.
- Model Information:** A vertical sidebar on the right with buttons for 'Model Information', 'Hardware Config' (highlighted with a red box), 'Signal Loss Config.', 'Channel Config.', 'MSTs Calibration', 'Setting End Band', and 'OK'.
- IMEI SVC&Repair Option:** A button at the bottom left.

9. Click 'Port Setting'



The 'Hardware Component Configuration' dialog box is shown. It has a title bar with a gear icon and the text 'Controller Type, IO Bus Type, Port Setting,...'. The dialog is divided into several sections:

- Phone:** Includes 'Count' (1), 'I/F - 1 Type' (Serial COM), 'I/F - 2 Type' (N/A), 'IF Jig Type' (AnyWayJig), and a checkbox for 'Use ID Check JIG'. A 'Port Setting' button is highlighted with a red box.
- MSTS Sharing Controller:** Includes 'Count' (0), 'Control Type' (N/A), 'I/F Type' (Serial COM), and buttons for 'Terminal' and 'Port Setting'.
- Robot / ShieldBox:** Includes 'Control Type' (N/A), 'I/F Type' (Serial COM), and a 'Port Setting' button.
- Power Supply:** Includes 'I/F Type' (GPIO) and a 'Port Setting' button.
- DBMS:** Includes 'Server' (HOME(GUMI)), 'Type' (Outside-Socket), and buttons for 'Port Setting'.
- Barcode Reader:** Includes 'Type' (N/A), 'I/F Type' (Serial COM), and a 'Port Setting' button.
- MES PN Sender:** Includes 'Type' (N/A) and a 'Port Setting' button.
- PBA F/T:** Includes 'Function Test Jig', 'NI-DAQ', 'Power Detector', and 'HDMI JIG', each with a 'Port Setting' button.
- SMD F/T:** Includes 'Type' (N/A), 'B'd Address' (5), and a 'Port Setting' button.
- MSTS:** Includes 'Count' (0), 'I/F Type' (GPIO), and a 'Port Setting' button.
- Buttons:** 'SAVE' and 'Cancel' buttons at the bottom right.

6. Level 1 Repair

10. Select Port Number and SAVE

Set IO BUS Configuration

Phone IO Bus Setting

Common

BaudRate: 115200
Data Bit: 8
Parity: No
Stop Bit: 1

No.	Port #1
1	1

SAVE

Cancel

11. Click OK to proceed

Set System Configuration

Set System Configuration Dialog...

Test Process

[Process]	[Master]	[Slave]
SMD F/T	☐	☐
PBA F/T	☐	☐
Calibration	☐	☐
Final Auto	☐	☐
Final Manual	☐	☐

IMEI Process

IMEI Write	☒	☐
IMEI Check	☒	☐
MDL +2nd Check	☐	☐
MDL Rework	☐	☐
IMEI Read	☐	☐

WLAN ☐
Power Off-On before WLAN ☐
Bluetooth ☐

Test Condition

Calibration
Real CAL Cycle: on every 20 default CALs
Calibration Mode: Dynamic

Final
Supply RF Signal by: Conduction

Test Signal Mode: Signaling

Developer Mode ☐

IMEI

Use RFSM ☐
Use Second PC ☐
Save ODS ☐

IMEI SVC&Repair Option

System Config.

Language: English
Line Name: LINE(temp)
Line Type: Block Cell
of Phone: 1
Start Number of Jig: 1
IP Address: 10.244.114.62

Operation Condition

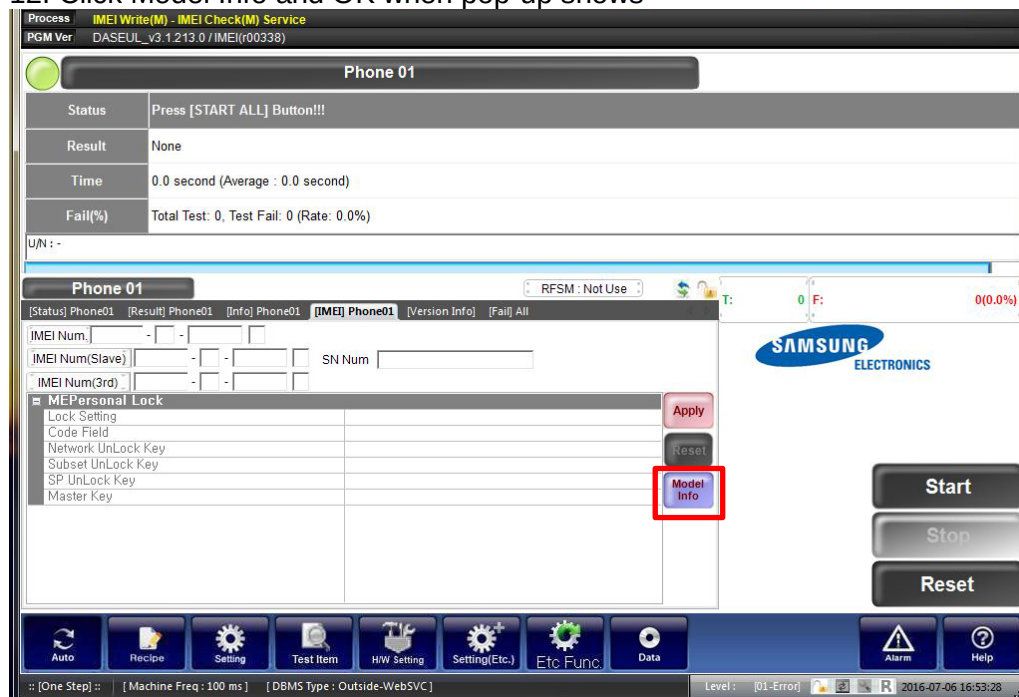
Operation Condition

Model Information
Hardware Config
Signal Loss Config.
Channel Config.
UART Calibration
Setting End Band

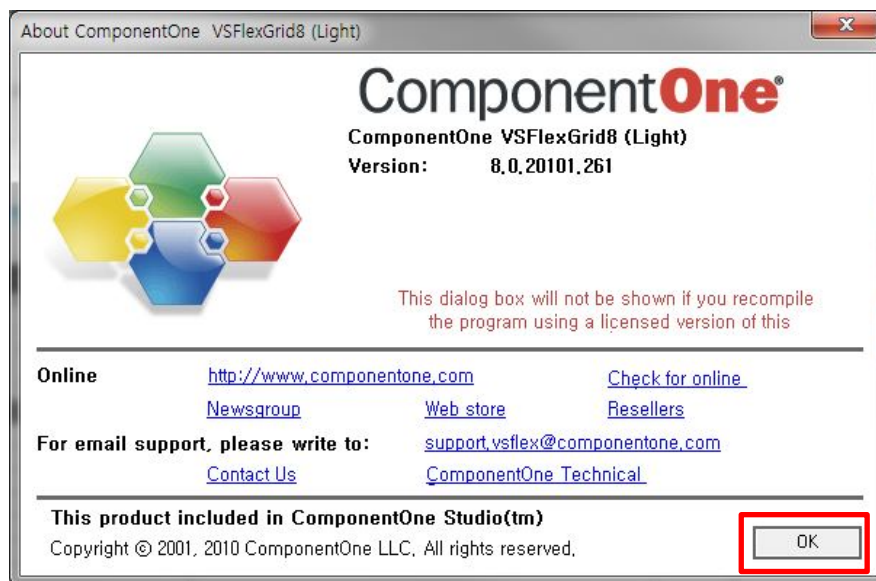
OK

6. Level 1 Repair

12. Click Model Info and OK when pop-up shows



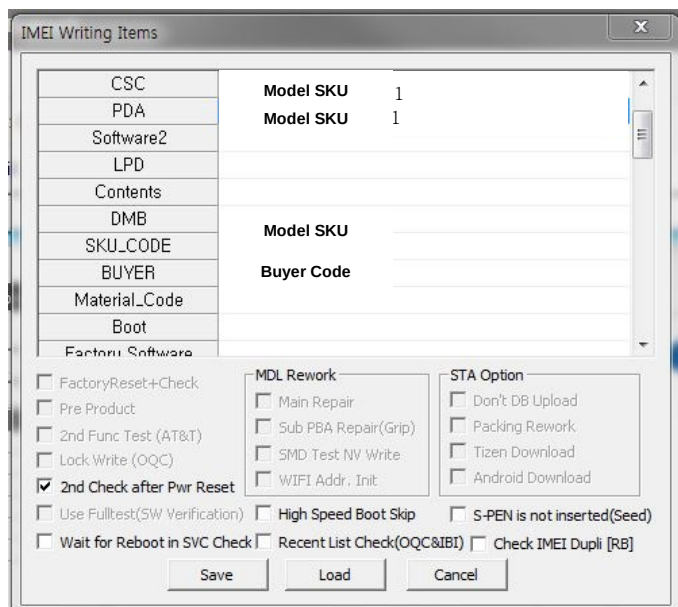
13. Click OK



6. Level 1 Repair

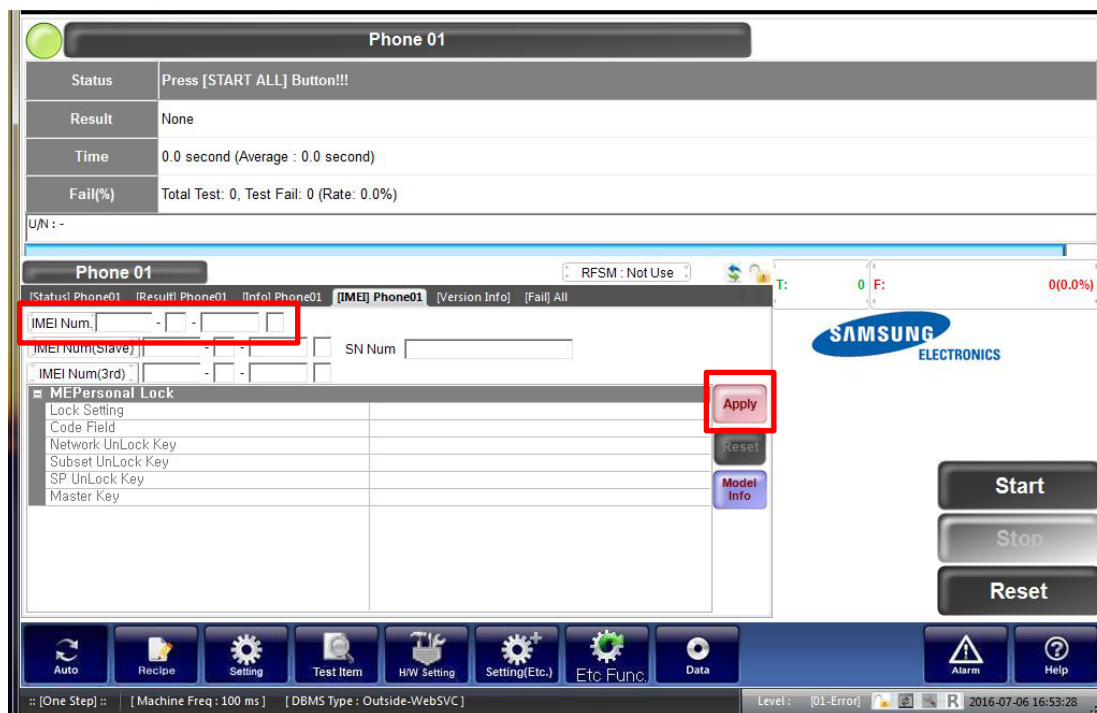
14. Input SKU_CODE and BUYER, then click Save button.

※ Refer to HHPsvc→IMEI Review to check SKU Code and buyer



The 'IMEI Writing Items' dialog box contains a list of items on the left and input fields on the right. The items listed are CSC, PDA, Software2, LPD, Contents, DMB, SKU_CODE, BUYER, Material_Code, Boot, and Factory Software. The input fields for Model SKU and Buyer Code are currently empty. Below the list, there are several checkboxes for various options: FactoryReset+Check, Pre Product, 2nd Func. Test (AT&T), Lock Write (OQC), 2nd Check after Pwr Reset (checked), Use Fulltest(SW Verification), Wait for Reboot in SVC Check, MDL Rework (Main Repair, Sub PBA Repair(Grip), SMD Test NV Write, WIFI Addr. Init), High Speed Boot Skip, Recent List Check(OQC&IBI), STA Option (Don't DB Upload, Packing Rework, Tizen Download, Android Download), S-PEN is not inserted(Seed), and Check IMEI Dupli [RB]. At the bottom are Save, Load, and Cancel buttons.

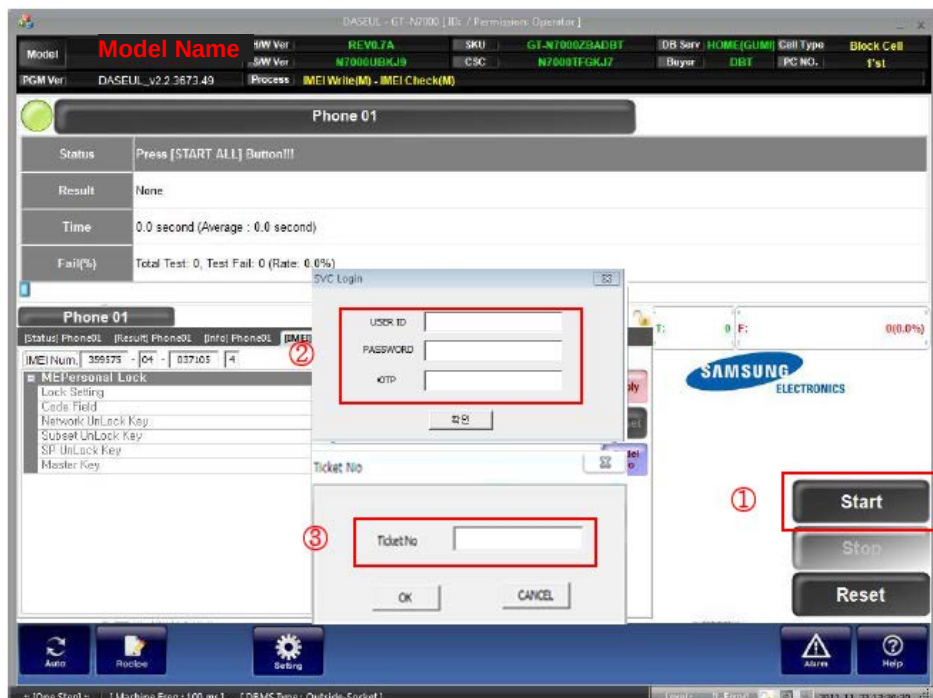
15. Input IMEI Number and click Apply



The 'Phone 01' interface shows a status bar at the top with 'Status: Press [START ALL] Button!!!', 'Result: None', 'Time: 0.0 second (Average : 0.0 second)', and 'Fail(%): Total Test: 0, Test Fail: 0 (Rate: 0.0%)'. Below this is a table with columns for Status, Result, Time, and Fail(%). The table has one row with 'Status: Press [START ALL] Button!!!', 'Result: None', 'Time: 0.0 second (Average : 0.0 second)', and 'Fail(%): Total Test: 0, Test Fail: 0 (Rate: 0.0%)'. The interface also includes a 'Phone 01' section with a 'IMEI Num.' field (highlighted with a red box) and an 'Apply' button (highlighted with a red box). Other fields include 'IMEI Num(Slave)', 'IMEI Num(3rd)', 'SN Num', and 'MEPersonal Lock'. The bottom of the interface features a toolbar with icons for Auto, Recipe, Setting, Test Item, HiW Setting, Setting(Etc.), Etc Func., Data, Alarm, and Help. The status bar at the very bottom shows 'Level : [01-Error]', 'DBMS Type : Outside-WebSVC', and the date '2016-07-06 16:53:28'.

6. Level 1 Repair

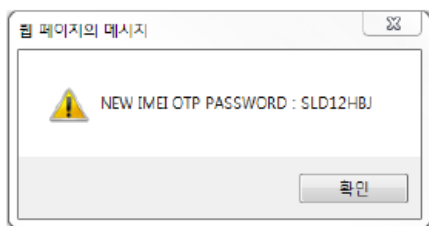
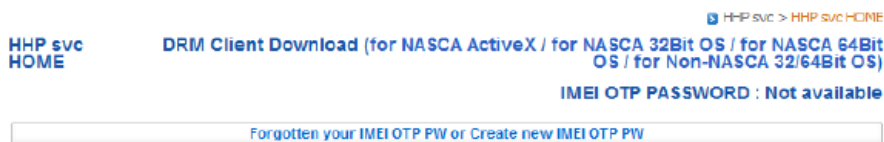
16. ① Click Start → ② Input IMEI writing ID and Password & OTP → ③ Input Ticket No



※ OTP(One time Password) : OTP is valid for 6 hours.

After that, you can get new OTP by click the “Forgotten your IMEI OTP PW or Create new IMEI OTP PW” button.

☞ OTP Location : GSPN → Knowledge → HHP svc → Home

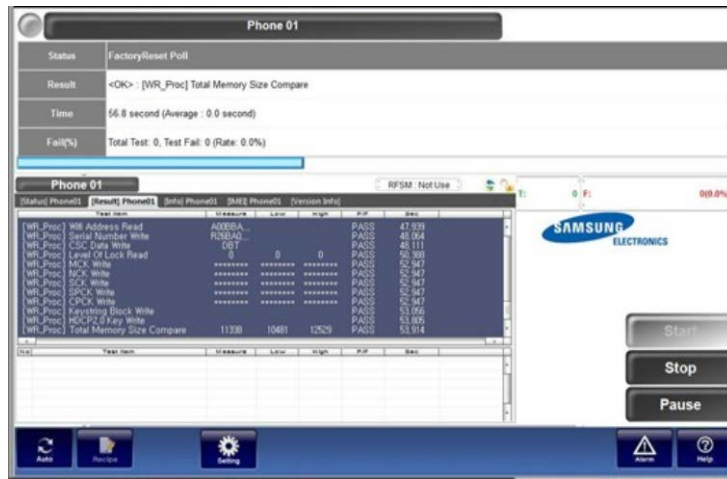


6. Level 1 Repair

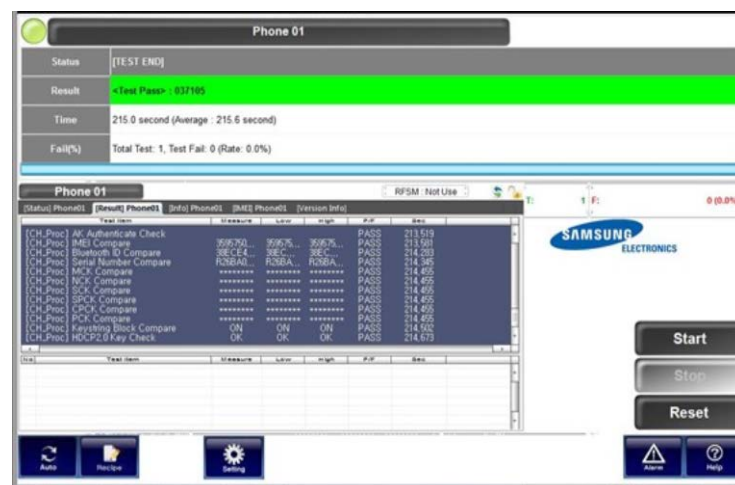
17. Connect the phone to Anyway JIG

- ※ When you connect the phone, the phone should be turned off.
After connecting the phone, the phone will be booted automatically.

18. IMEI Writing Proceeding



19. IMEI Writing Success



6. Level 1 Repair

6-4. RF Calibration

6-4-1. Required items in order to calibrate RF

- Installation program: RF Calibration Program
 - Daseul_Launcher_vx.x.xx.exe
 - Daseul_CAL_ALL_Runtime_x.x.xxx.x.CAB
 - Model File
- : **Model Name_OPEN_CALIBRATION_Ver_x.x.xxx.x.CAB**

※ It is required to use the latest program.

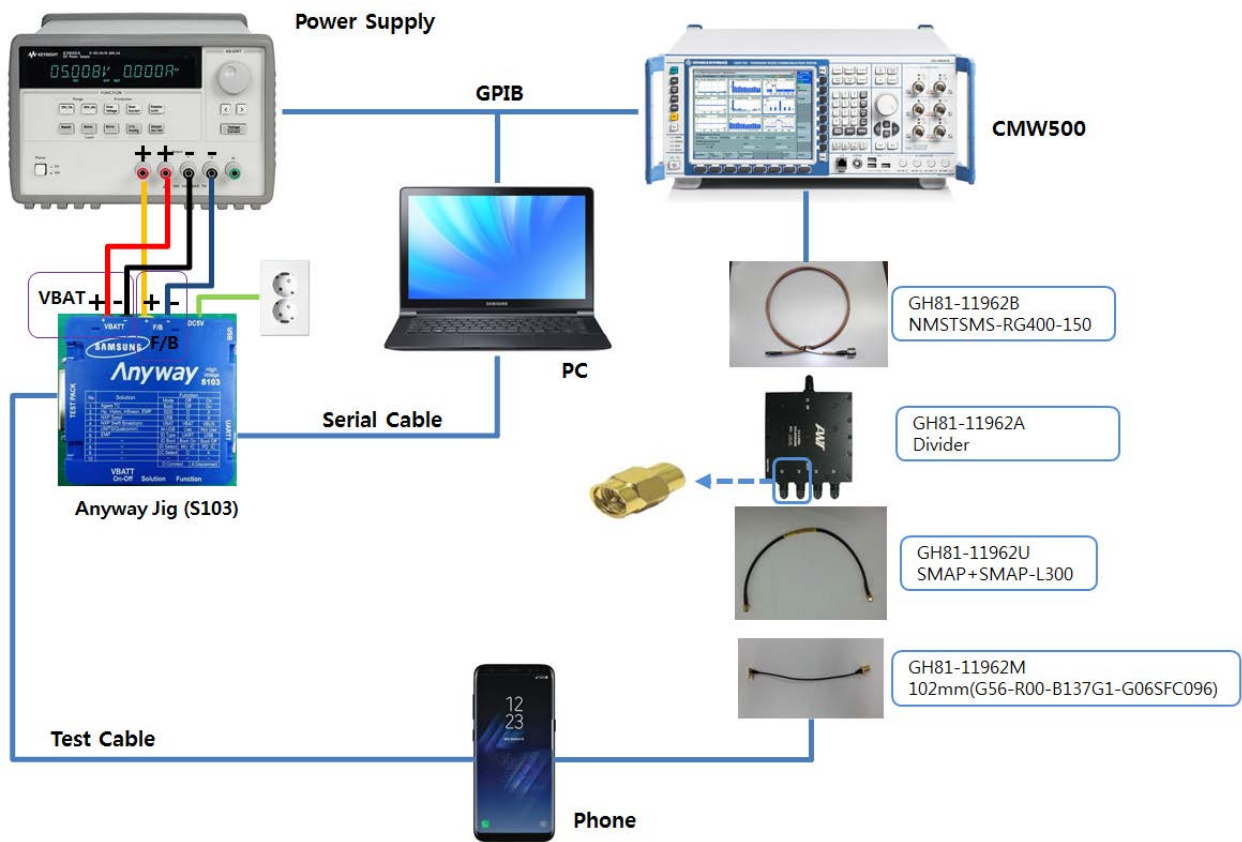
- Mobile Phone
- R&S CMW500
- E3632A Power Supply
- GPIB Cable (2ea)
- JIG BOX (S103)
- Adapter
- UART Serial Cable
- IF Cable (GH81-10962W)

❖ Table of test cables

RF Cable (Manual)	GH81-11962M (2ea)	GH81-11962U (2ea)	
	1.2T, 102mm 	1.2T, 102mm 	
4 Port Divider	GH81-11962A Divider 	GH81-11962B Divider Cable 	GH81-11962E 50Ω terminator 

6. Level 1 Repair

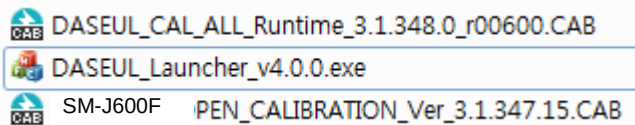
❖ Setting



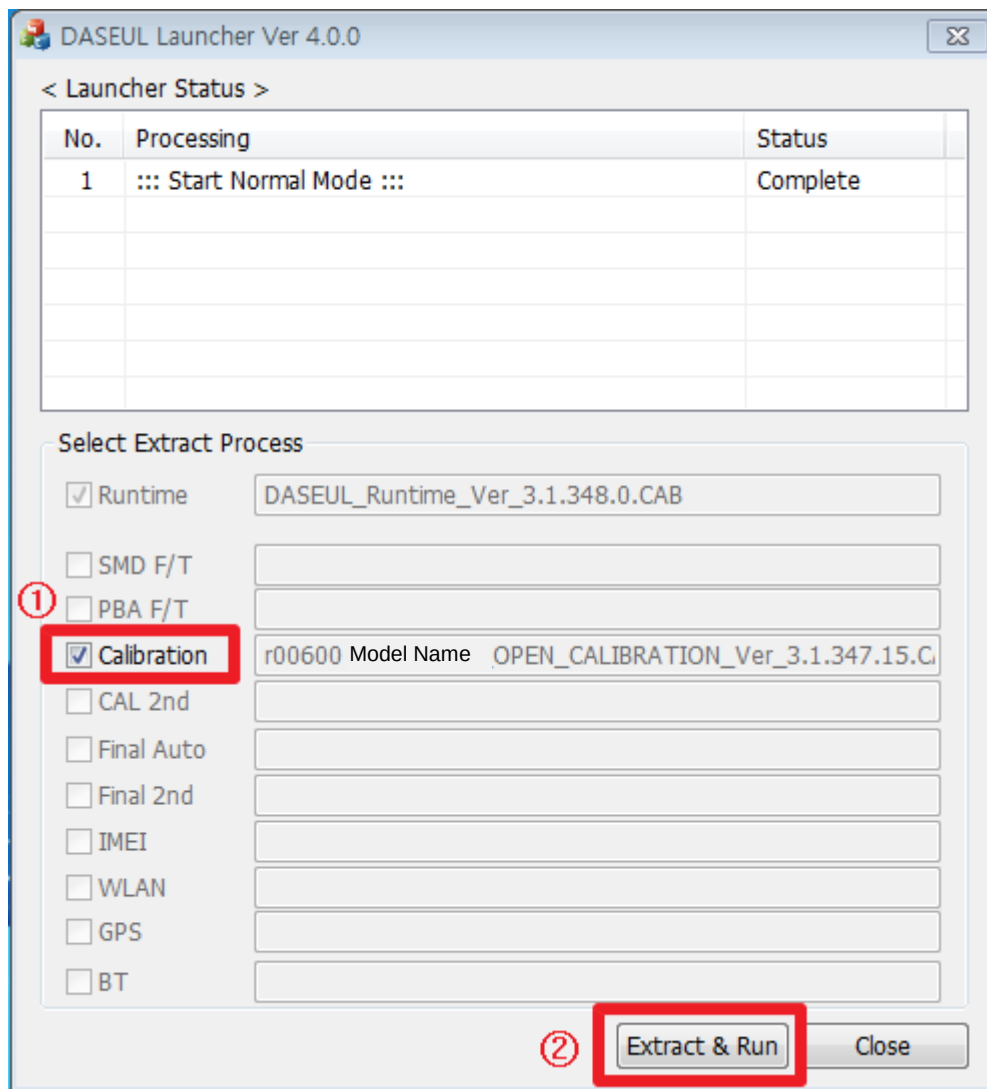
6. Level 1 Repair

6-4-2. RF Calibration Program

1. Run the RF Calibration Program Launcher, 'DASEUL_Launcher_vx.x.xx.exe'.

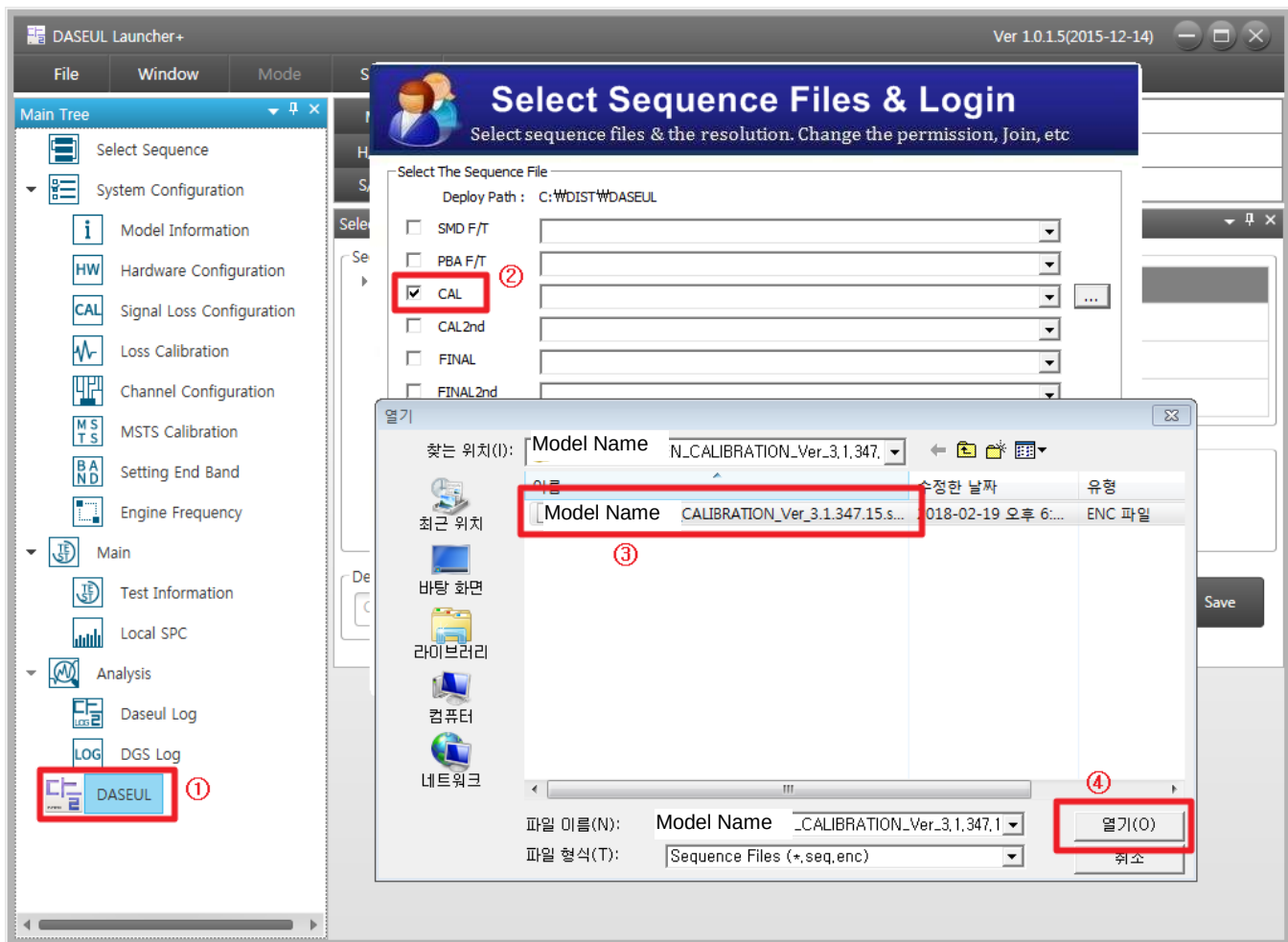


2. Check the 'Calibration' option and Click 'Extract & Run'.



6. Level 1 Repair

3. Check the 'CAL' and open the [model file](#), then select 'Start' button.



6. Level 1 Repair

4. Change the Line Type to 'Block Cell' and disable 'Smart Cloud Cell'.

Set System Configuration
Set System Configuration Dialog...

Test Process

[Process]	[Master]	[Slave]
SMD F/T	☐	☐
PBA F/T	☐	☐
Calibration	☒	☐
Calibration 2ND	☐	☐
Final Auto	☐	☐
Final Auto 2ND	☐	☐
Final Manual	☐	☐
IMEI Write	☐	☐
IMEI Check	☐	☐
MDL+2nd Check	☐	☐
MDL Rework	☐	☐
IMEI Read	☐	☐
STA Write	☐	☐
STA Check	☐	☐
STA Reset	☐	☐
WLAN	☐	☐
GPS	☐	☐
BT	☐	☐
WLAN	☐	☐
Power Off-On before WLAN	☐	☐
Bluetooth	☐	☐
LCIA	☐	☐
Merge 2G3G Block Rad.	☐	☐

Test Condition

Calibration
Real CAL Cycle: on every 20 default CALs
Calibration Mode : FDT
CAL2nd Mode : FDT

Final
Supply RF Signal by Conduction

Reset Loss Correction Count

Test Mode : Signaling

WLAN
Test Mode : WLAN

IMEI
Use RFSM
Use Second PC
Save ODS
Merge Felica Cal
OQC Reset
IBI Reset
OQC SKD USER D/L

System Config.

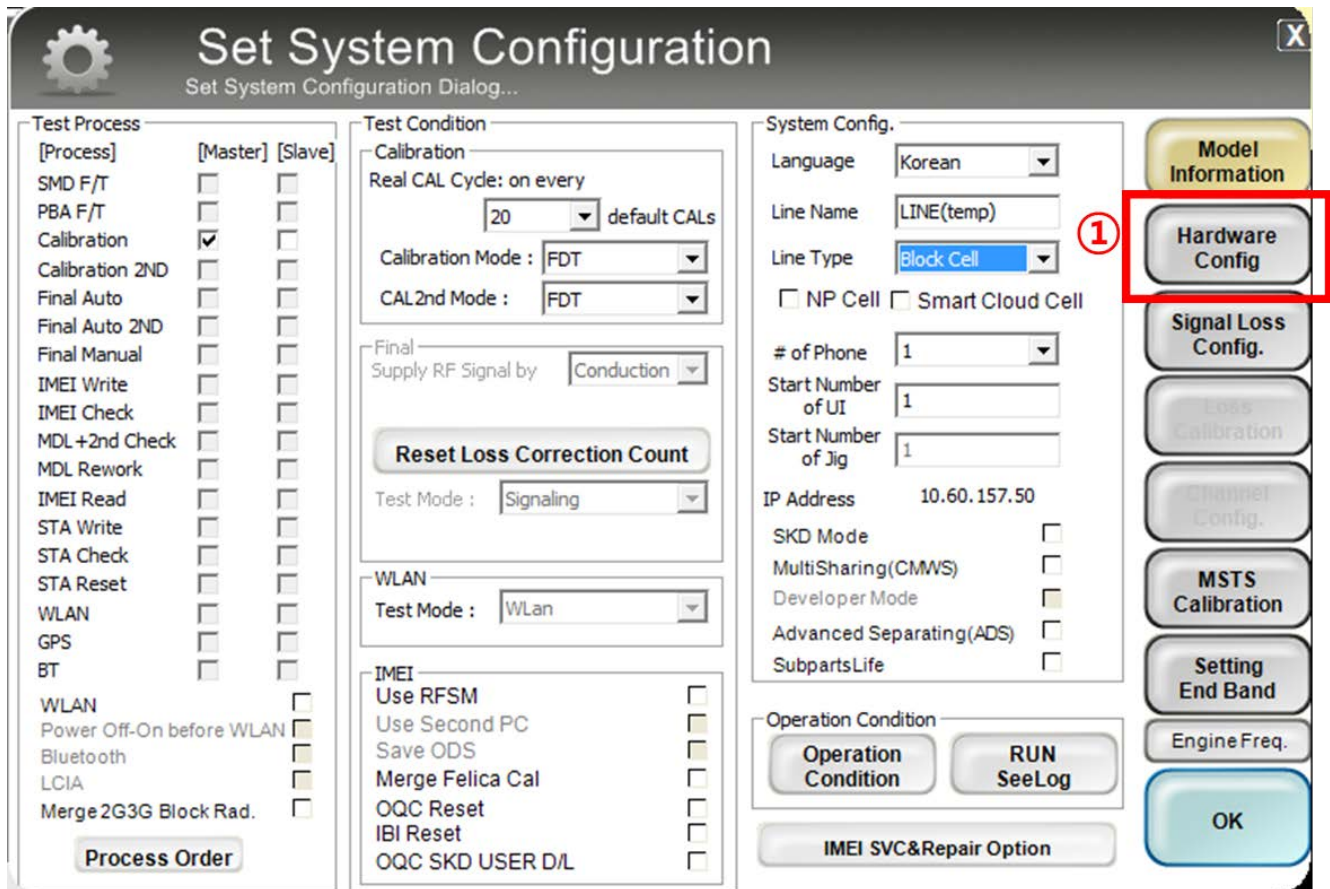
Language: Korean
Line Name: LINE(temp)
Line Type: Block Cell
☐ NP Cell ☐ Smart Cloud Cell
of Phone: 1
Start Number of UI: 1
Start Number of Jig: 1
IP Address: 10.60.157.50
SKD Mode
MultiSharing(CMWS)
Developer Mode
Advanced Separating(ADS)
SubpartsLife

Operation Condition
Operation Condition RUN SeeLog
IMEI SVC&Repair Option

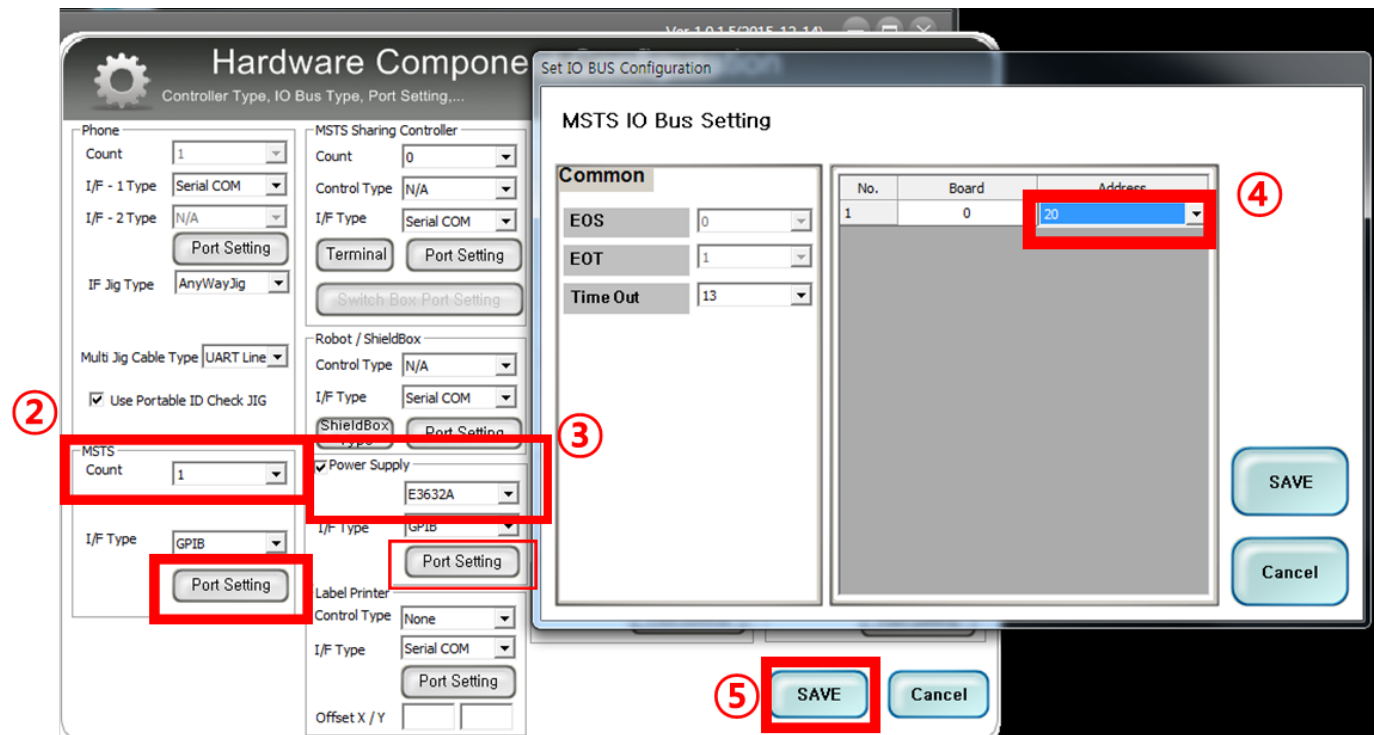
Model Information
Hardware Config.
Signal Loss Config.
Loss Calibration
Channel Config.
MSTS Calibration
Setting End Band
Engine Freq.
OK

6. Level 1 Repair

5. Set the GPIB address of MSTS(CMW500) and Power Supply(E3632A) to enter 'Hardware Config' and 'Save'. (Check the GPIB address of equipments in advance)



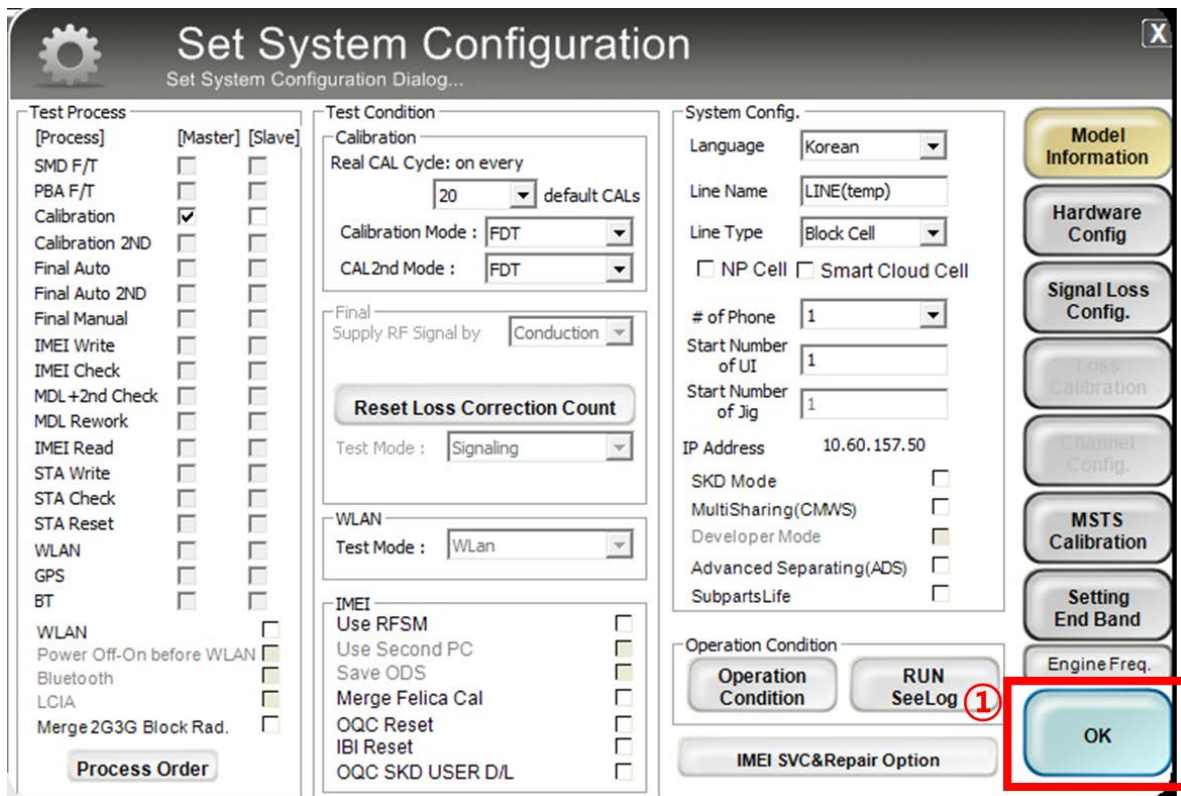
The 'Set System Configuration' dialog box is shown. It has several tabs: 'Test Process', 'Test Condition', 'System Config.', and 'Model Information'. The 'Model Information' tab is selected, and the 'Hardware Config' button is highlighted with a red box and a red circle with the number 1. Other buttons in the 'Model Information' tab include 'Signal Loss Config.', 'Loss Calibration', 'Channel Config.', 'MSTS Calibration', 'Setting End Band', 'Engine Freq.', and 'OK'.



The 'Hardware Component' dialog box is shown, with the 'MSTS IO Bus Setting' sub-dialog box open. The 'MSTS IO Bus Setting' dialog box has a 'Common' tab and a table for 'MSTS IO Bus Setting'. The 'Common' tab is selected, and the 'EOS' field is set to 0, 'EOT' to 1, and 'Time Out' to 13. The 'MSTS IO Bus Setting' table has columns for 'No.', 'Board', and 'Address'. The first row has 'No.' 1, 'Board' 0, and 'Address' 20. The 'Address' field is highlighted with a red box and a red circle with the number 4. The 'SAVE' button is highlighted with a red box and a red circle with the number 5. The 'MSTS Count' field in the 'Hardware Component' dialog box is set to 1, and the 'I/F Type' is set to 'GPIB'. The 'Port Setting' button is highlighted with a red box and a red circle with the number 3.

6. Level 1 Repair

6. Press 'OK' to start RF Calibration after completing all settings.



The 'Set System Configuration' dialog box is shown. It contains several sections: 'Test Process' with checkboxes for various tests; 'Test Condition' with settings for calibration cycles and modes; 'System Config.' with dropdowns for language, line name, and type, and checkboxes for various features; and a 'Process Order' section. A red box highlights the 'OK' button at the bottom right.

Test Process

[Process]	[Master]	[Slave]
SMD F/T	☐	☐
PBA F/T	☐	☐
Calibration	☒	☐
Calibration 2ND	☐	☐
Final Auto	☐	☐
Final Auto 2ND	☐	☐
Final Manual	☐	☐
IMEI Write	☐	☐
IMEI Check	☐	☐
MDL+2nd Check	☐	☐
MDL Rework	☐	☐
IMEI Read	☐	☐
STA Write	☐	☐
STA Check	☐	☐
STA Reset	☐	☐
WLAN	☐	☐
GPS	☐	☐
BT	☐	☐
WLAN	☐	☐
Power Off-On before WLAN	☐	☐
Bluetooth	☐	☐
LCIA	☐	☐
Merge 2G3G Block Rad.	☐	☐

Test Condition

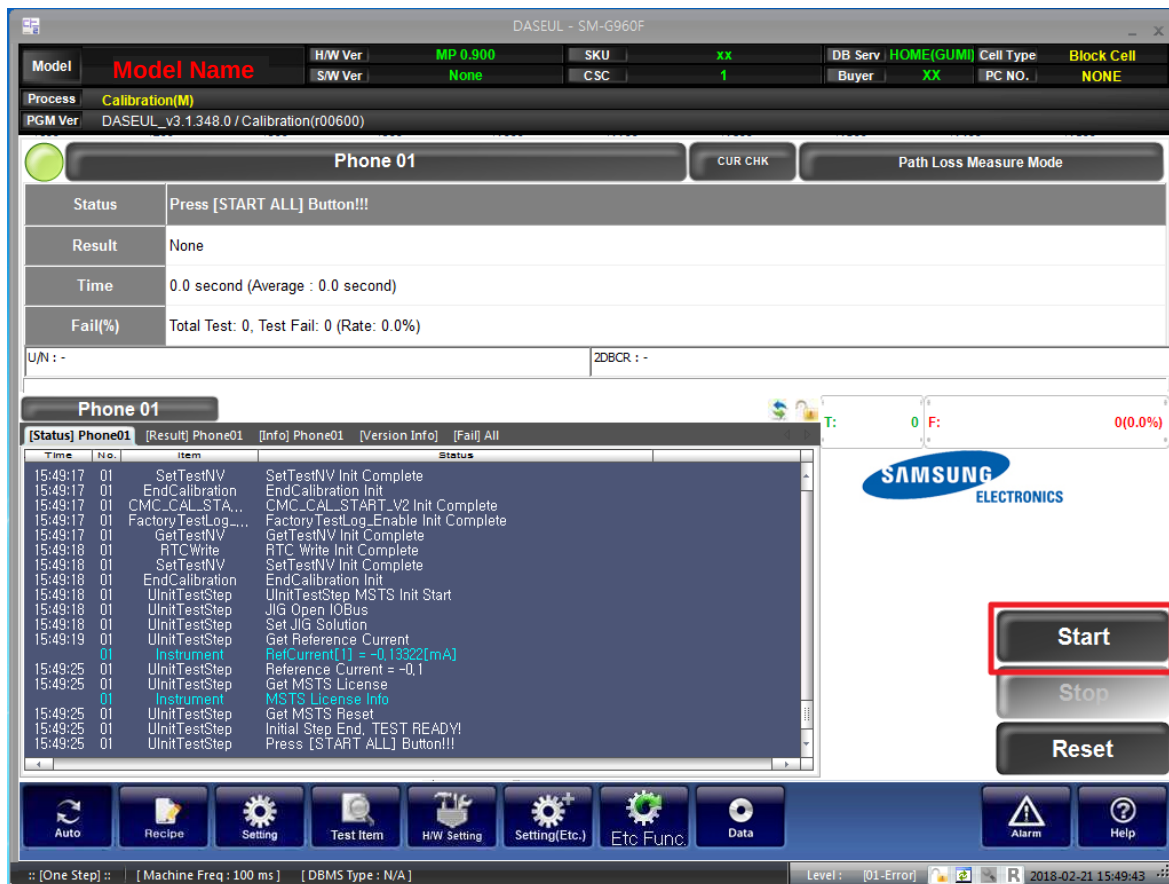
Calibration
Real CAL Cycle: on every 20 default CALs
Calibration Mode : FDT
CAL2nd Mode : FDT
Final Supply RF Signal by : Conduction
Reset Loss Correction Count
Test Mode : Signaling
WLAN Test Mode : WLAN
IMEI Use RFSM
Use Second PC
Save ODS
Merge Felica Cal
OQC Reset
IBI Reset
OQC SKD USER D/L

System Config.

Language : Korean
Line Name : LINE(temp)
Line Type : Block Cell
☐ NP Cell ☐ Smart Cloud Cell
of Phone : 1
Start Number of UI : 1
Start Number of Jig : 1
IP Address : 10.60.157.50
SKD Mode
MultiSharing(CMWS)
Developer Mode
Advanced Separating(ADS)
SubpartsLife

Operation Condition

Operation Condition RUN SeeLog 1 OK



The 'DASEUL - SM-G960F' calibration screen is shown. It displays various status information and a list of test steps. A red box highlights the 'Start' button.

Model Model Name H/W Ver MP 0.900 SKU xx IDB Serv HOME(GUMI) Cell Type Block Cell
SW Ver None CSC 1 Buyer XX PC NO. NONE

Process Calibration(M)
PGM Ver DASEUL_v3.1.348.0 / Calibration(r00600)

Phone 01 CUR CHK Path Loss Measure Mode

Status Press [START ALL] Button!!!
Result None
Time 0.0 second (Average : 0.0 second)
Fail(%) Total Test: 0, Test Fail: 0 (Rate: 0.0%)
U/N : - ZBCR : -

Phone 01

Time	No.	Item	Status
15:49:17	01	SetTestNV	SetTestNV Init Complete
15:49:17	01	EndCalibration	EndCalibration Init
15:49:17	01	CMC_CAL_STA...	CMC_CAL_START_V2 Init Complete
15:49:17	01	FactoryTestLog....	FactoryTestLog_Enable Init Complete
15:49:17	01	GetTestNV	GetTestNV Init Complete
15:49:18	01	RTCWrite	RTC Write Init Complete
15:49:18	01	SetTestNV	SetTestNV Init Complete
15:49:18	01	EndCalibration	EndCalibration Init
15:49:18	01	UnitTestStep	UnitTestStep MSTs Init Start
15:49:18	01	UnitTestStep	JIG Open IOBus
15:49:18	01	UnitTestStep	Set JIG Solution
15:49:19	01	UnitTestStep	Get Reference Current
15:49:25	01	UnitTestStep	RefCurrent[1] = -0.13322[mA]
15:49:25	01	UnitTestStep	Reference Current = -0.1
15:49:25	01	UnitTestStep	Get MSTs License
15:49:25	01	UnitTestStep	MSTs License Info
15:49:25	01	UnitTestStep	Get MSTs Reset
15:49:25	01	UnitTestStep	Initial Step End, TEST READY!
15:49:25	01	UnitTestStep	Press [START ALL] Button!!!

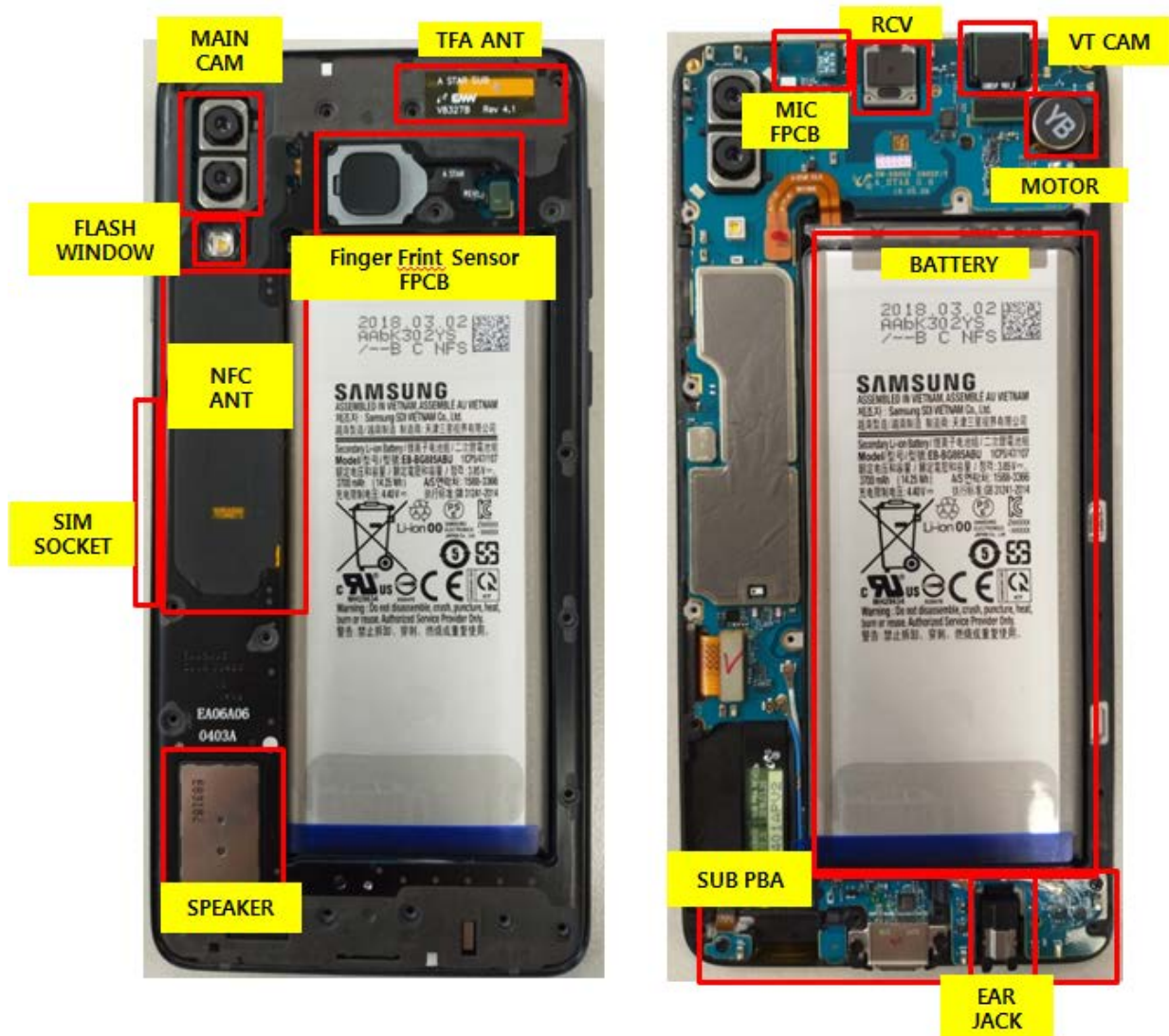
Start
Stop
Reset

Auto Recipe Setting Test Item H/W Setting Setting(Etc.) Etc Func. Data Alarm Help

Level : [01-Error] 2018-02-21 15:49:43

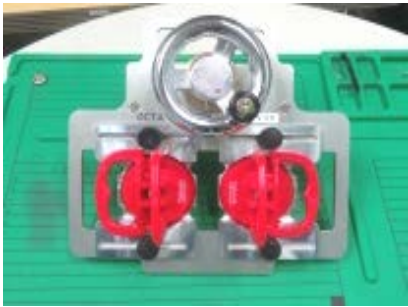
7. Level 2 Repair

7-1. Components on the Rear Case



7. Level 2 Repair

7-2. Pre-requisite

	
Tweezers / Disass'y Stick / Screw Driver	Anti-static Gloves
	
Anti-static Mat	Hot Plate
	
A OCTA Disassembly Holder	OCTA Disassembly Upper
	
Ethyl Alcohol	Cotton Swab

7. Level 2 Repair

7-3. Parts which must be changed after repair

BOM description & part code	Image	Remark
A/S-SVC TAPE BG OUTER (Common) [GH81-15673A]		Replace for Back Glass repair

7. Level 2 Repair

7-4. Disassembly

1

1) Put the device in the chamber as following below heating condition
 - SOC 68%↓: 70℃/10~20 minute
 - SOC 68%↑: 70℃/10~20 minute
 ※ Please confirm the heating condition released lastly, and follow it.



2

Detach the Back Glass. And Finger sensor
 1) Detach the left side of Back glass
 2) Detach the Finger sensor connector



※ Caution

1) Be care of scratch

※ Caution

1) Be care of scratch

3

Unscrew 19 Point and disassemble SIM Tray from device



4

Disassemble Rear.



※ Caution

1) Be care of Rear damage

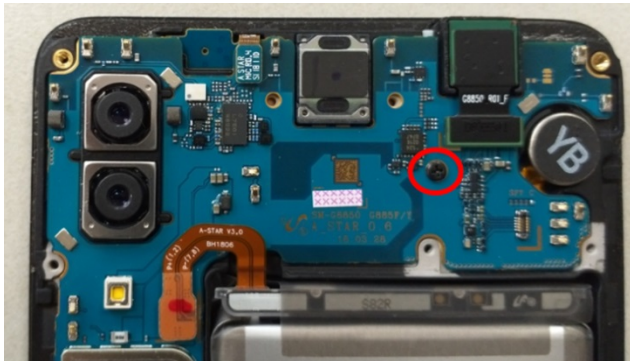
※ Caution

1) Be care of scratch
 2) Be care of Rear and connector damage

7. Level 2 Repair

5

Unscrew PBA 1points.

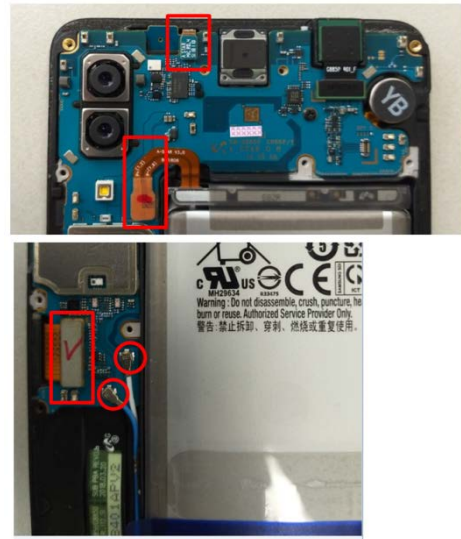


※ Caution

- 1) Be care of scratch
- 2) Be care of Rear and connector damage
- 3) Be careful not to damage the PBA

6

Detach 2 coaxial cable, Battery connect and the other connectors. (2 Point- Display and MIC FPCB)



※ Caution

- 1) Be careful not to damage the PBA

7

Detach SUB PBA connector

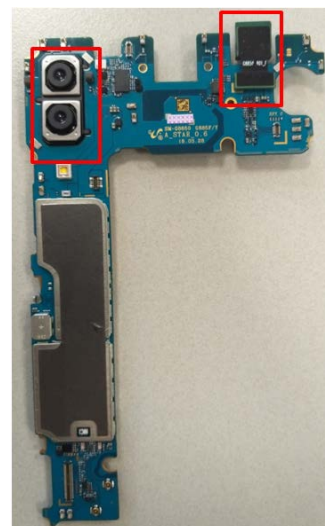


※ Caution

- 1) Be care of scratch
- 2) Be care of connector/cable damage

8

Detach other components.(Main CAM and SUB CAM)



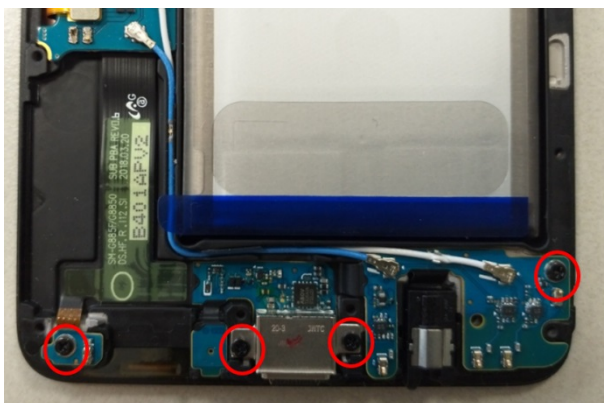
※ Caution

- 1) Be care of FPCB damage
- 2) Be care of SUB PBA connector

7. Level 2 Repair

9

Unscrew SUB PBA 4points

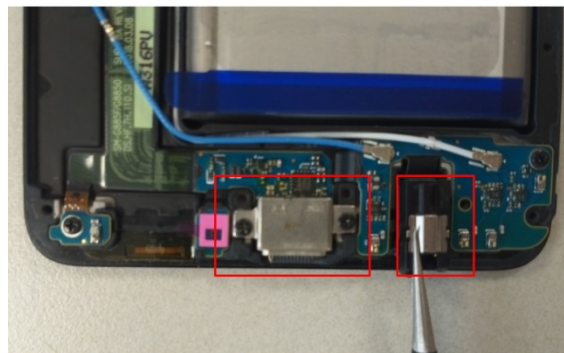


⚠ Caution

1) Be care of several kinds of damage

10

Detach Ear Jack and USB (SUB PBA)
Detach SUB PBA

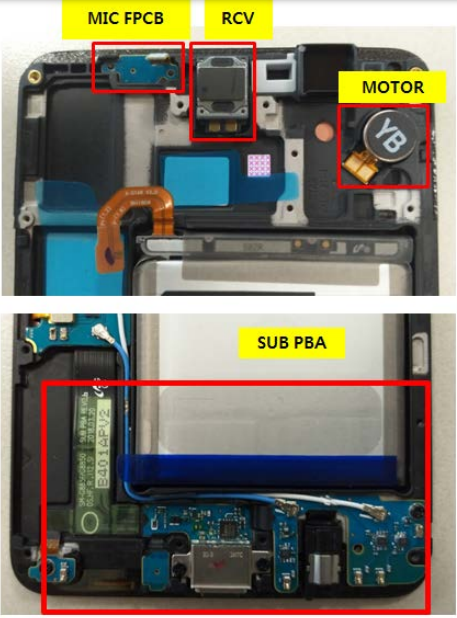
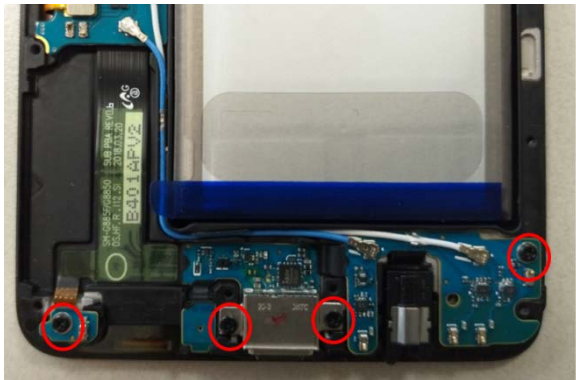
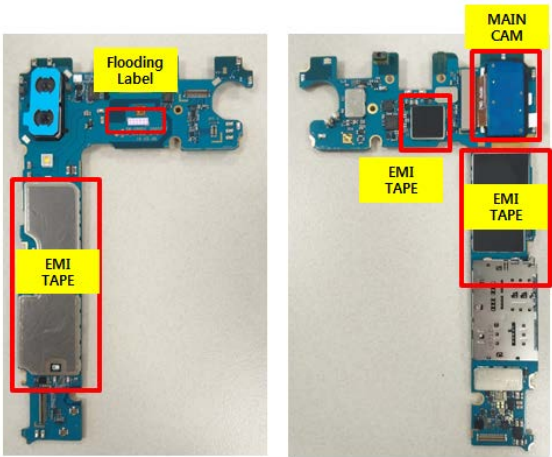
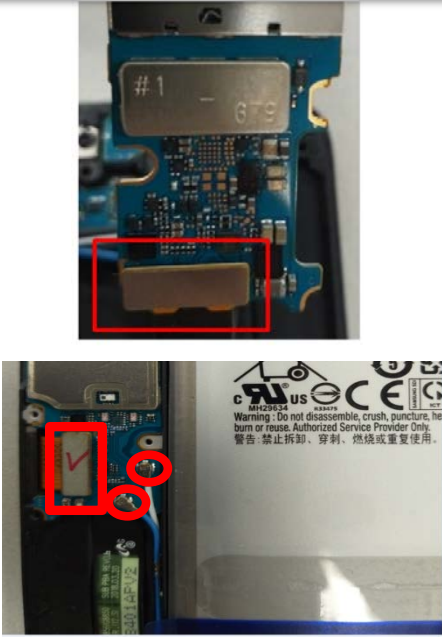


⚠ Caution

1) Be careful not to damage the SUB PBA

7. Level 2 Repair

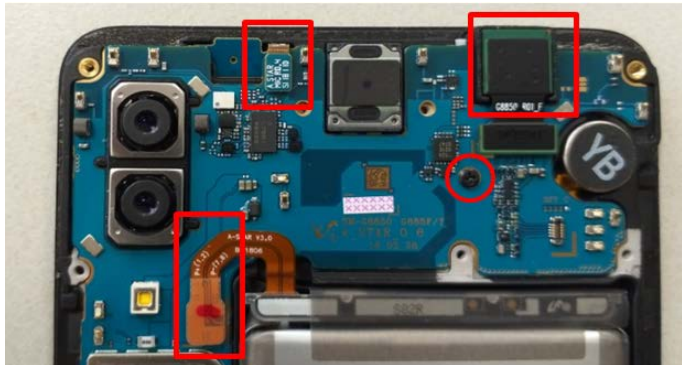
7-5. Assembly

1 Attach components on Front. (RCV module, Sensor FPCB, Motor and SUB PBA) 	2 Attach 2 coaxial cable, Screw 4point in SUB PBA. 
※ Caution 1) Be care of coaxial cable damage	※ Caution 1) Be care of chip damage nearby screw point.
3 Attach EMI Tape(3), Flooding label and Main CAM in PBA 	4 Attach PBA assy in SUB PBA. And attach OCTA Connector and 2 coaxial cables. 
※ Caution 1) Be care of FPCB damage and tilt 2) Be care of press power/time	※ Caution 1) Be care of Push SUB PBA connector. 2) Be care of components FPCB.

7. Level 2 Repair

5

Attach Battery connect, VT CAM and MIC FPCB. Screw 1 point in PBA



6

Assemble the REAR.



※ Caution

1) Be care of press power/time

※ Caution

1) Be care of FPCB and Y-OCTA damage

7

Screw 19 Point and assemble SIM Tray.
* Torque value : 1.2 N · m



8

Attach the finger sensor on the REAR and
Assemble the finger sensor on the PBA



※ Caution

1) Be care of scratch and REAR damage

※ Caution

1) Be care of scratch and REAR damage

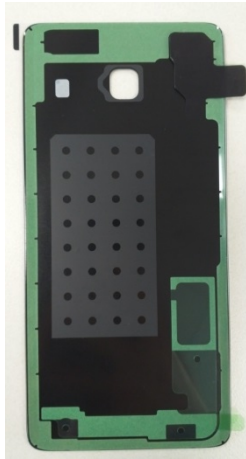
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7. Level 2 Repair

9

Attach the back glass inner tape on the rear case



10

Attach the back glass on the REAR side.



※ Caution

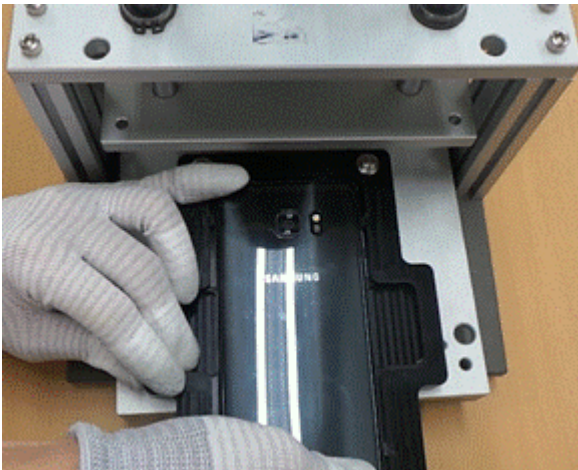
1) Be care of scratch and REAR damage

※ Caution

1) Be care of scratch and tilt.

11

Press Back Glass with pressing jig for SVC.
Pressing force : 1N
Pressing time : 1 minute

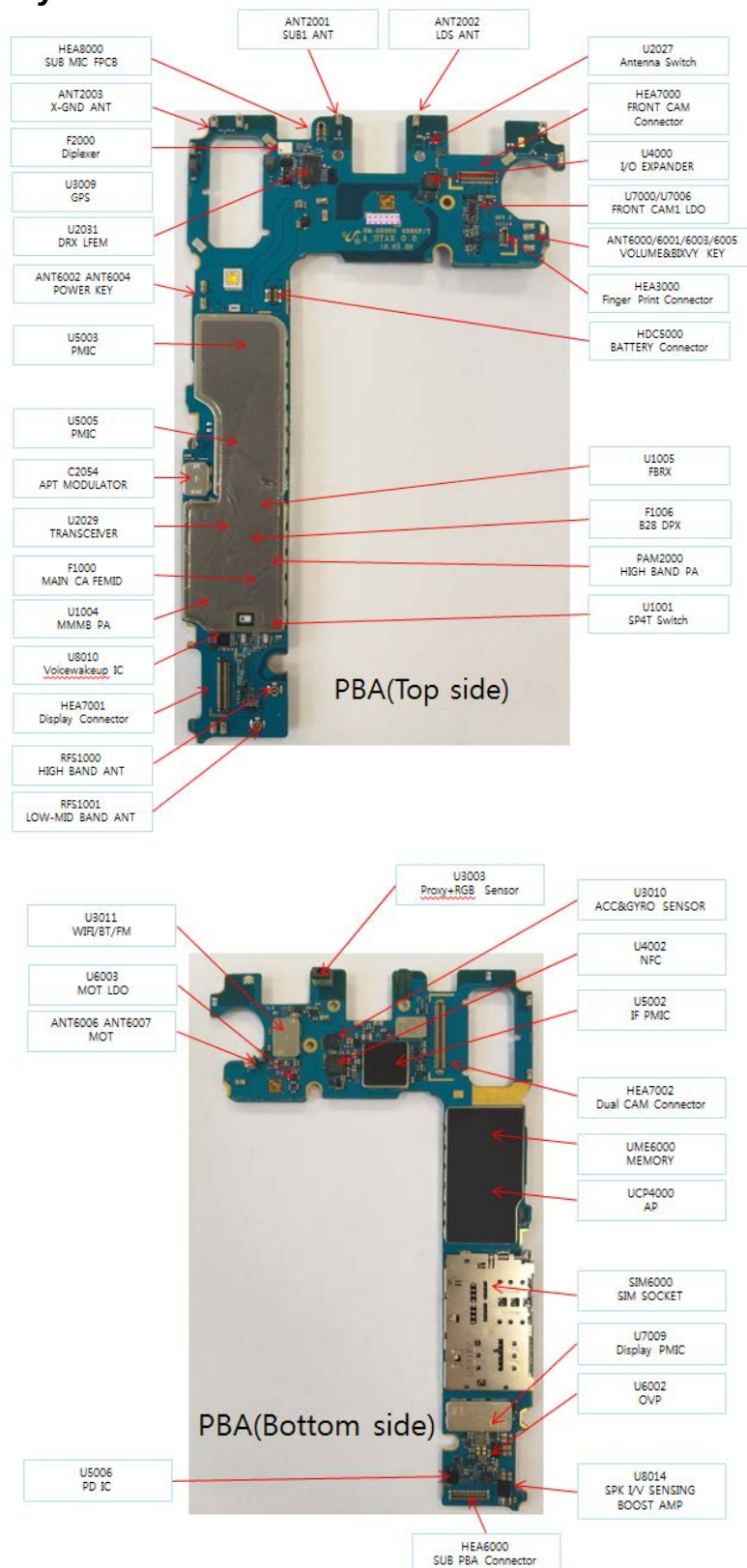


※ Caution

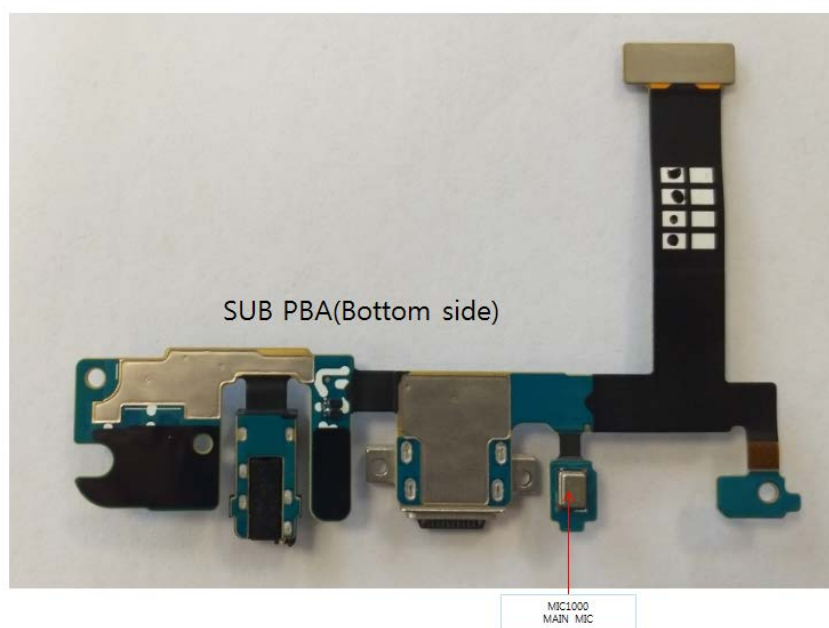
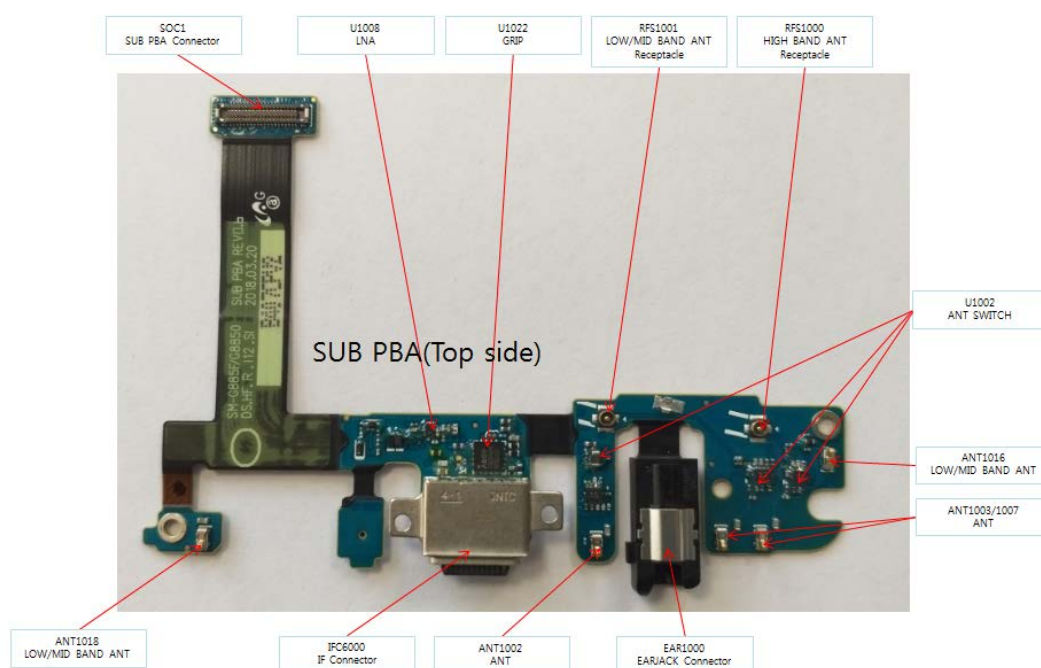
1) Be care of scratch and tilt.

8. Level 3 Repair

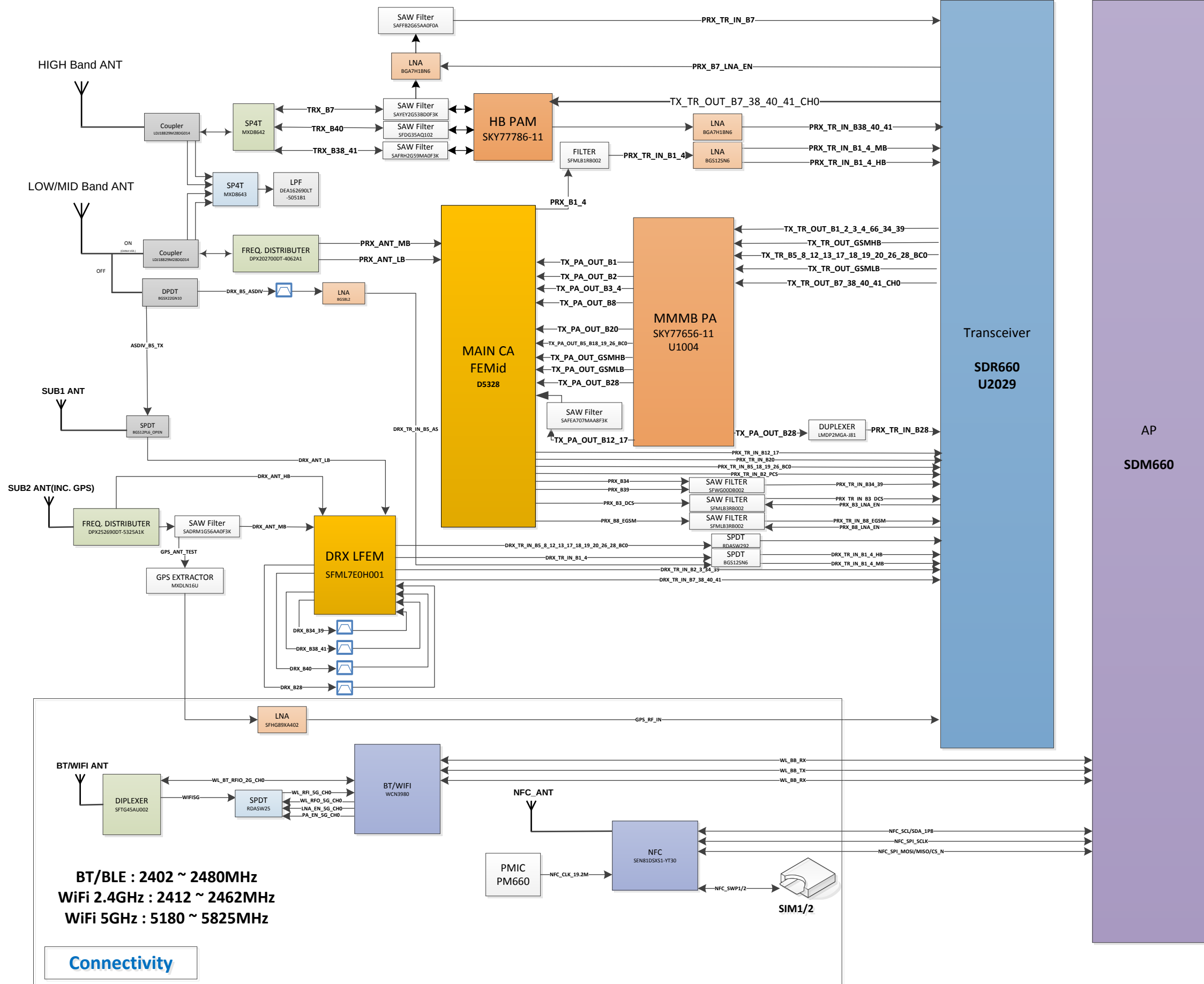
8-1. Components Layout

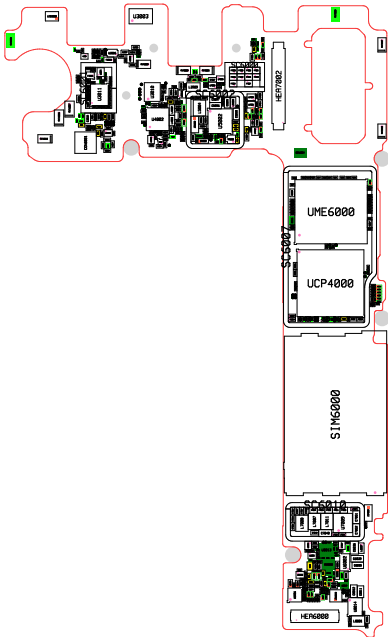


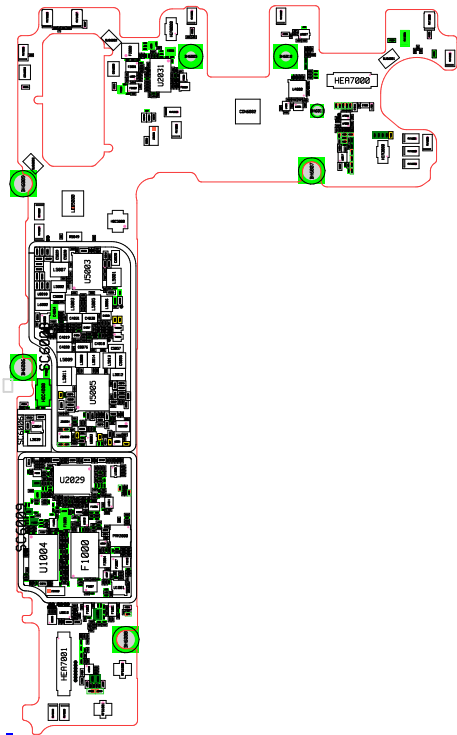
8. Level 3 Repair



A-STAR Block Diagram (RF Part)

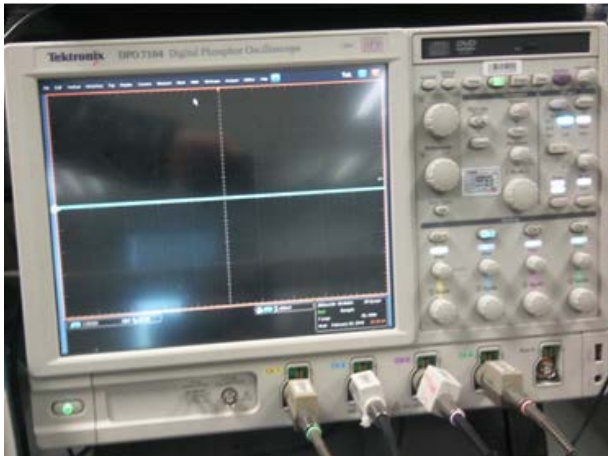






8. Level 3 Repair

8-3. Flow chart of Troubleshooting.



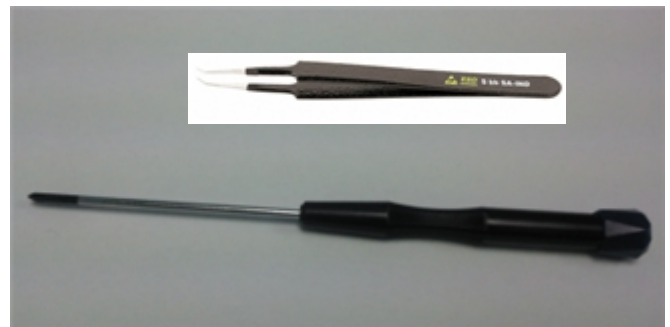
Oscilloscope



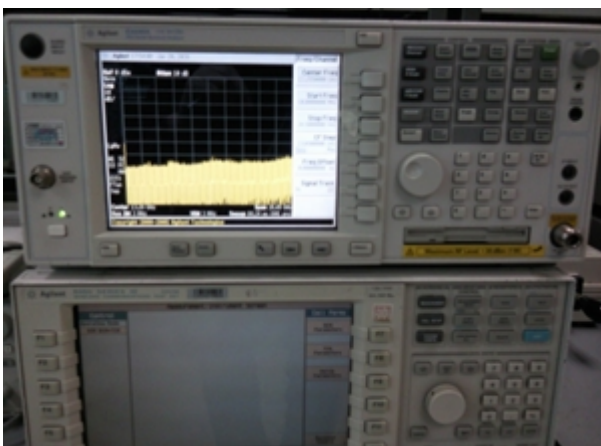
Digital Multimeter



Power Supply



+ driver, ESD Safe Tweezer



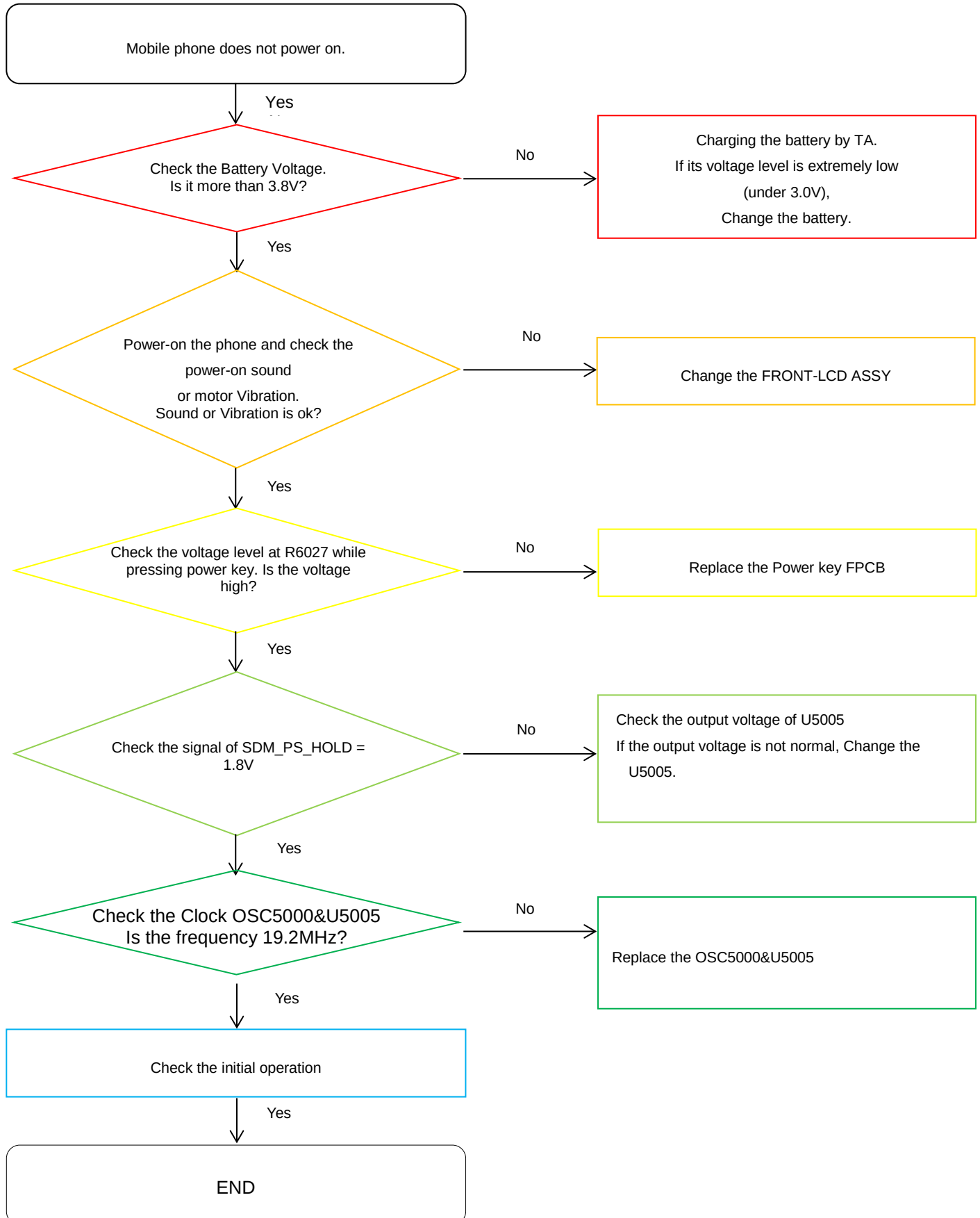
8960 & Spectrum Analyzer



Soldering iron

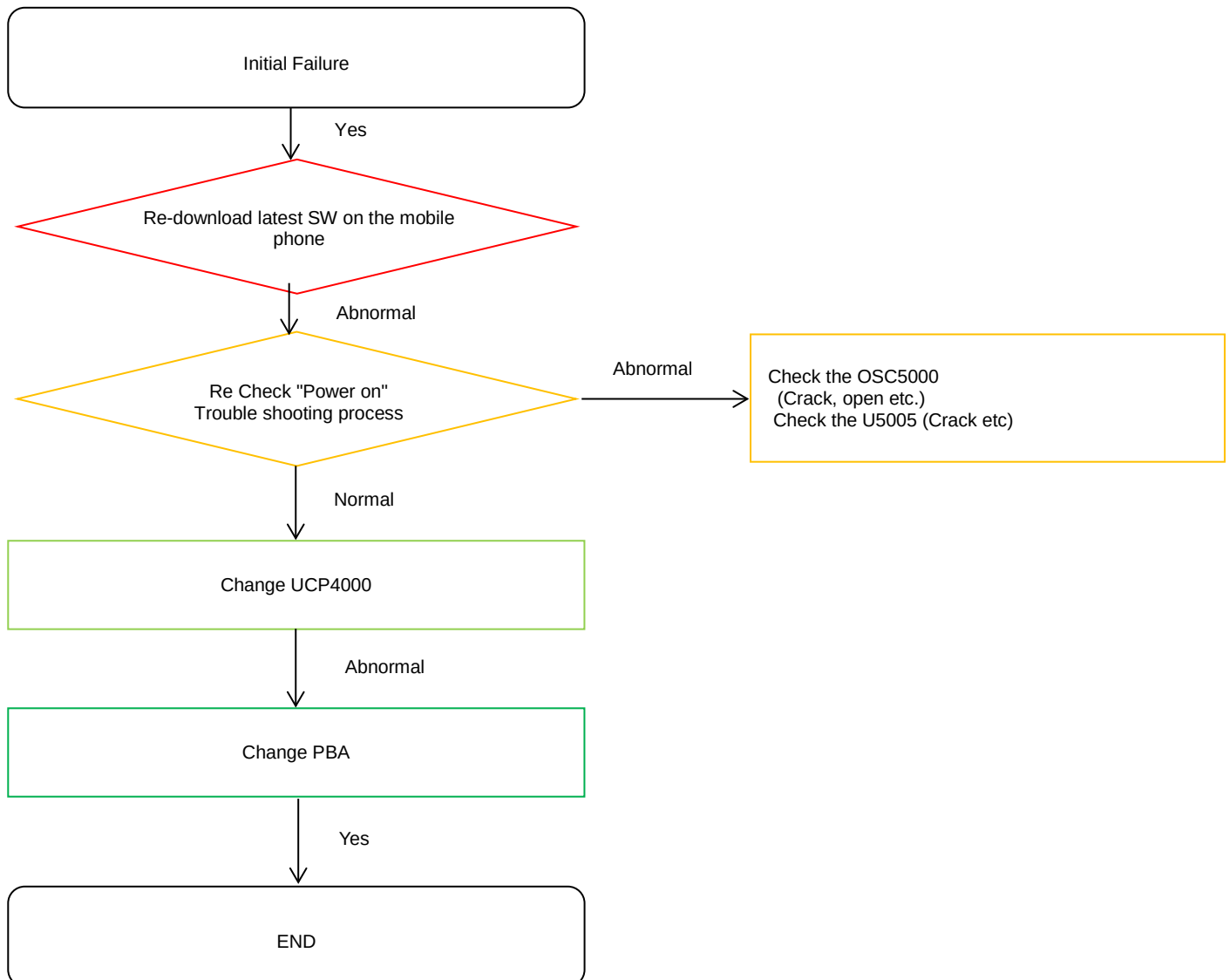
8. Level 3 Repair

8-4-1. Power On



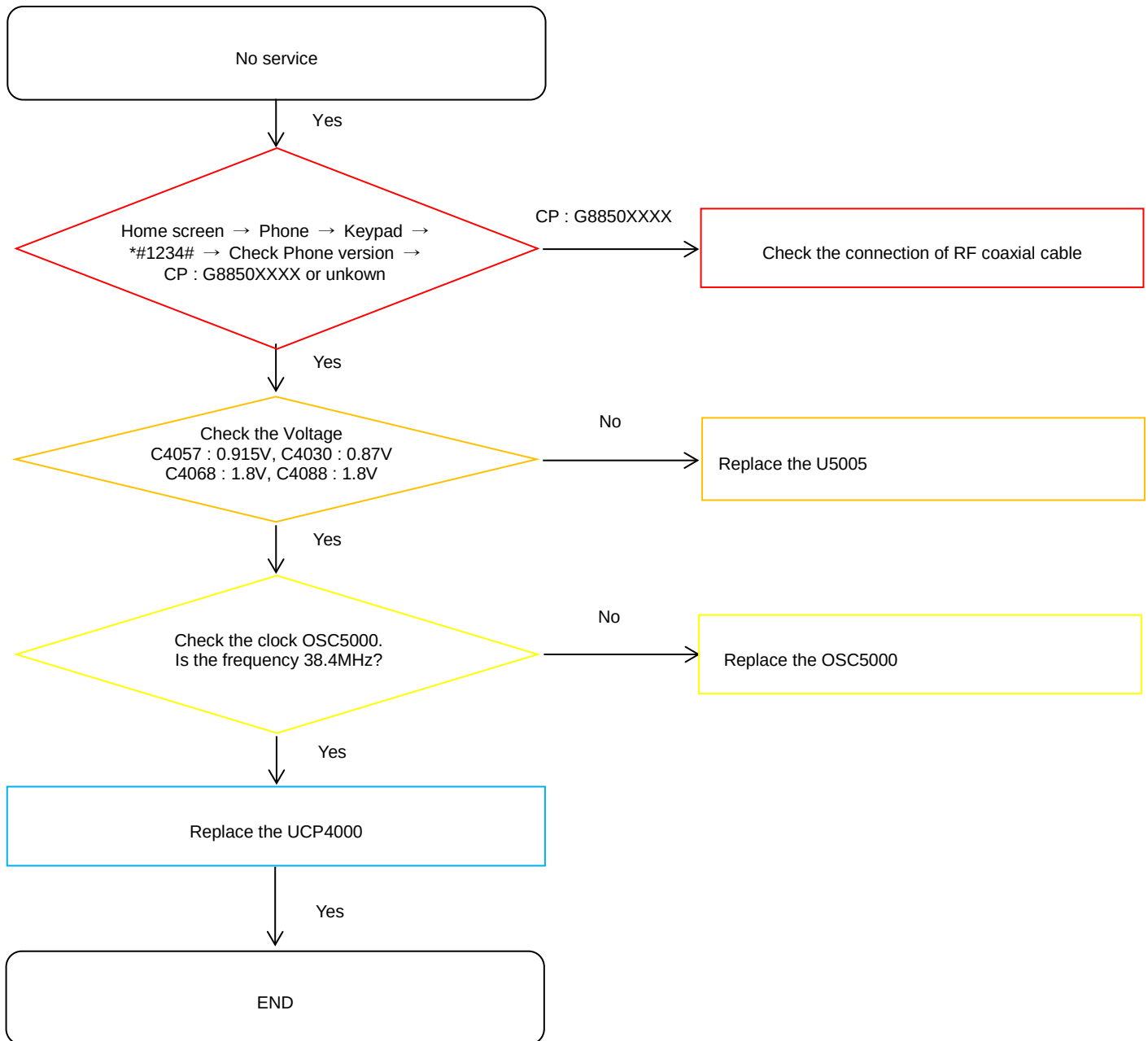
8. Level 3 Repair

8-4-2. Initial



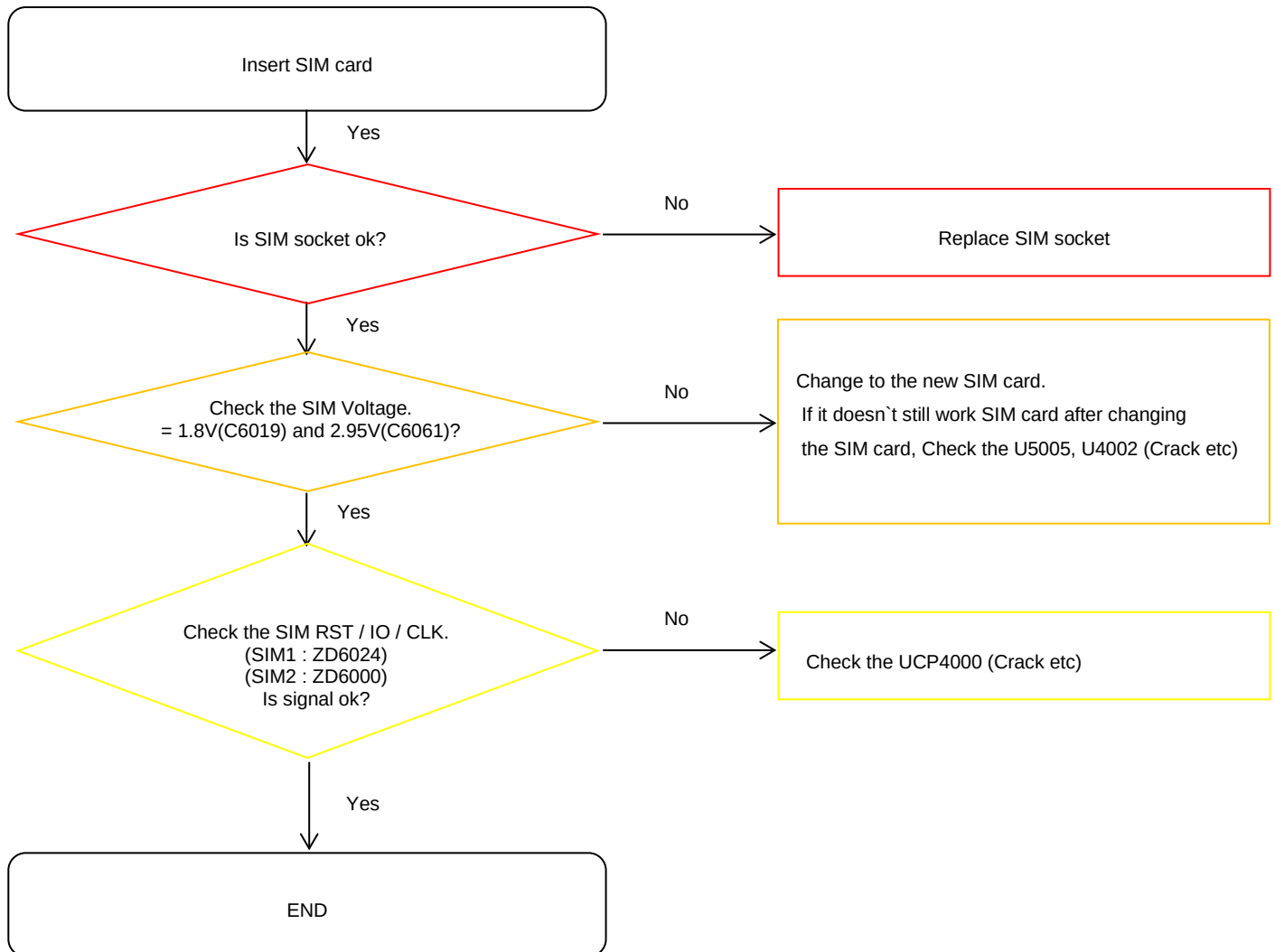
8. Level 3 Repair

8-4-3. No Service



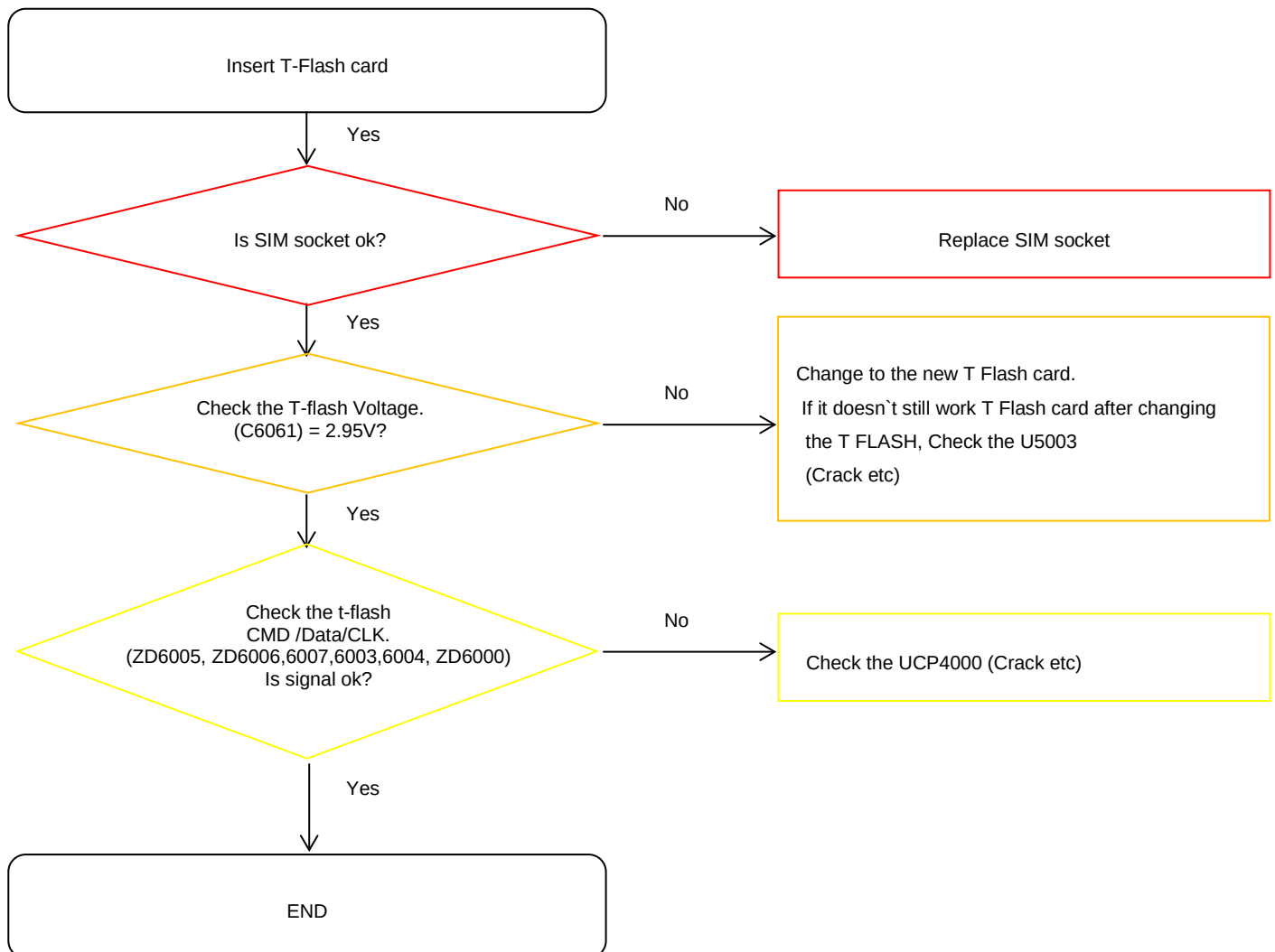
8. Level 3 Repair

8-4-4. SIM Part



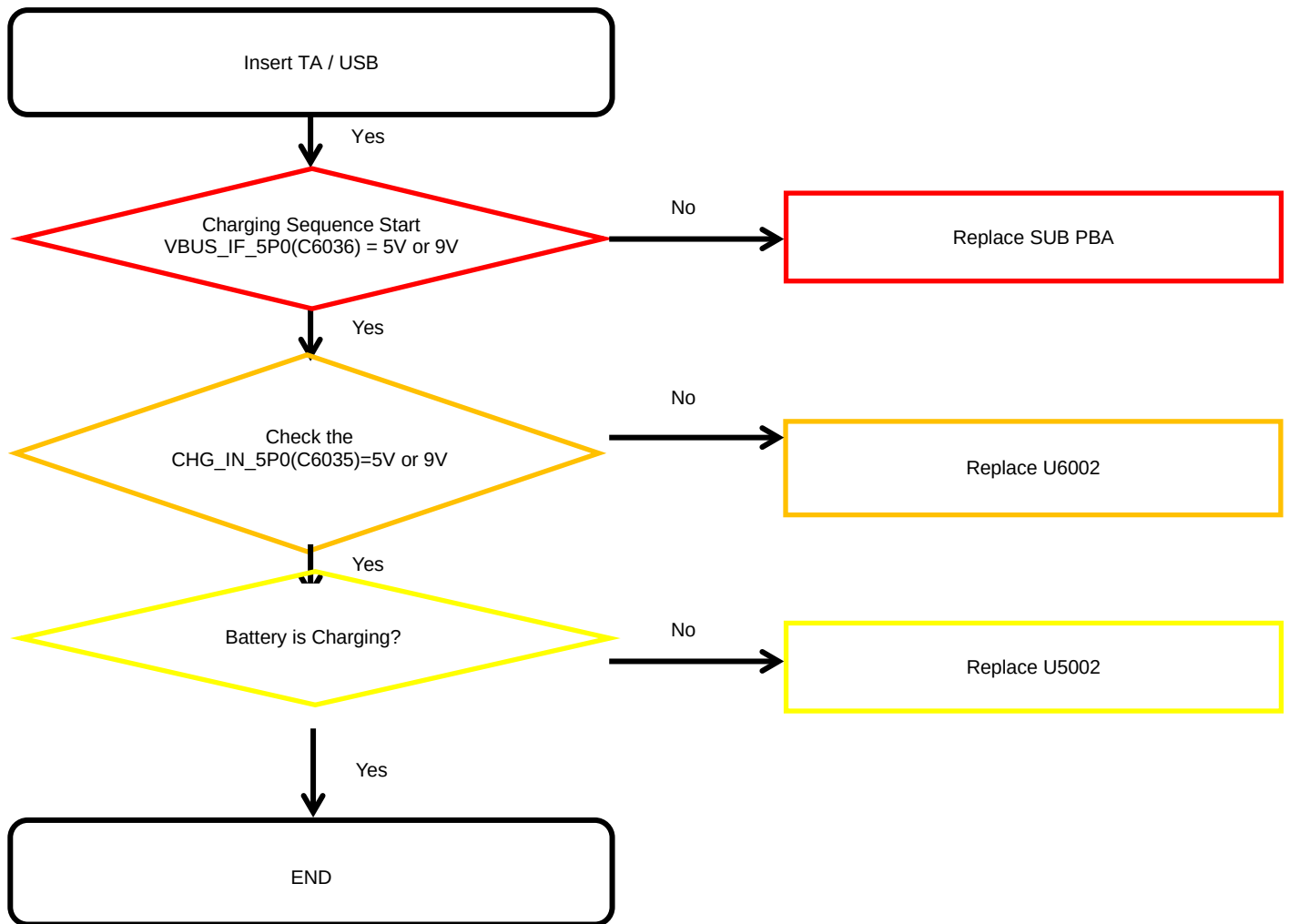
8. Level 3 Repair

8-4-5. T-Flash Part



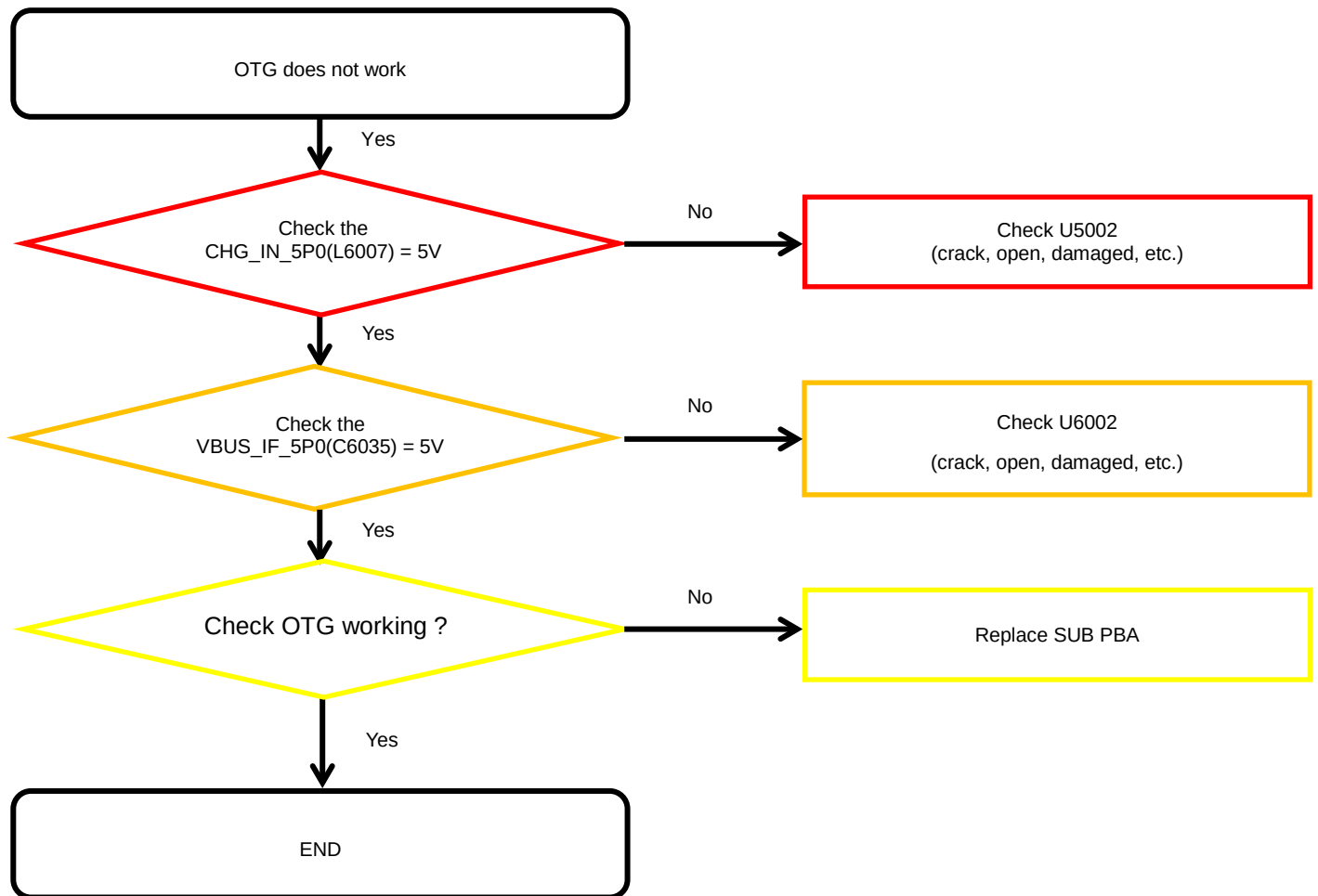
8. Level 3 Repair

8-4-6. Cable Charging Part (Normal/Fast)



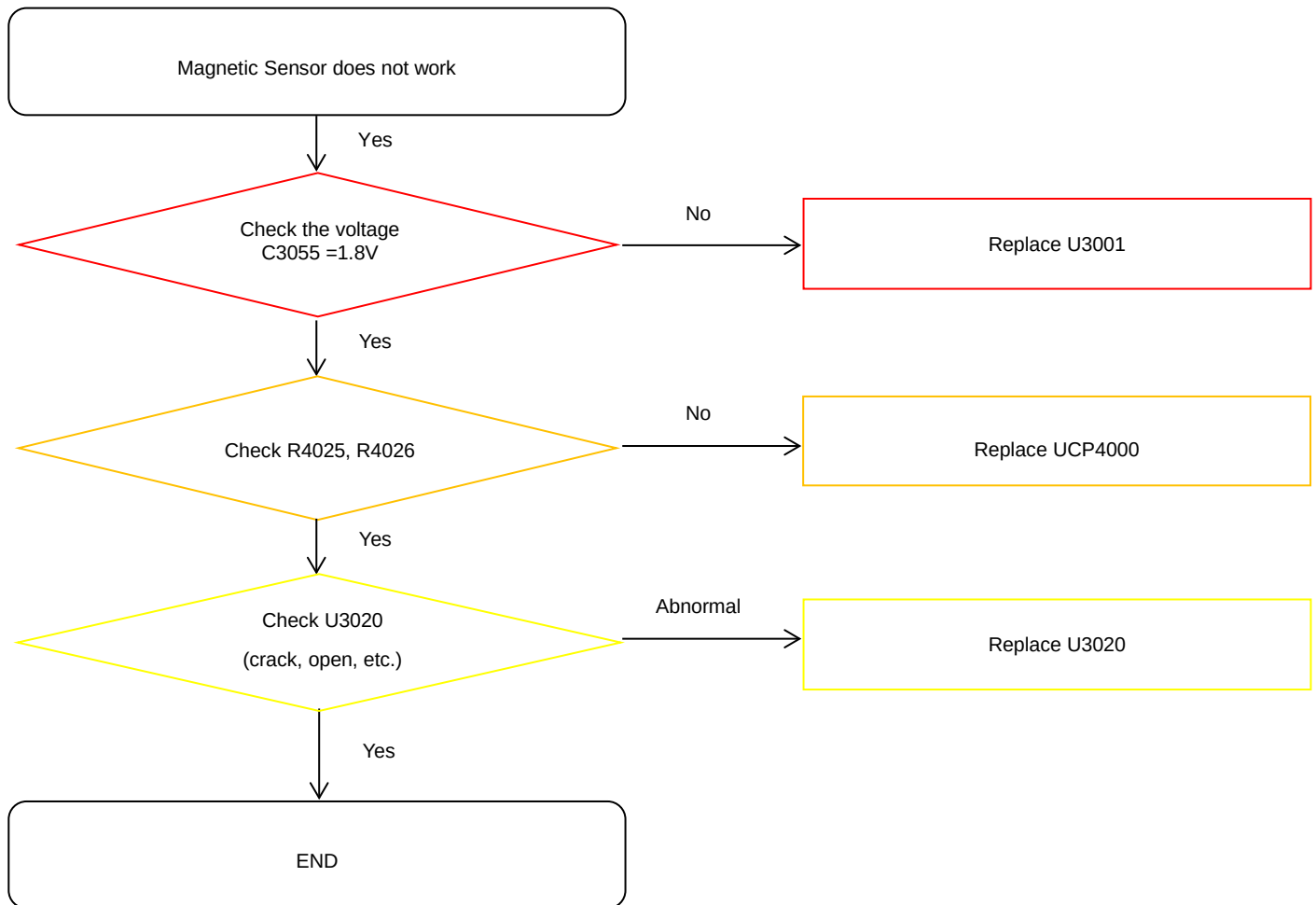
8. Level 3 Repair

8-4-7. OTG



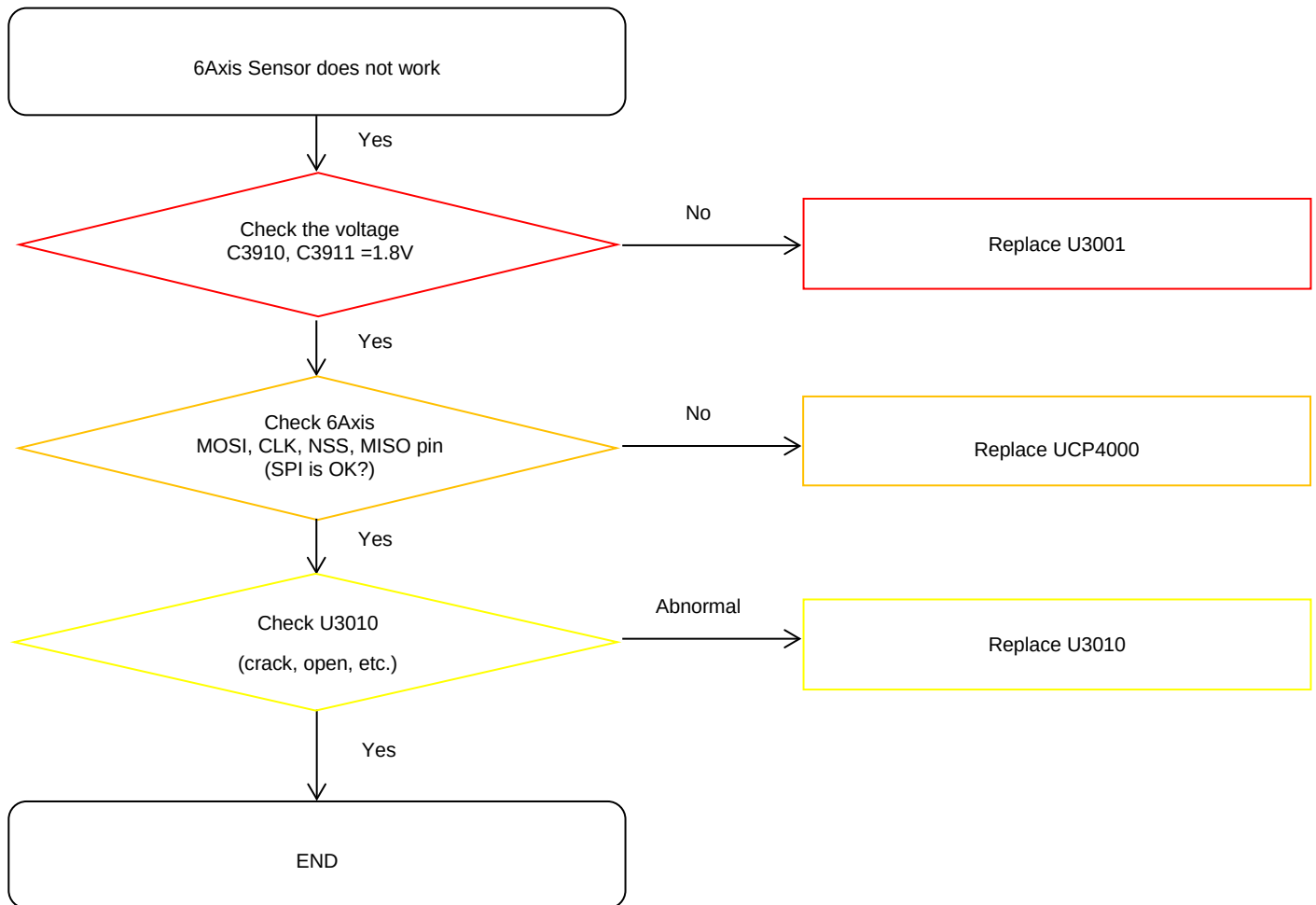
8. Level 3 Repair

8-4-8-1. Magnetic Sensor



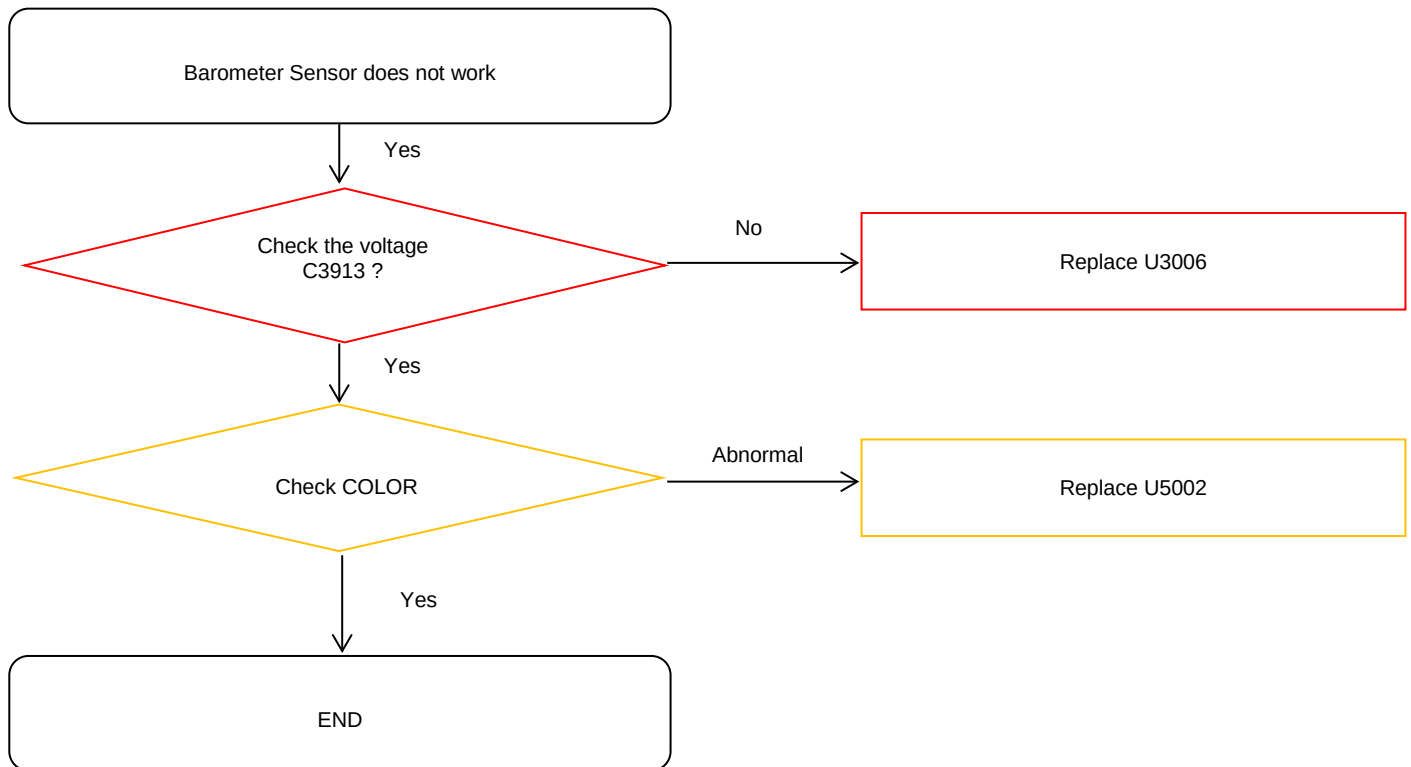
8. Level 3 Repair

8-4-8-2. 6Axis Sensor



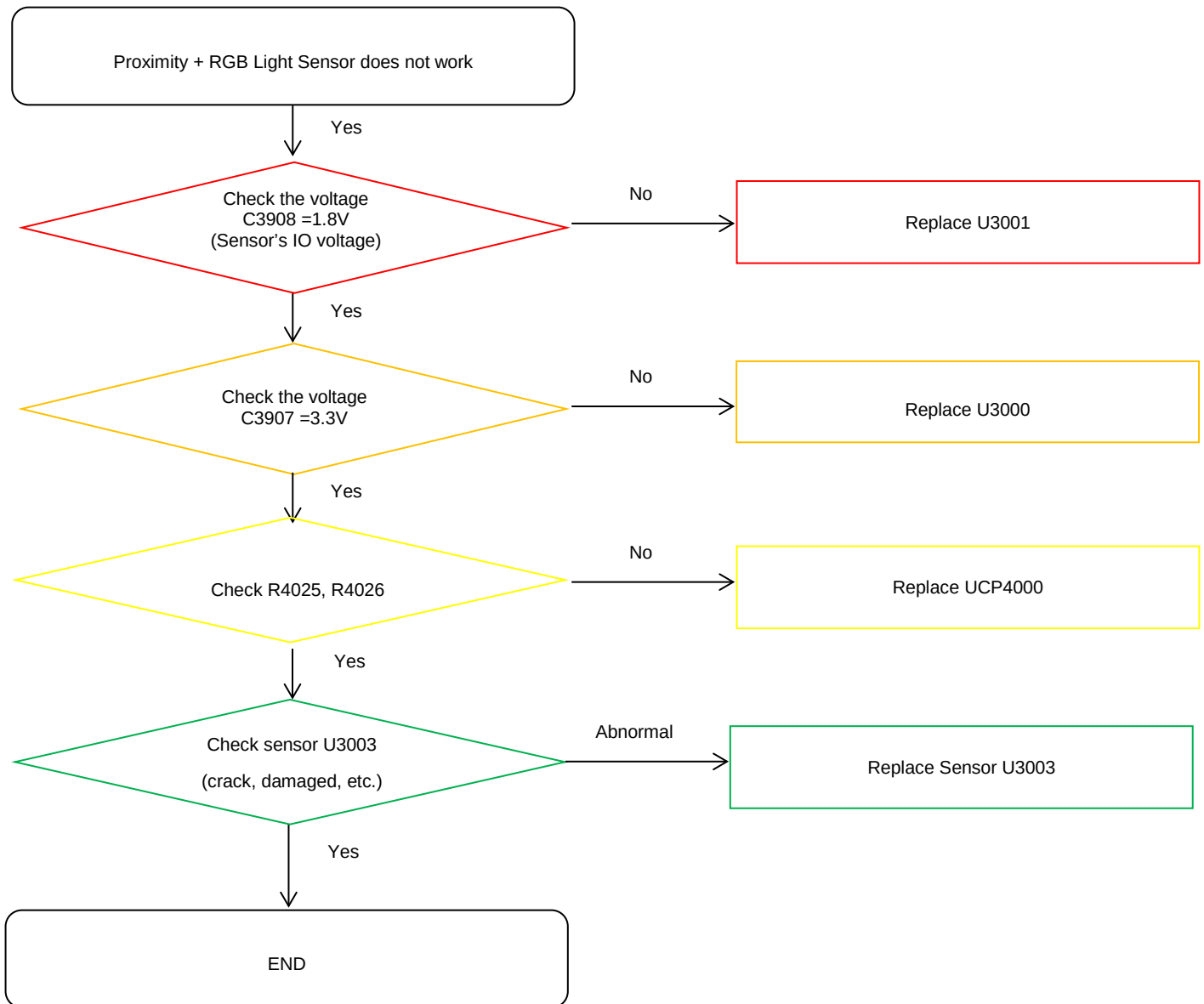
8. Level 3 Repair

8-4-8-3. SVC sensor



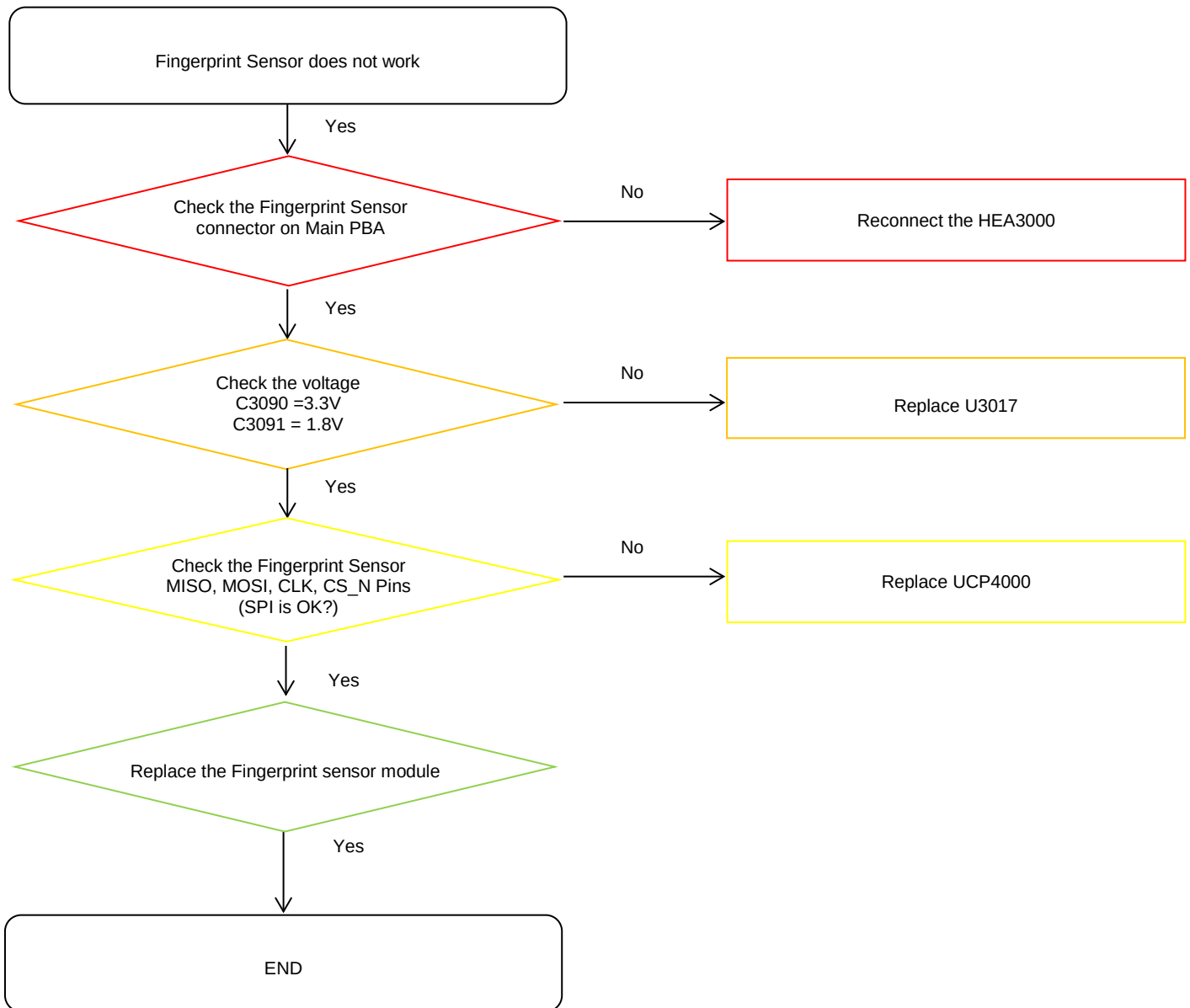
8. Level 3 Repair

8-4-8-4. Proximity + RGB Light sensor



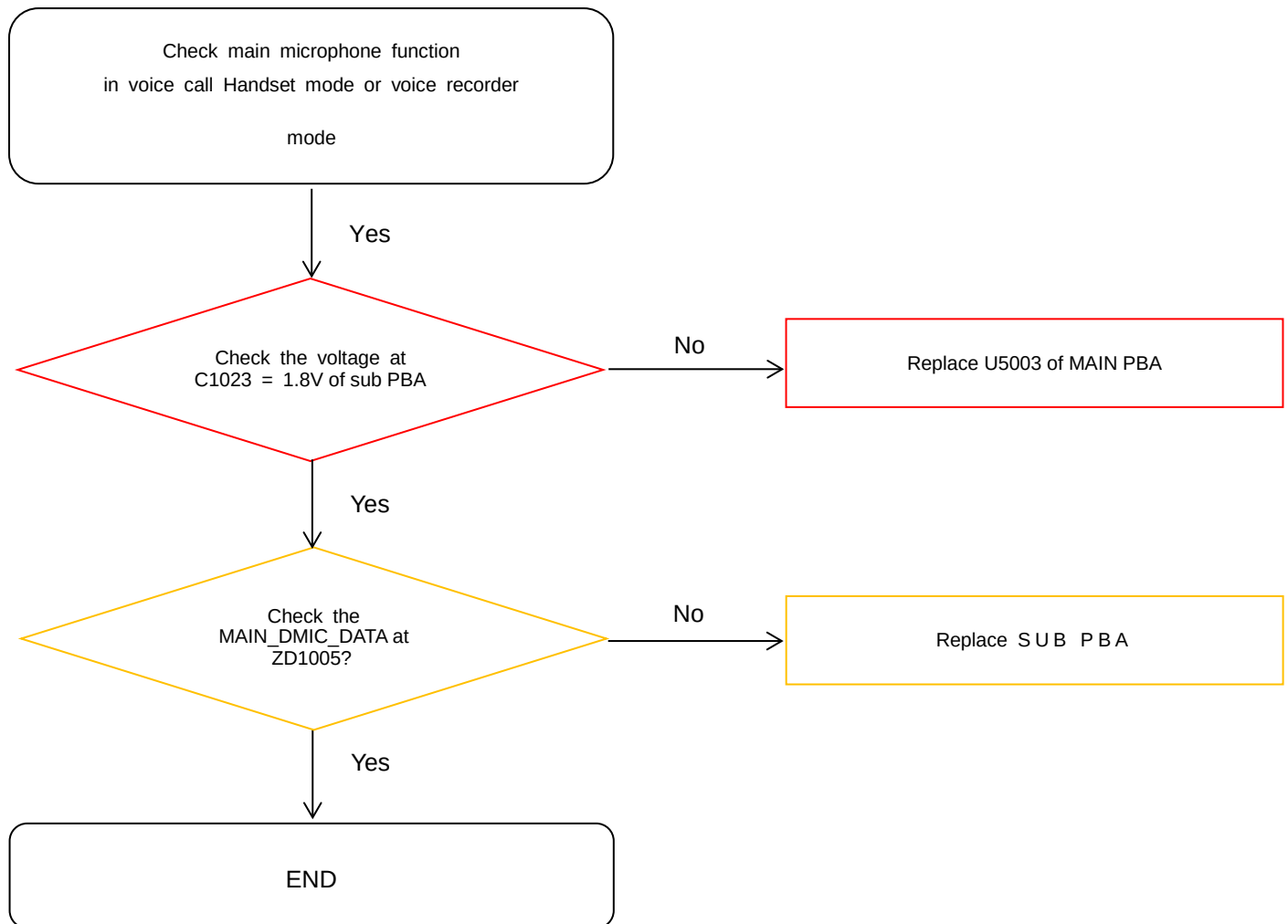
8. Level 3 Repair

8-4-8-5. Fingerprint Sensor



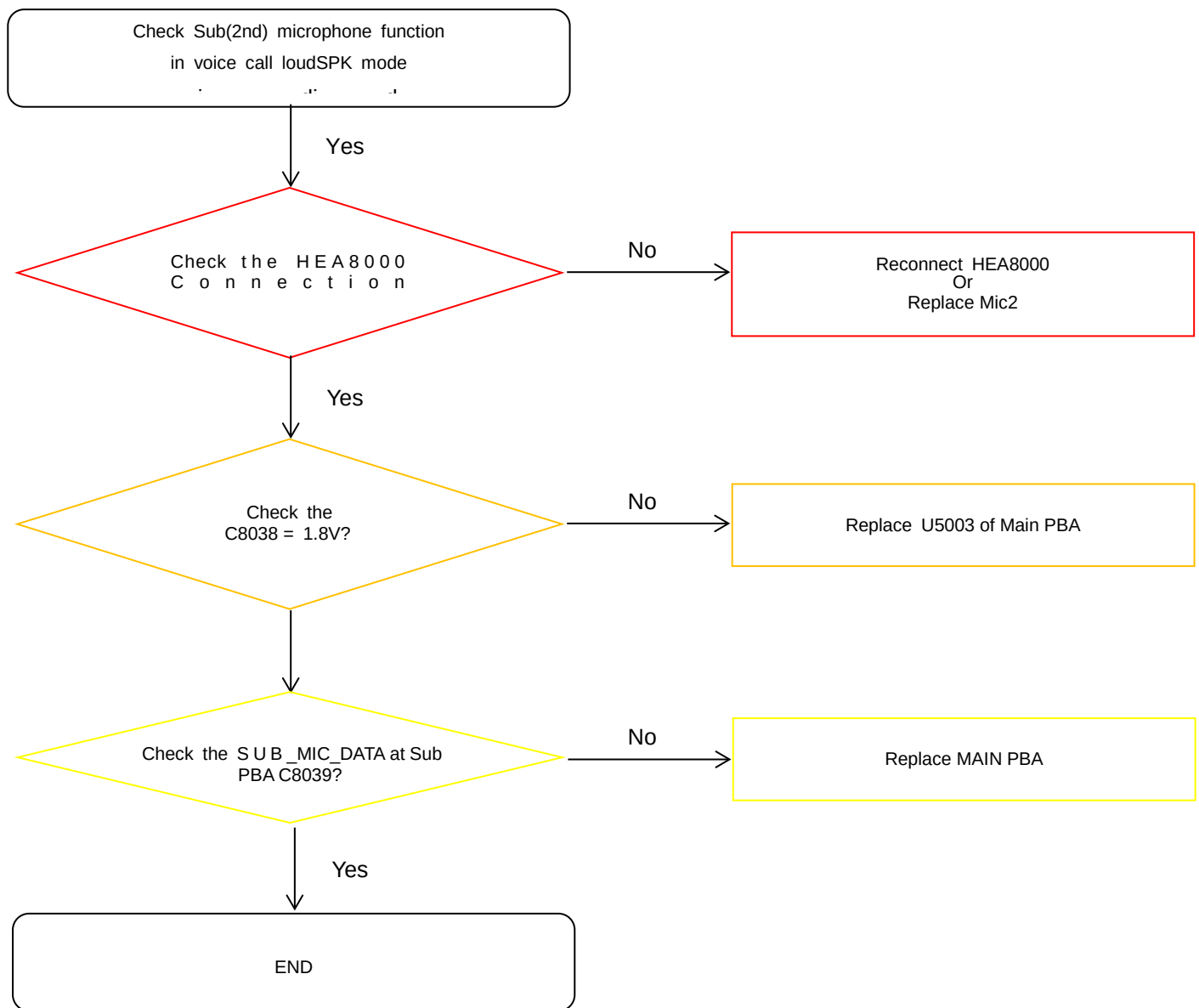
8. Level 3 Repair

8-4-9-1. Microphone Part - Main MIC



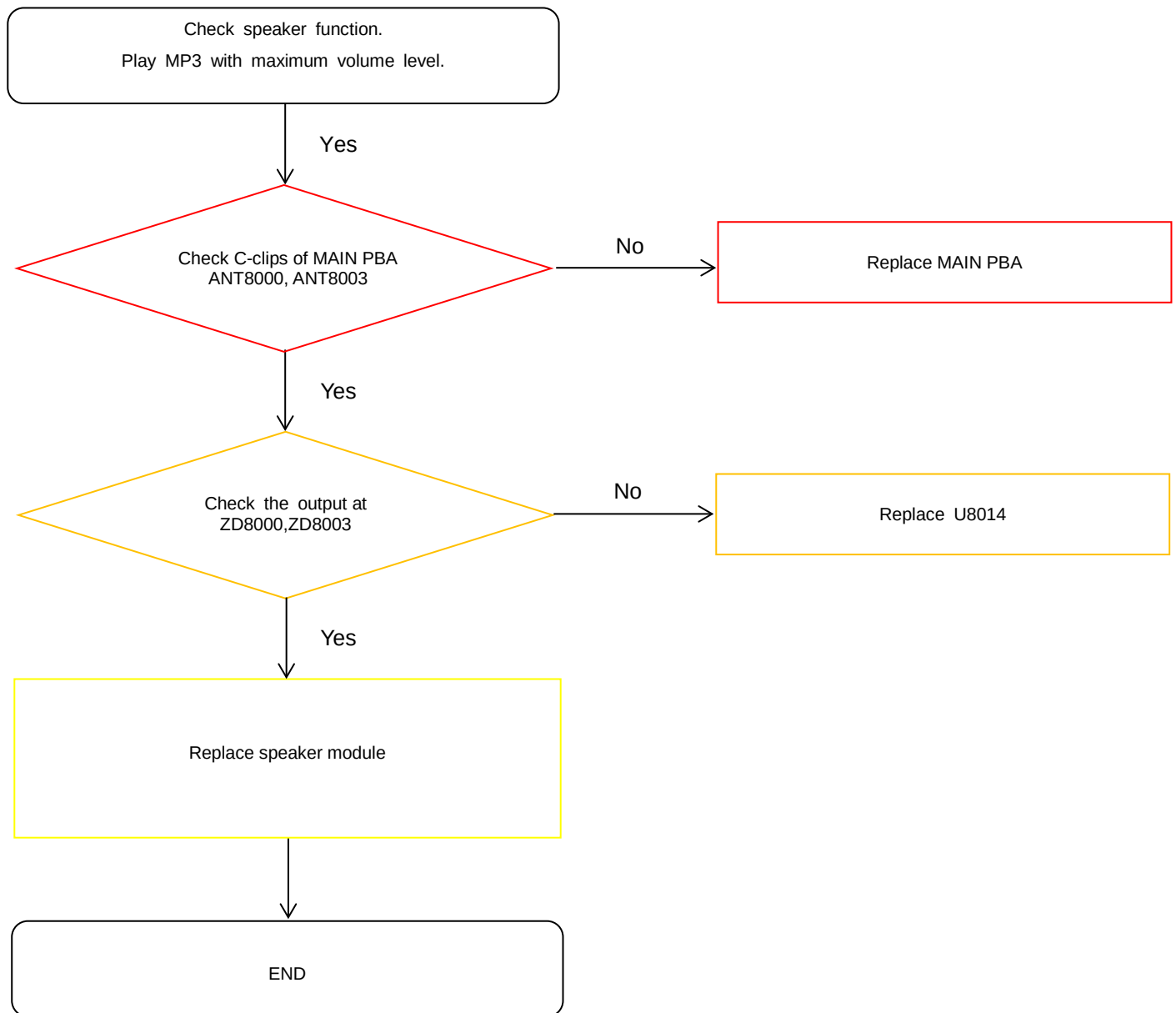
8. Level 3 Repair

8-4-9-2. Microphone Part - Sub(2nd) MIC



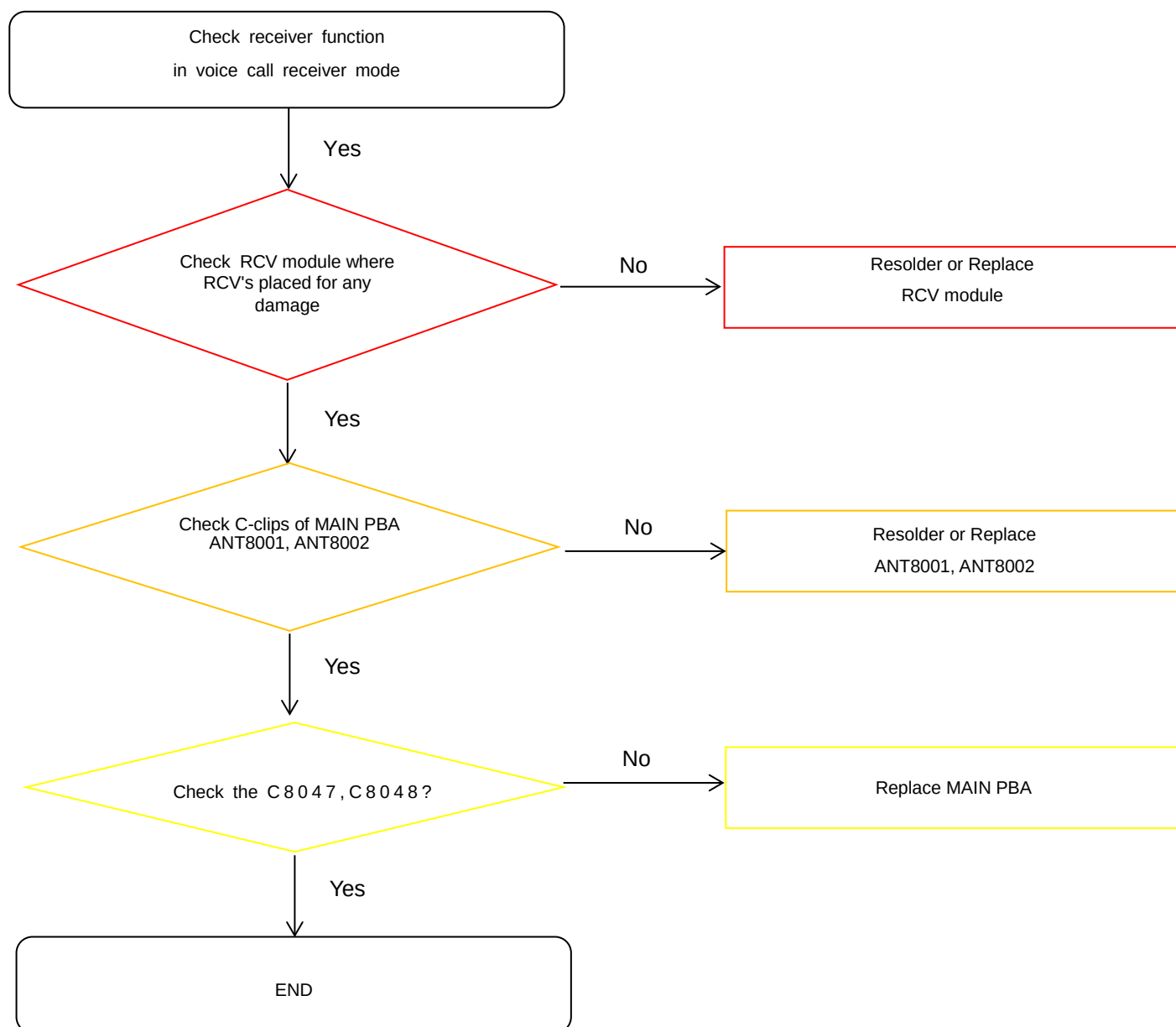
8. Level 3 Repair

8-4-10. Speaker Part



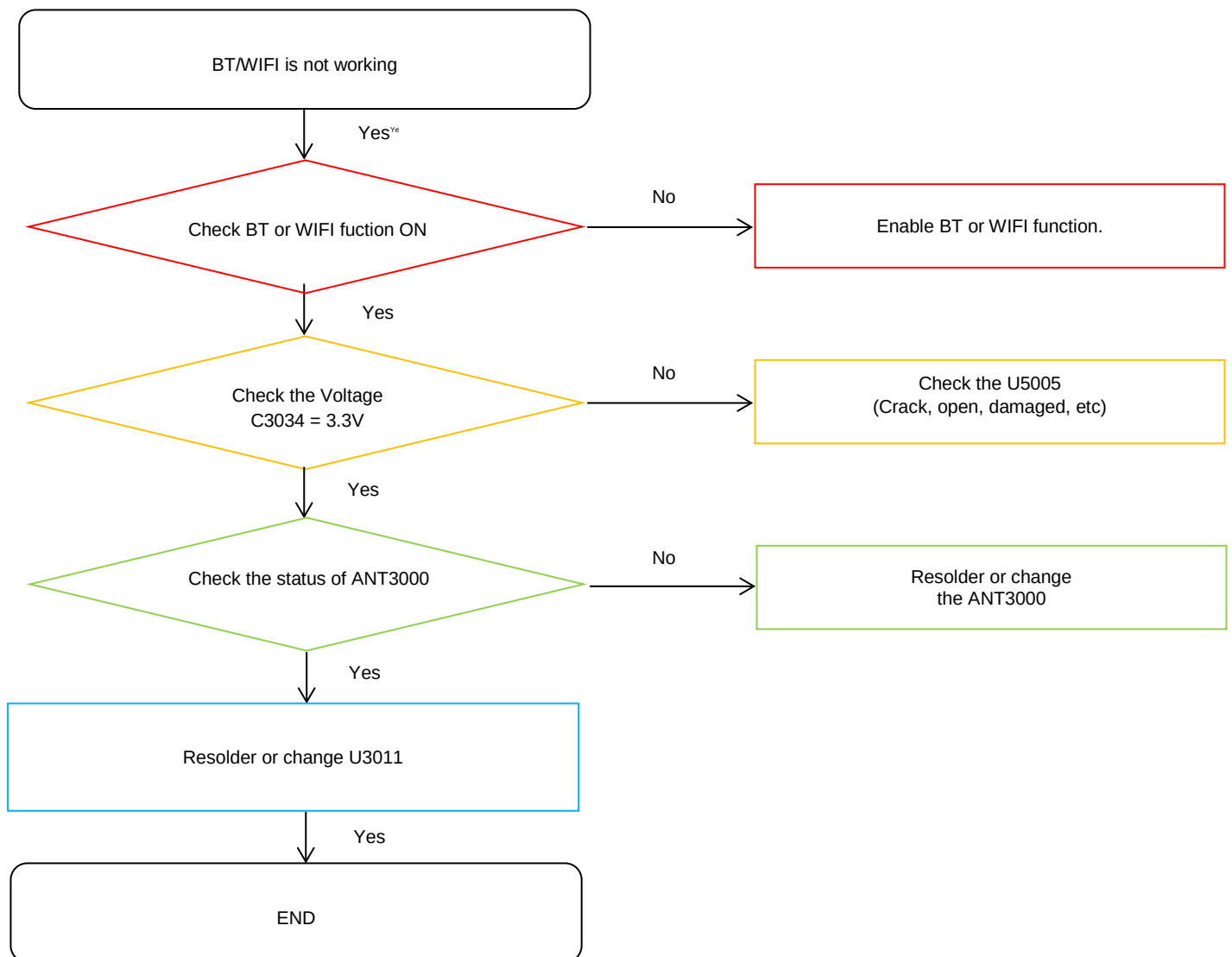
8. Level 3 Repair

8-4-11. Receiver Part



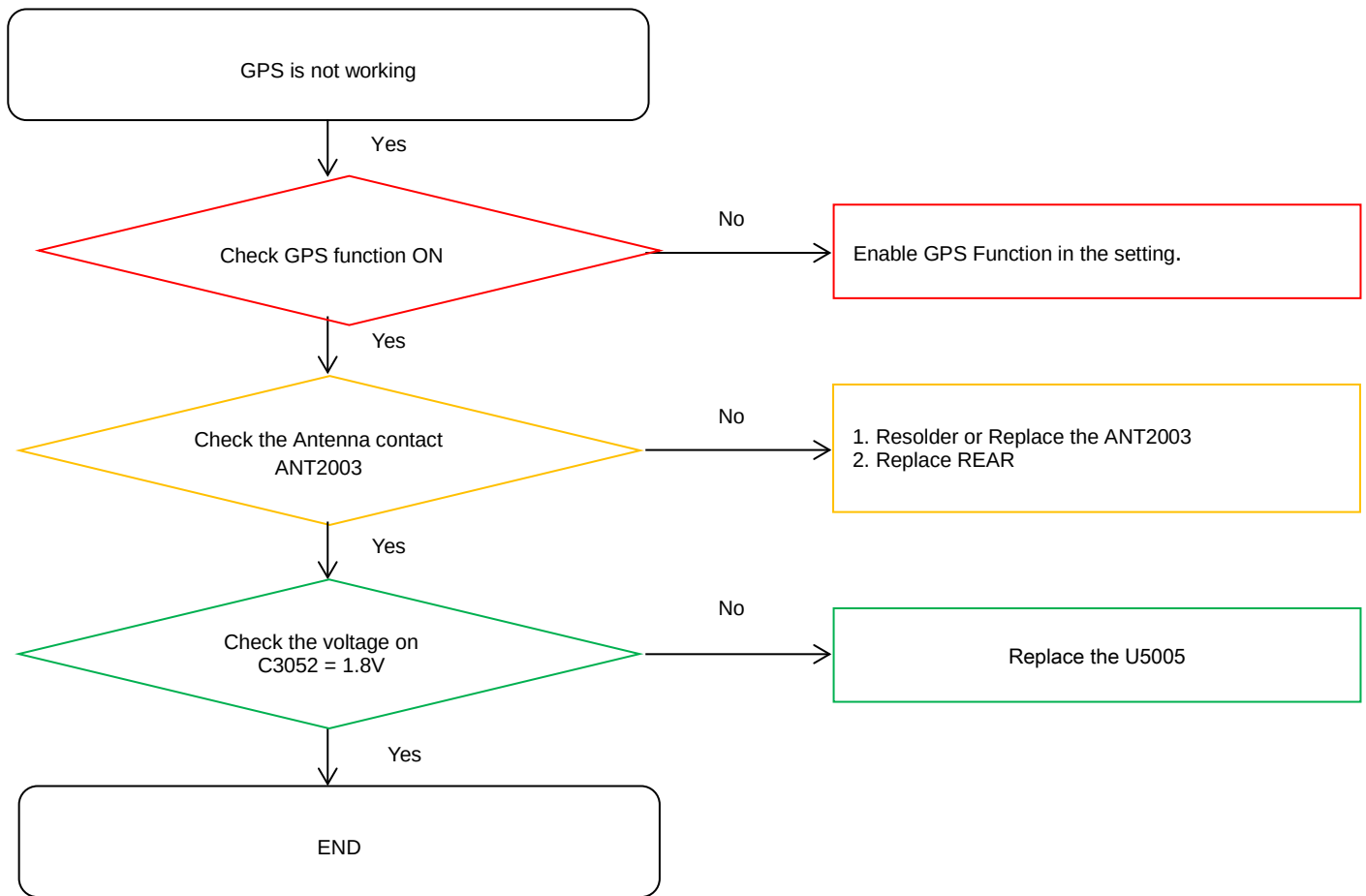
8. Level 3 Repair

8-4-12. BT/WIFI



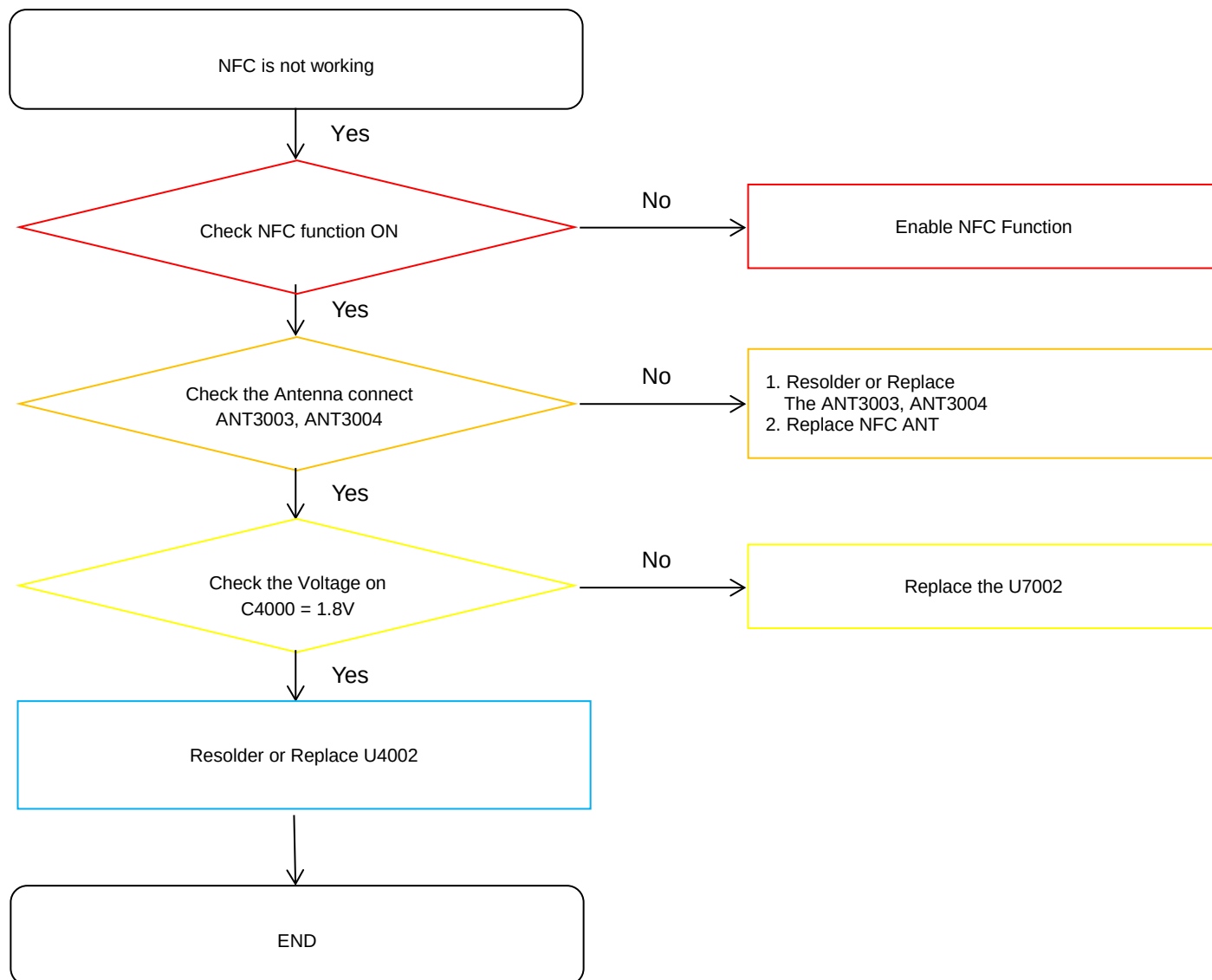
8. Level 3 Repair

8-4-13. GPS



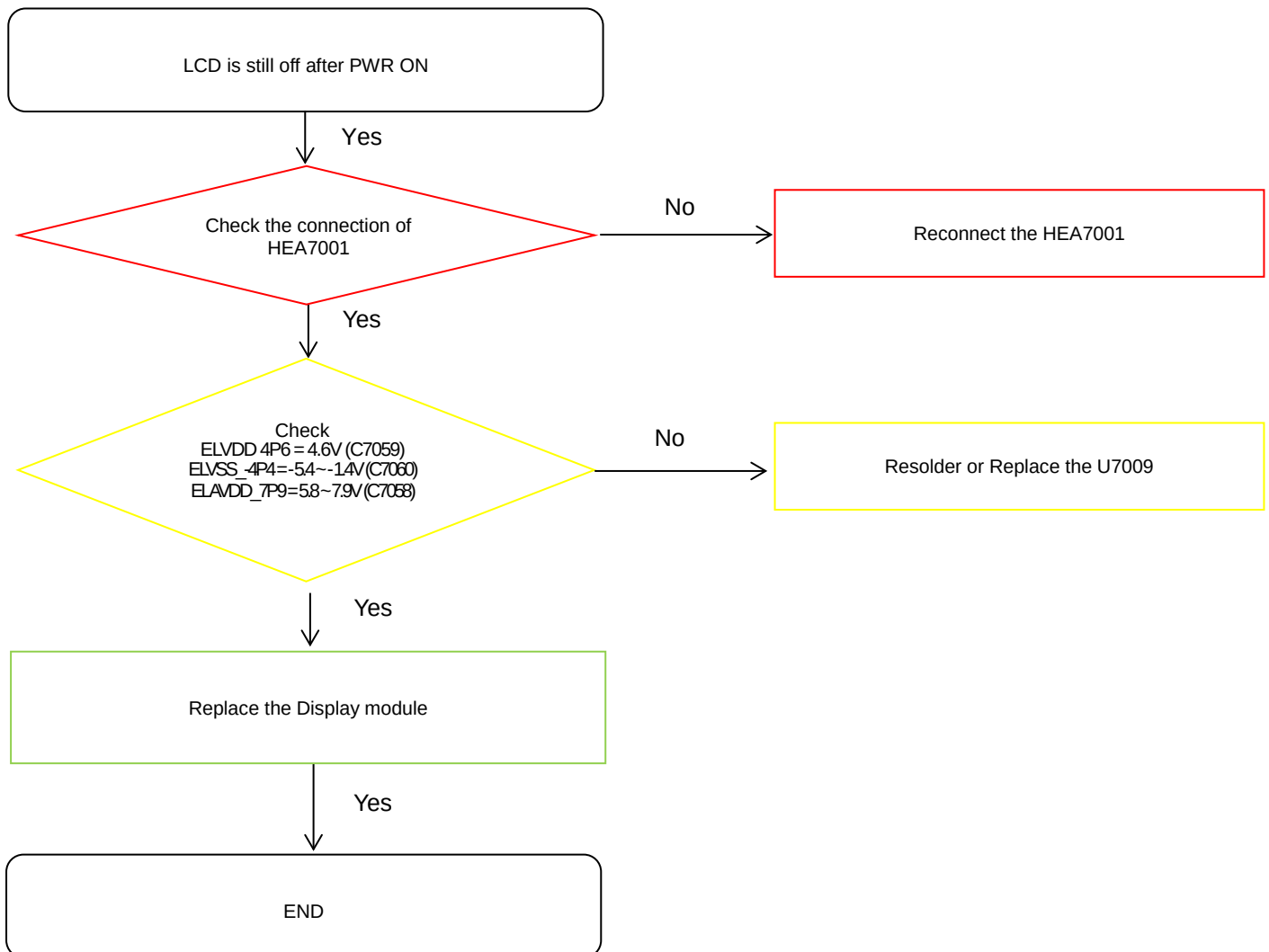
8. Level 3 Repair

8-4-14. NFC



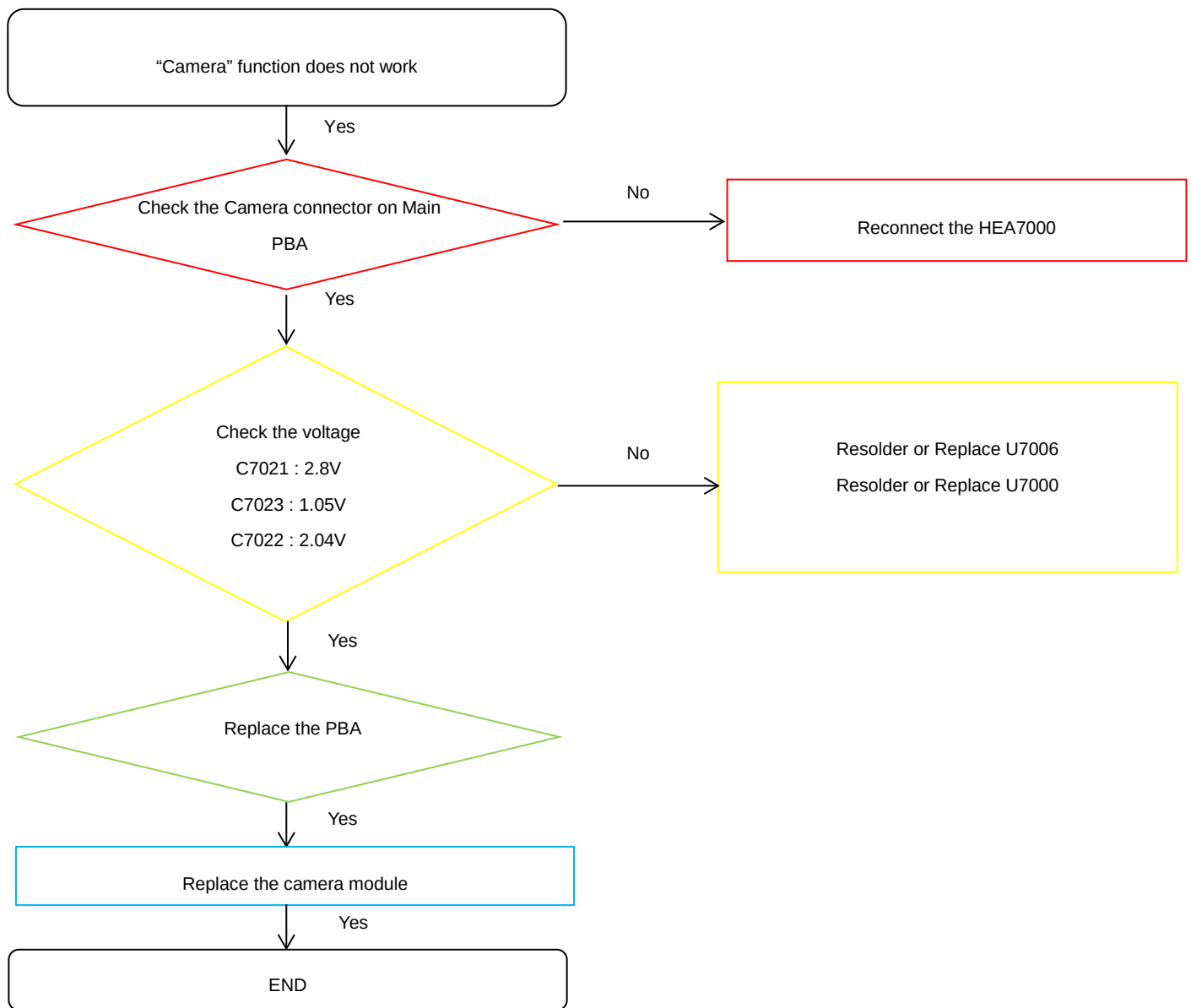
8. Level 3 Repair

8-4-15. Display



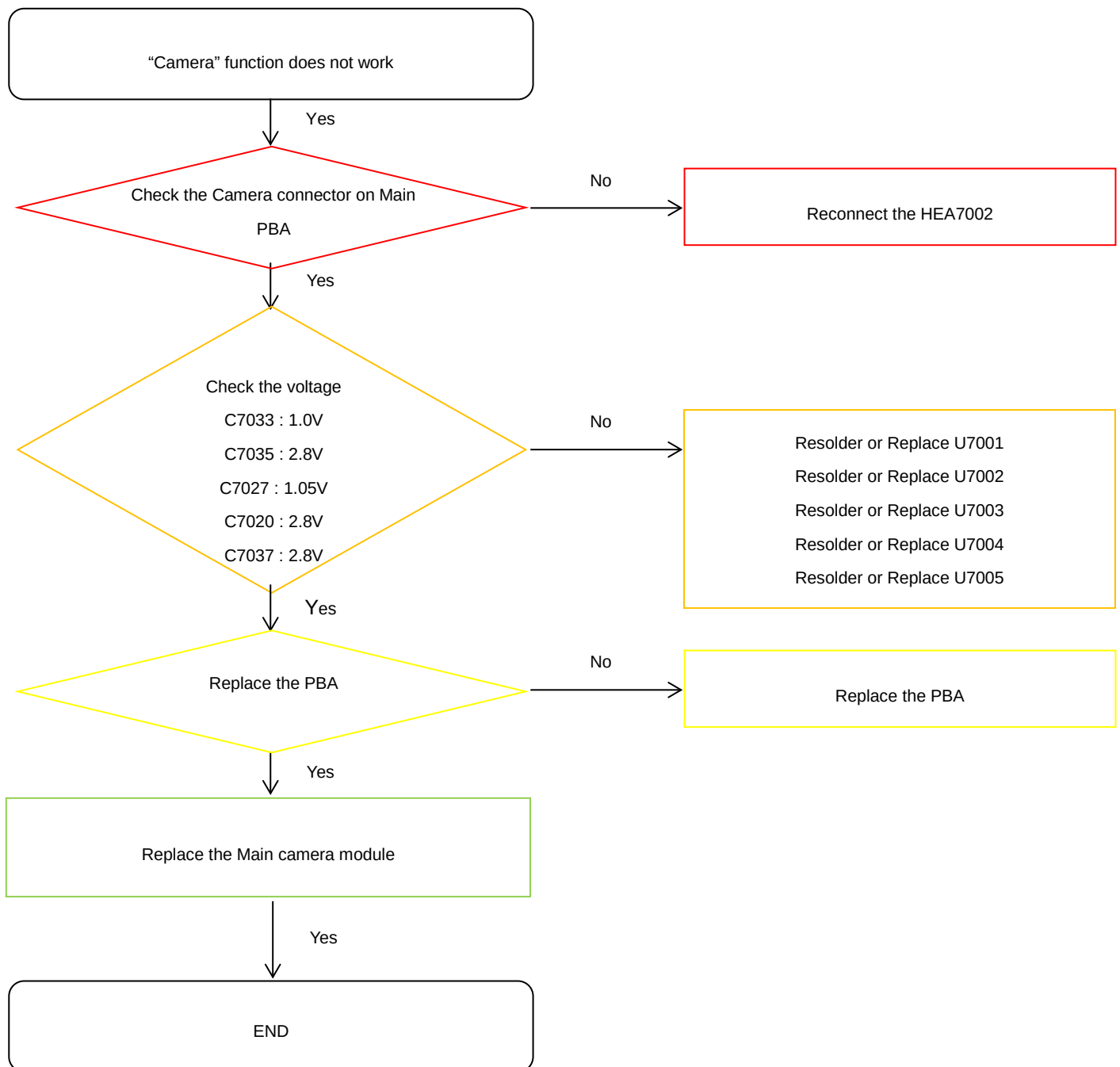
8. Level 3 Repair

8-4-16. 24M CAMERA(FRONT CAMERA)



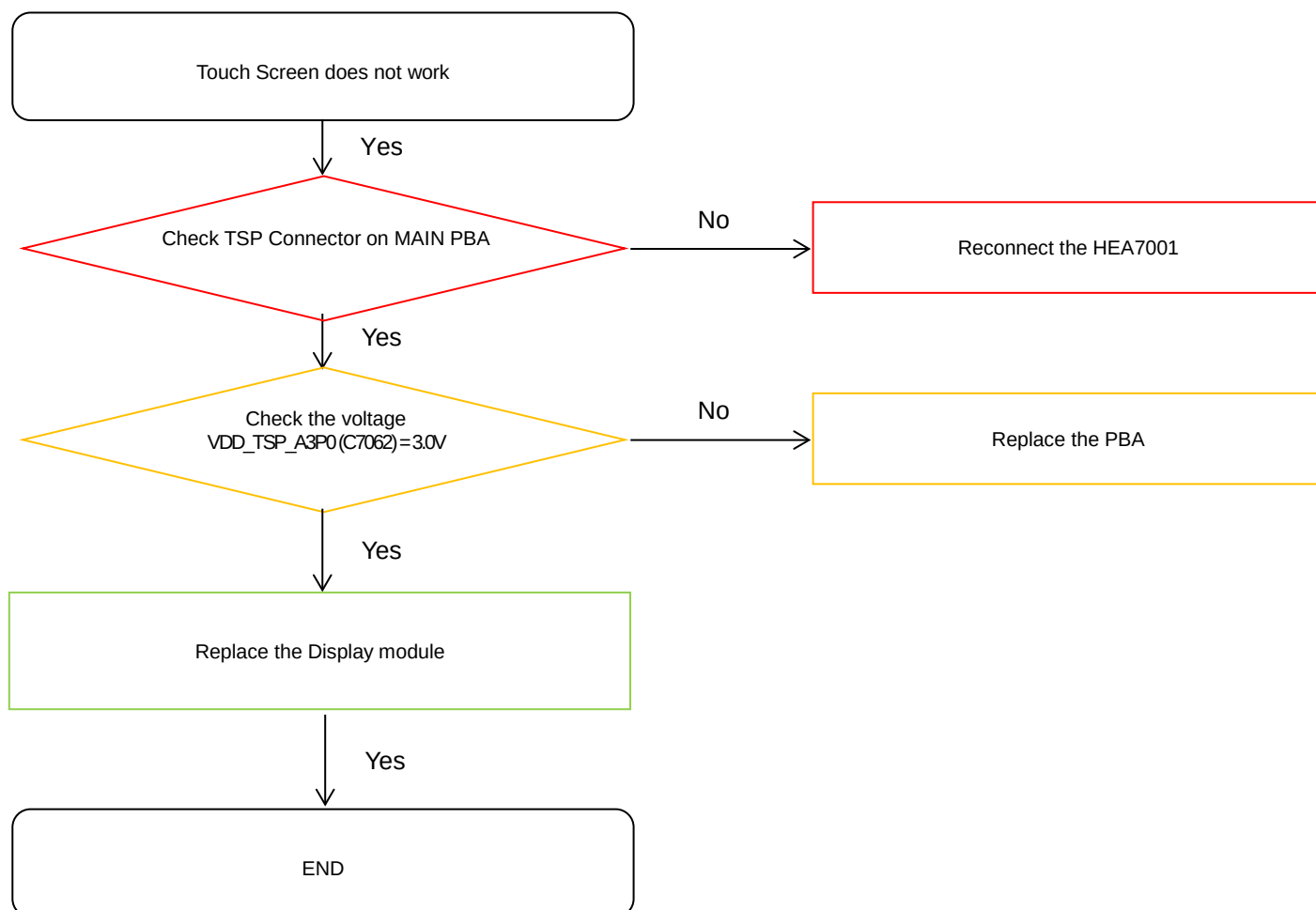
8. Level 3 Repair

8-4-17. DUAL CAMERA(MAIN CAMERA)



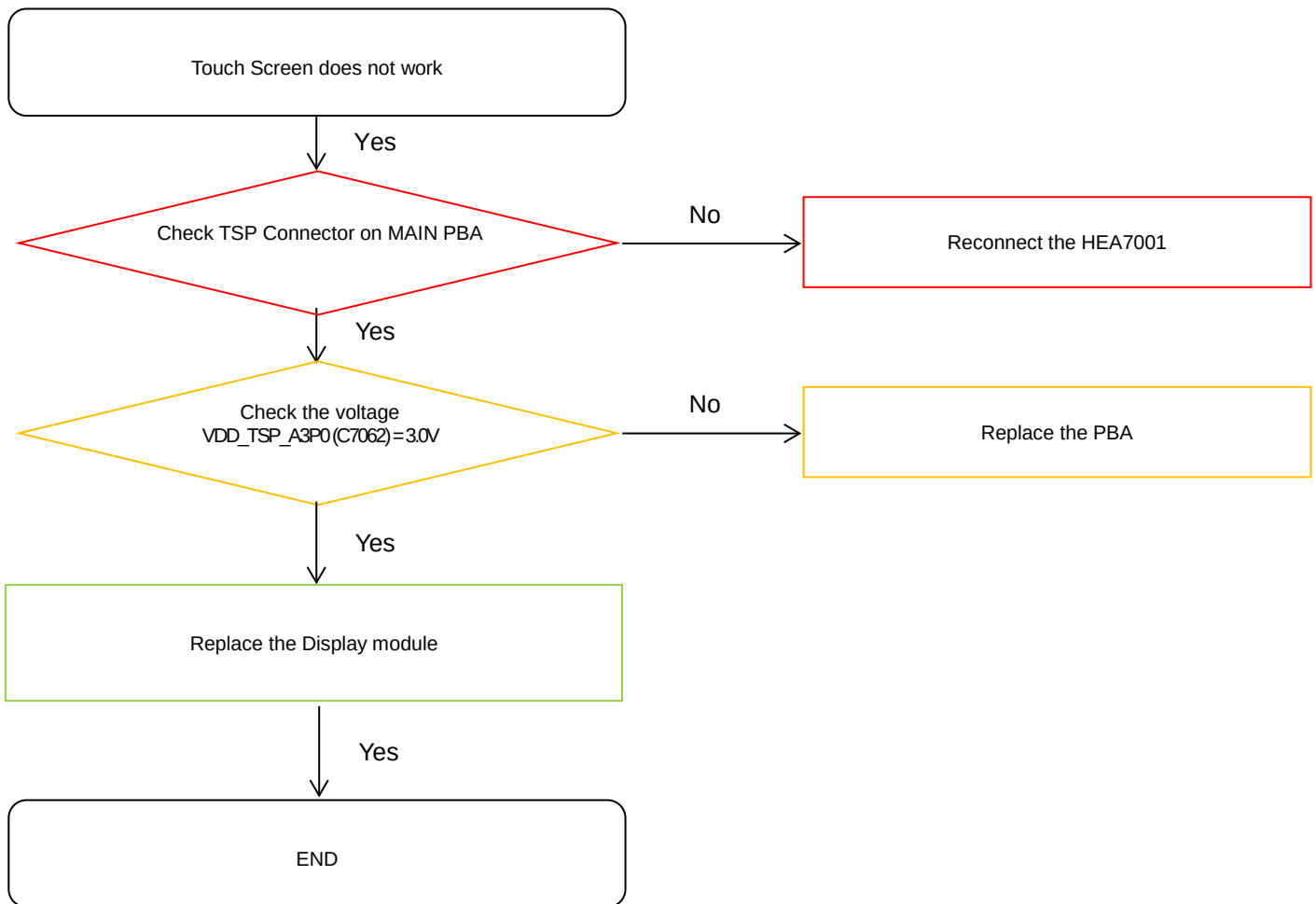
8. Level 3 Repair

8-4-18. TSP



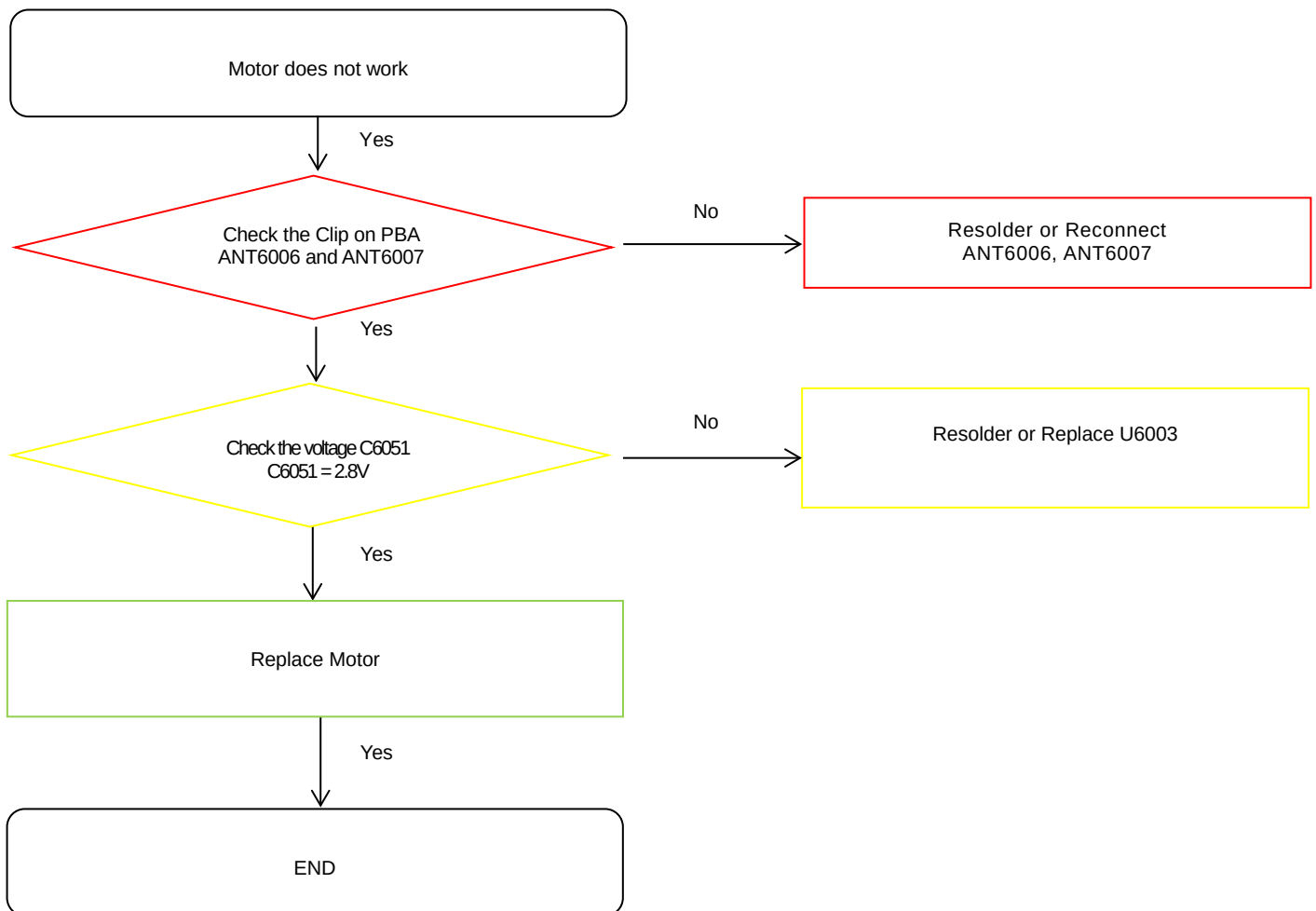
8. Level 3 Repair

8-4-19. Pressure Touch (Home Button function)



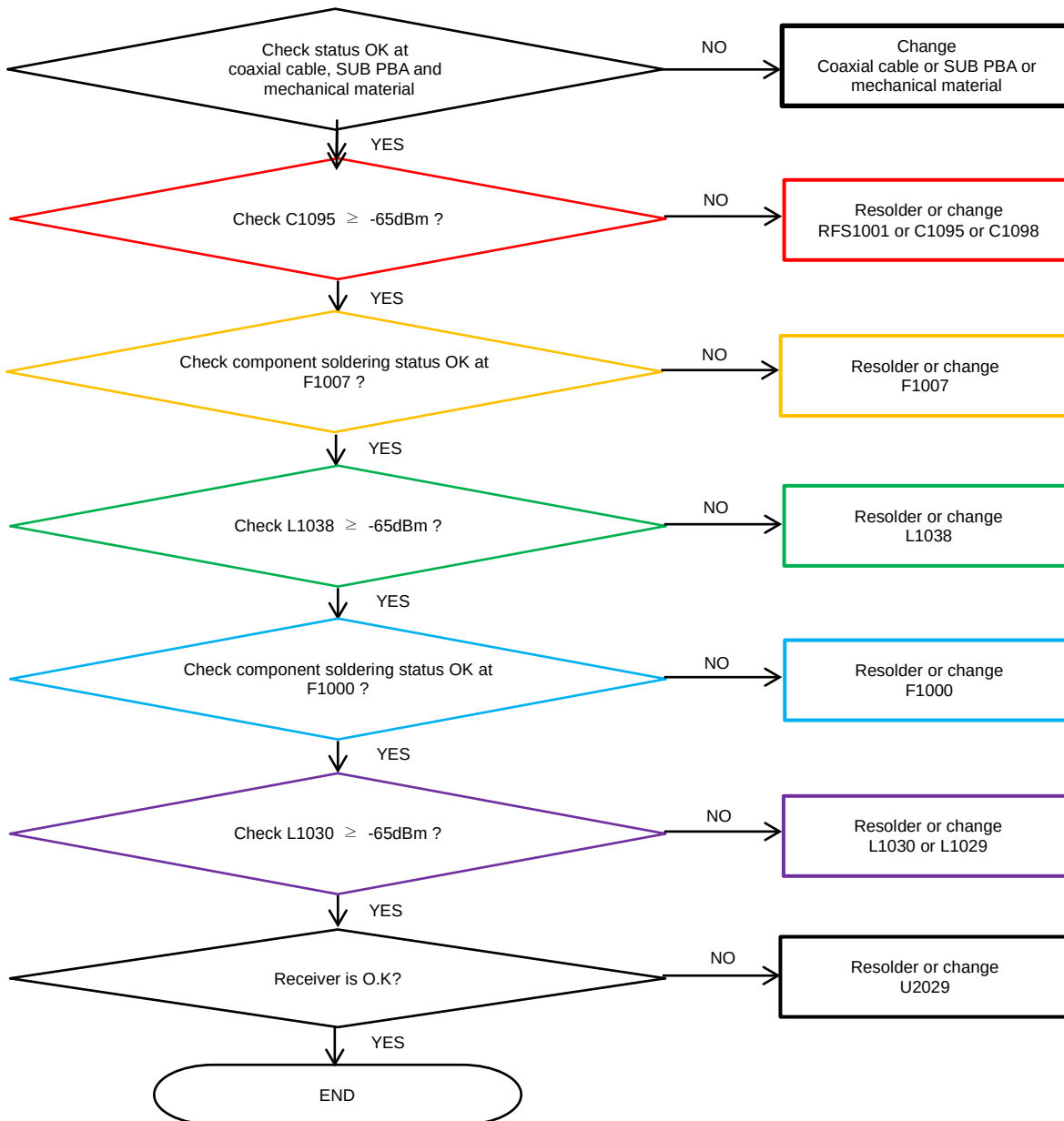
8. Level 3 Repair

8-4-20. Motor



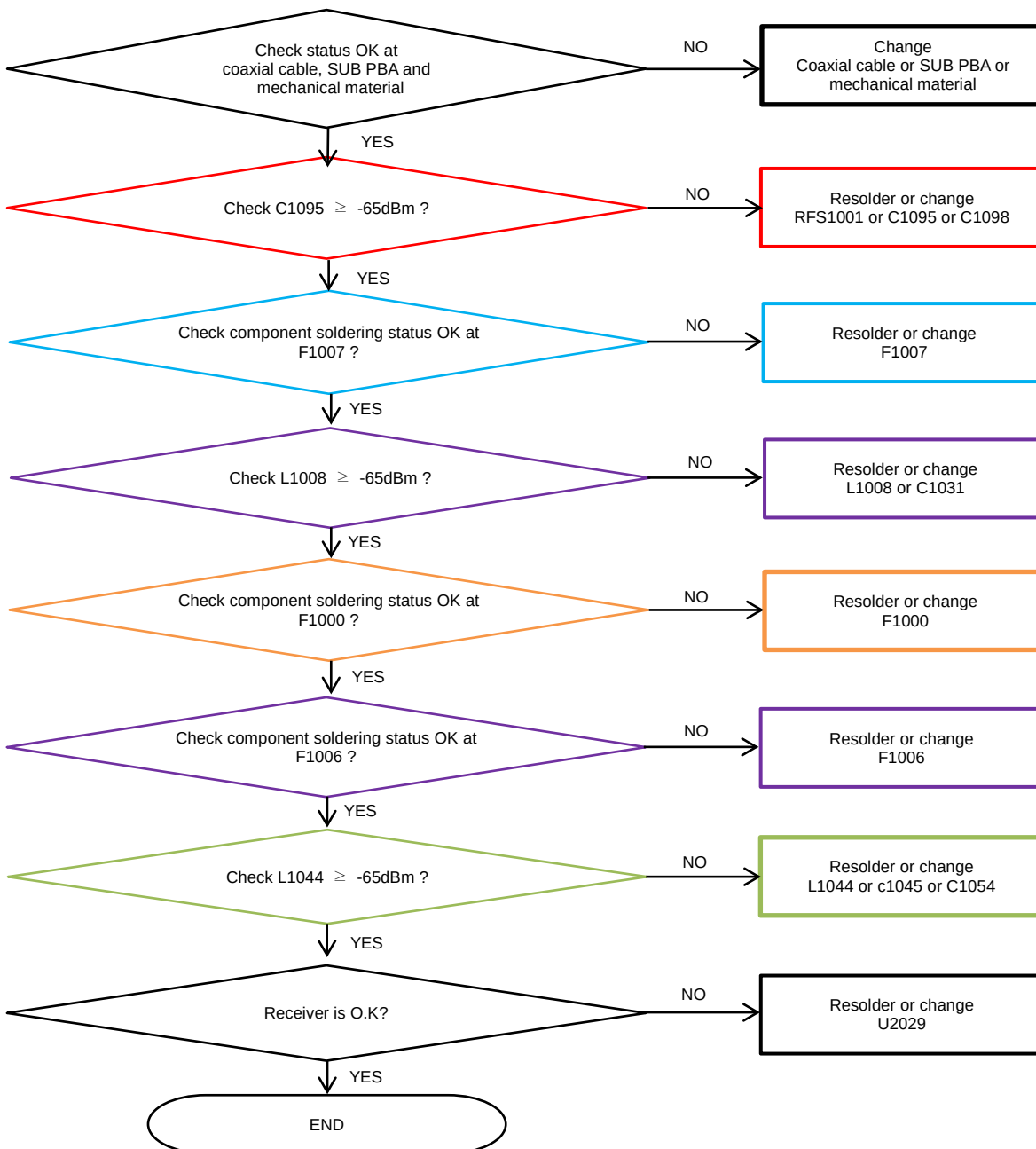
8. Level 3 Repair

8-4-21. GSM 850/900, WCDMA B5/8, LTE B5/8 Rx



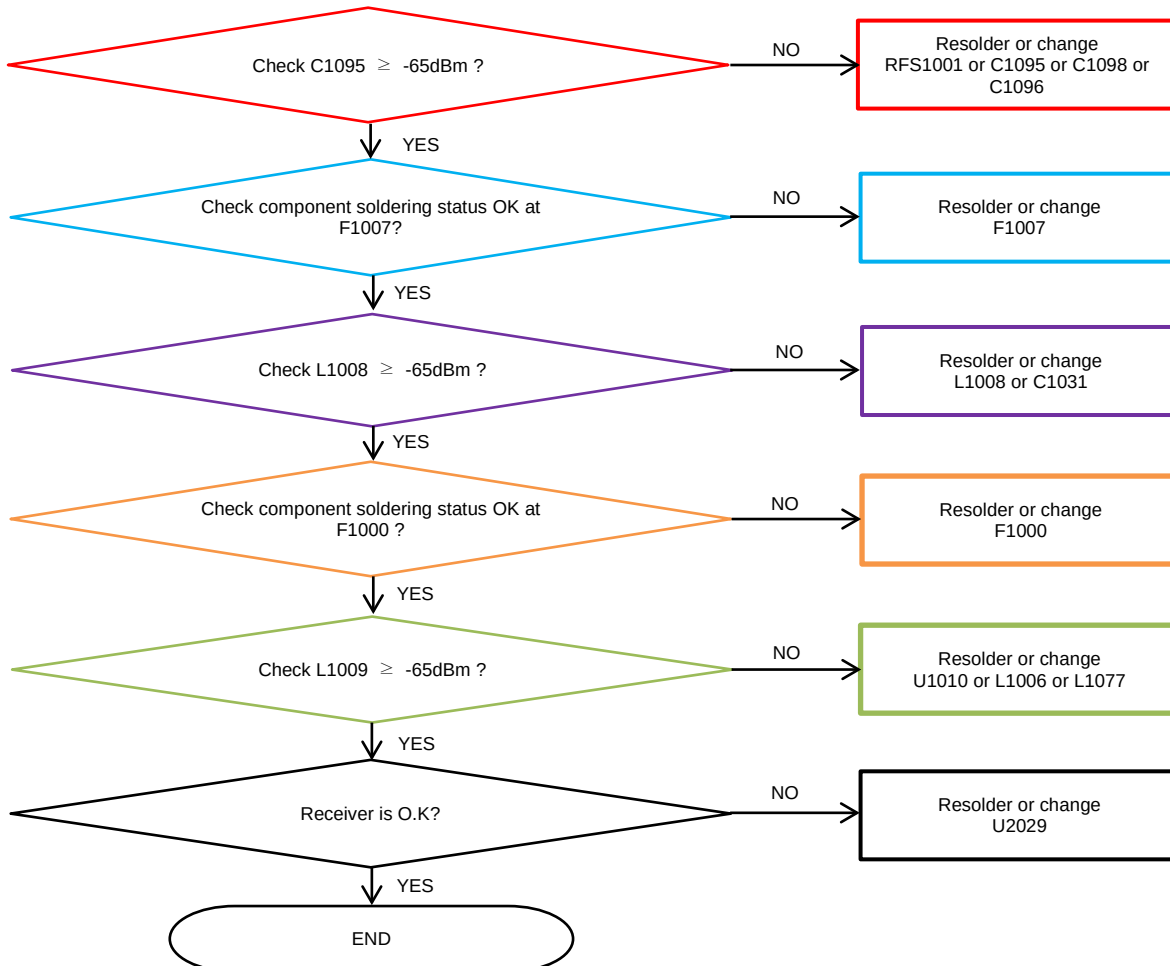
8. Level 3 Repair

8-4-22. TD-SCDMA B34/39, LTE B34/39 Rx



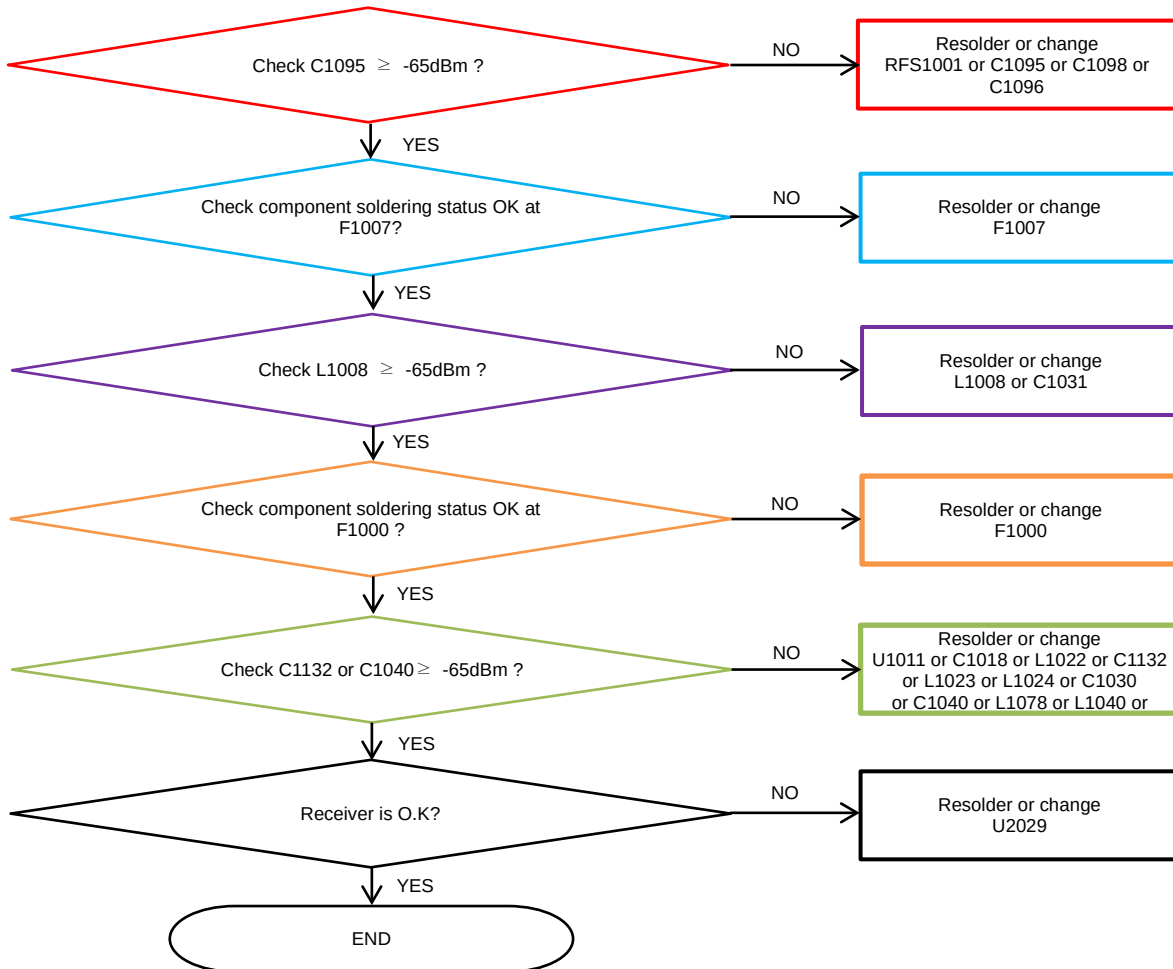
8. Level 3 Repair

8-4-23. WCDMA B1, LTE B1/4 Rx



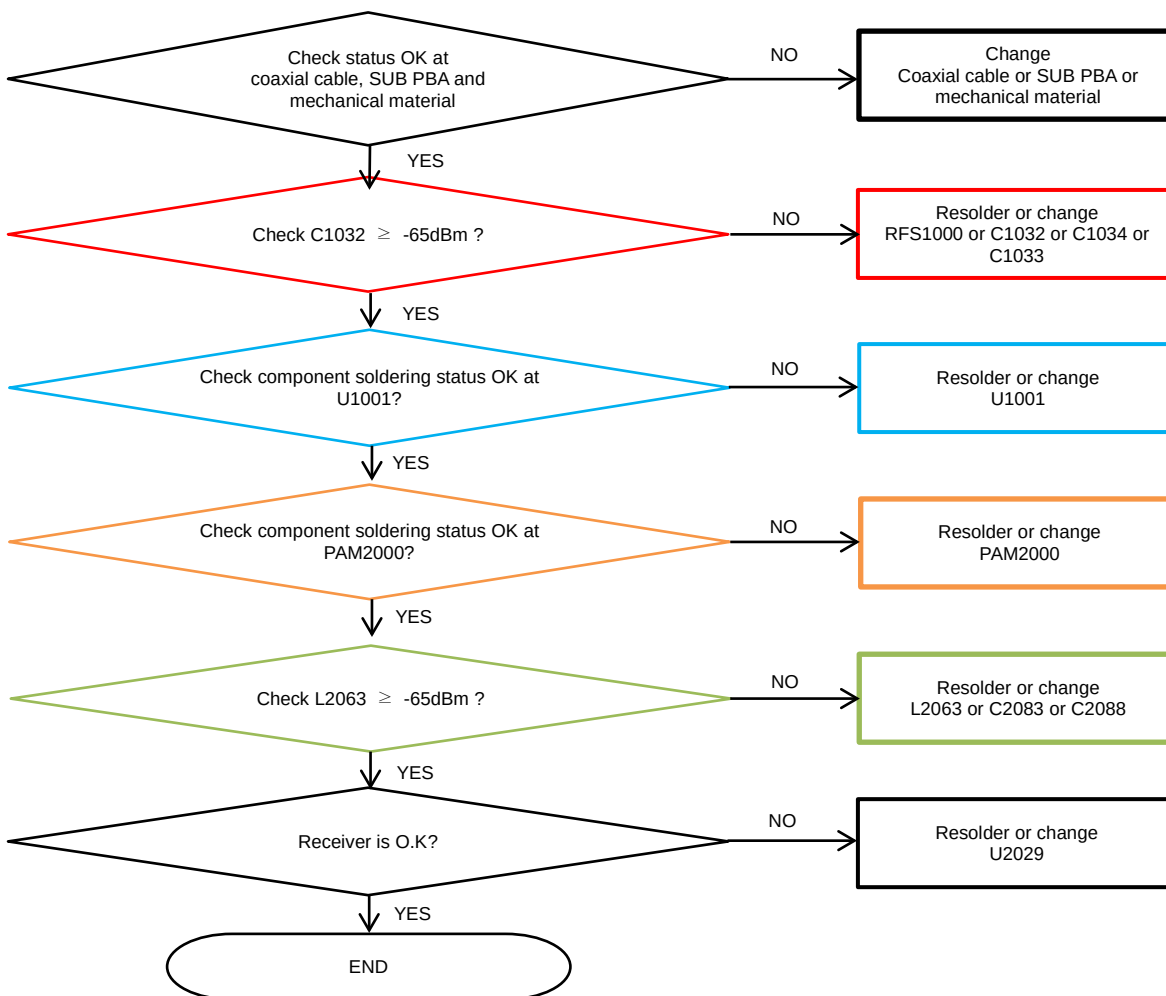
8. Level 3 Repair

8-4-24. DCS1800, LTE B3 Rx



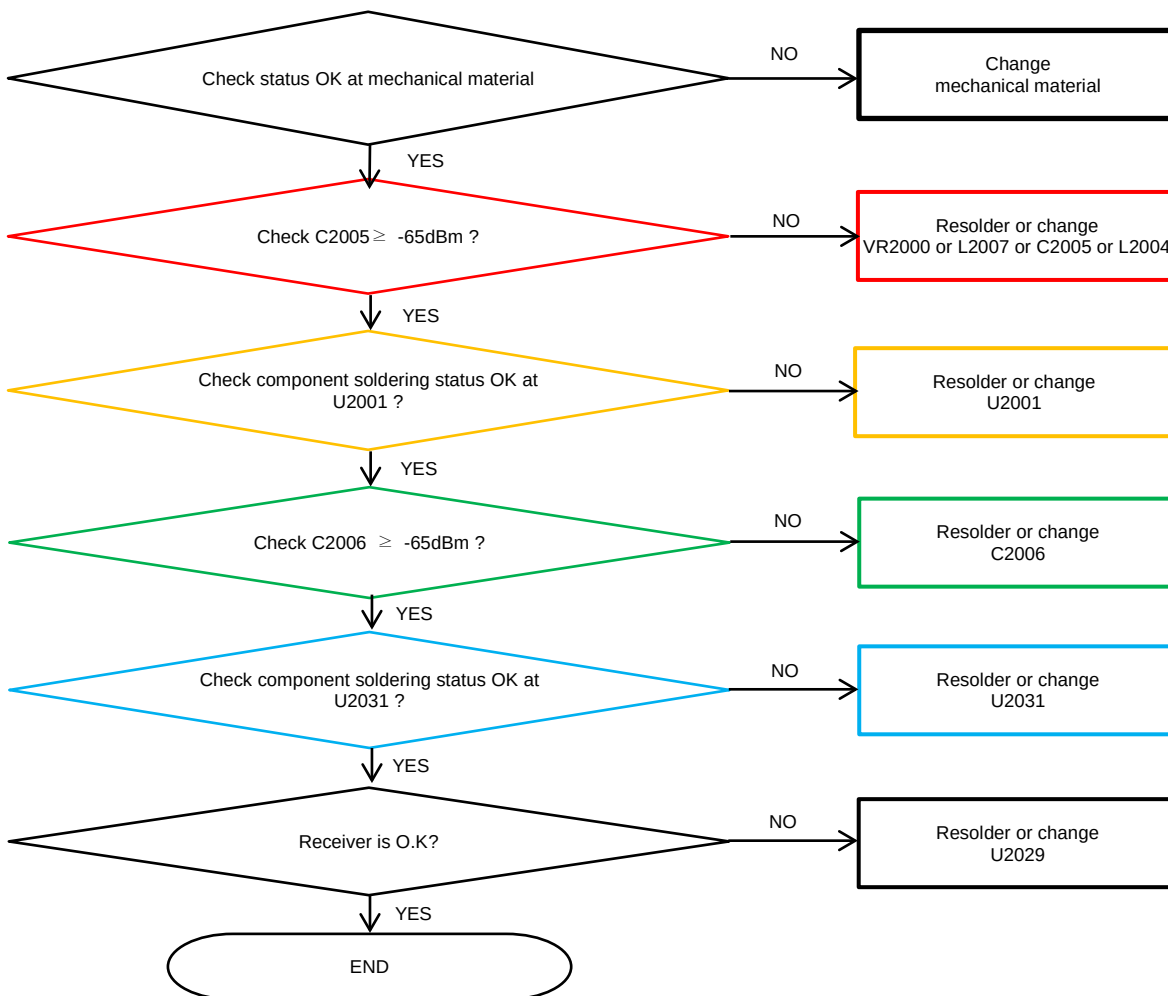
8. Level 3 Repair

8-4-25. LTE B7/38/40/41 Rx



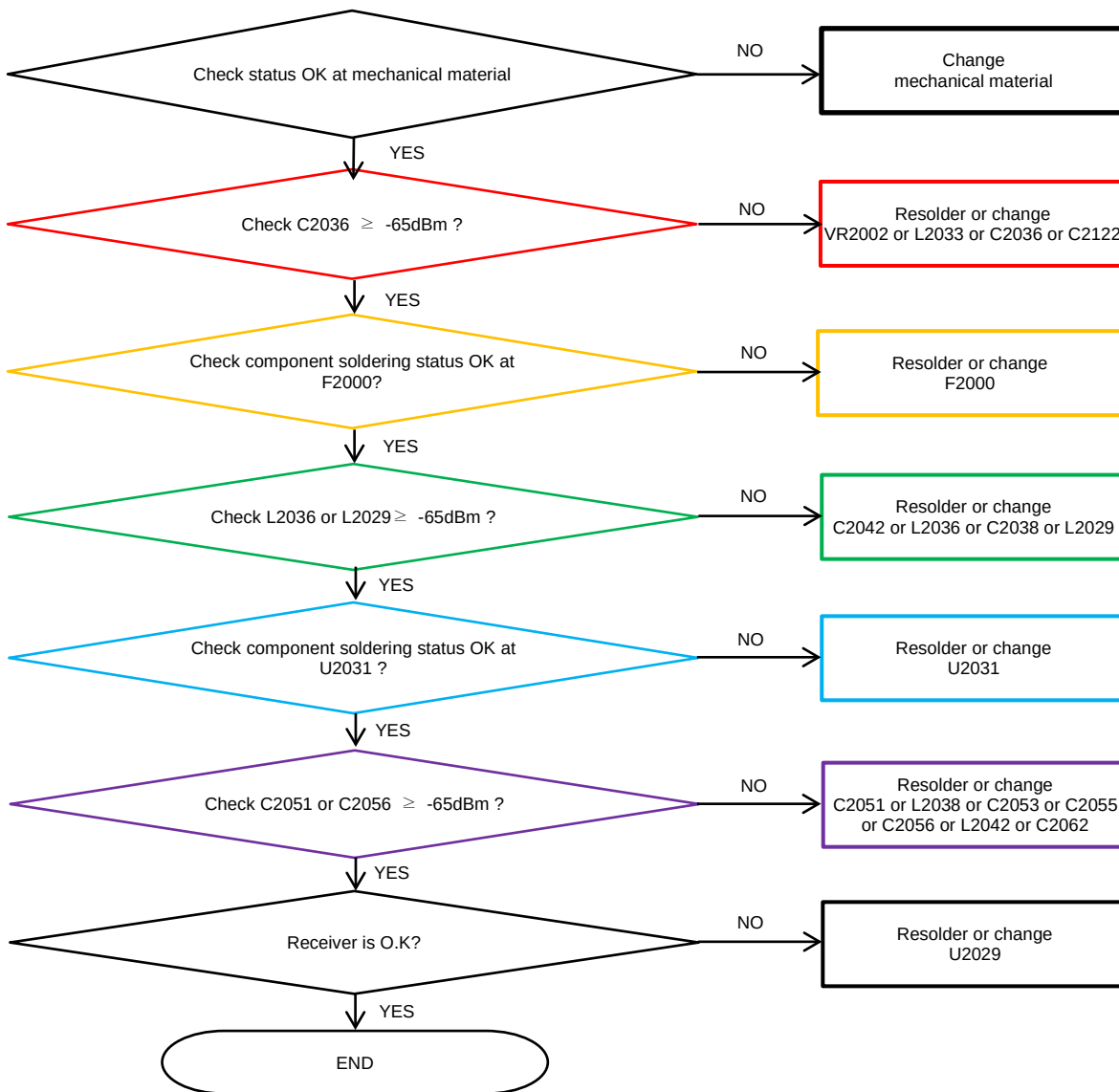
8. Level 3 Repair

8-4-26. LTE B5/8, WCDMA B5/8 , GSM850/900 DRx



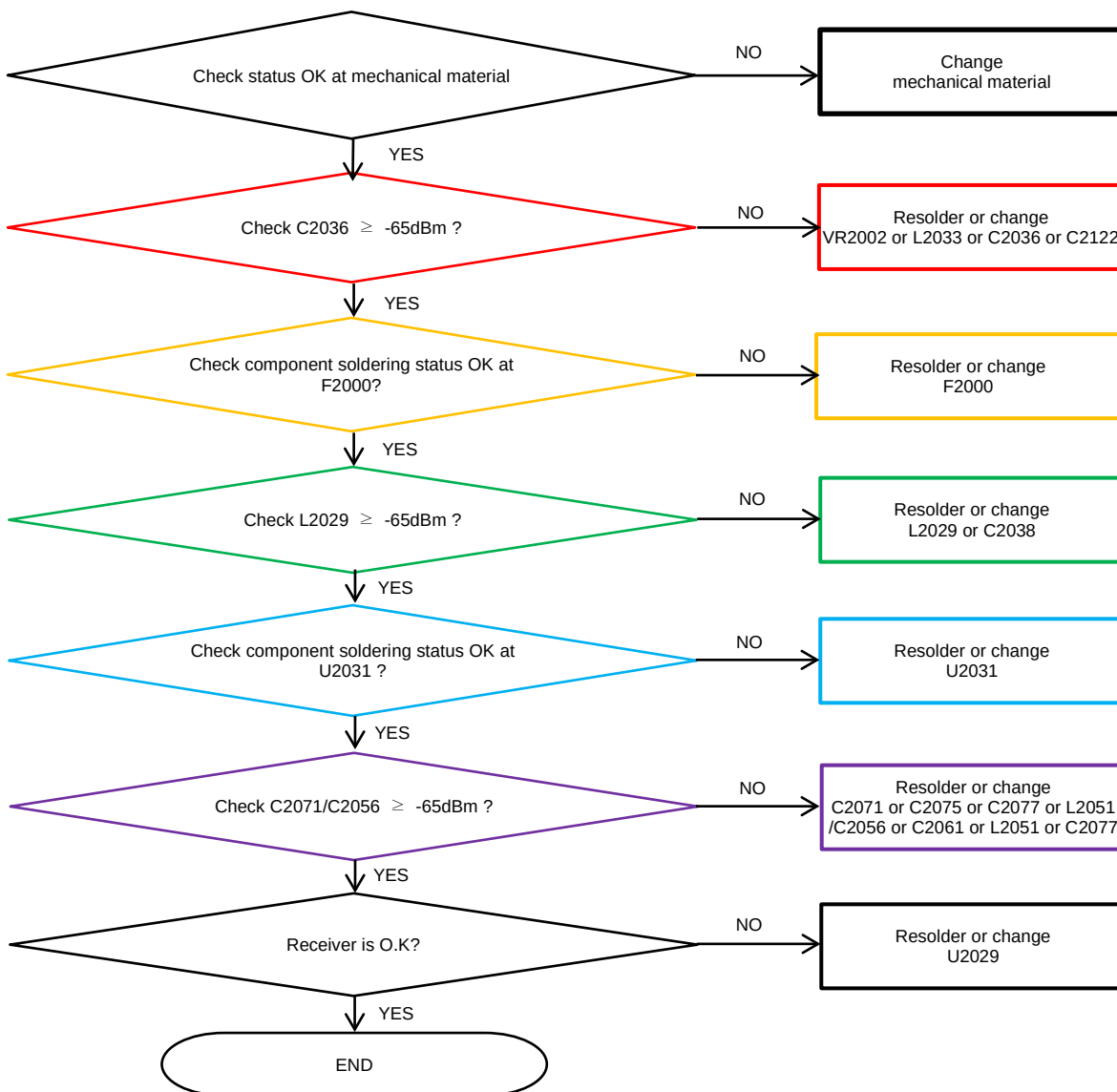
8. Level 3 Repair

8-4-27. LTE B7/38/40/41 , DRx



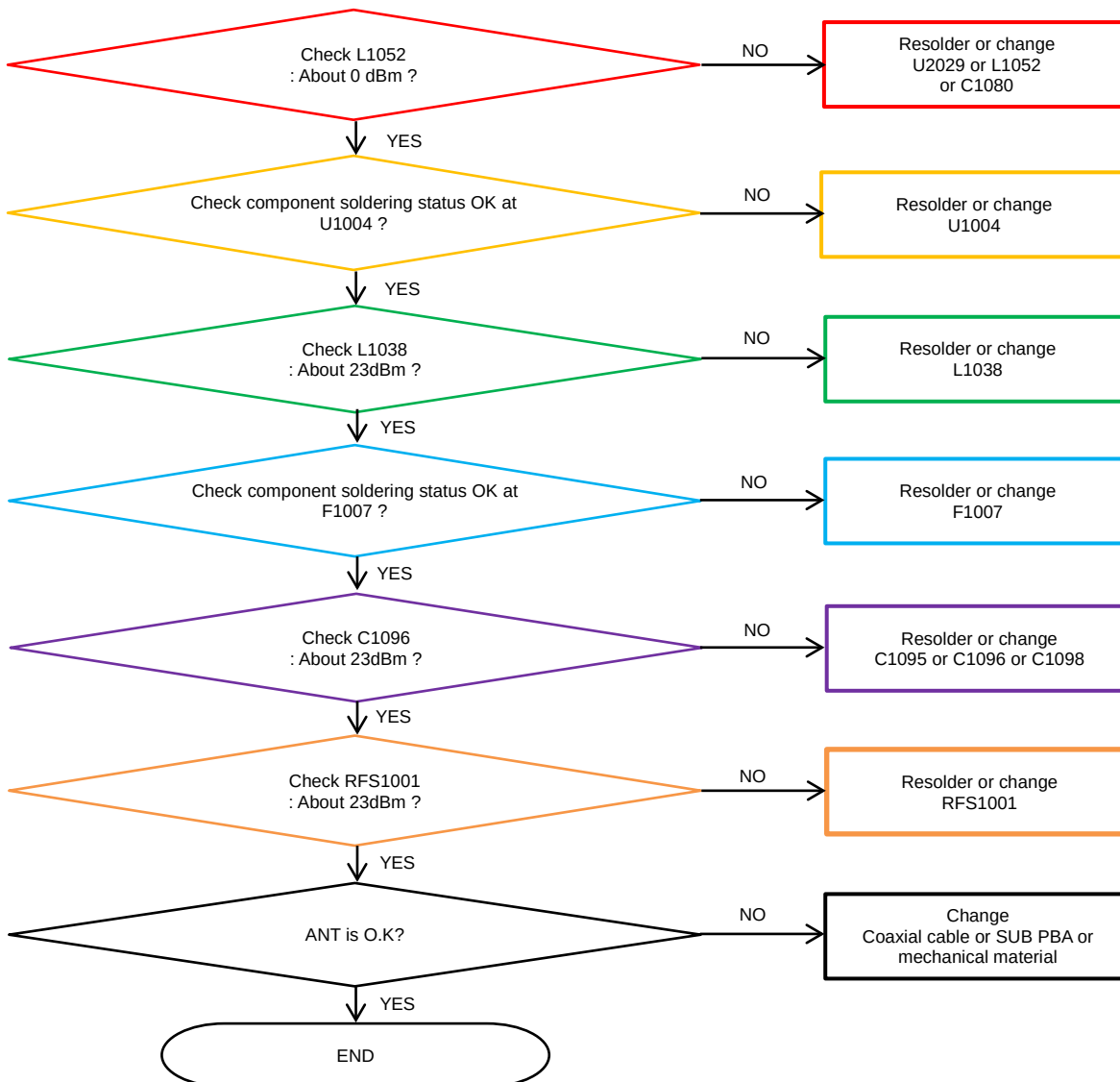
8. Level 3 Repair

8-4-28. LTE B1/3/4/34/39 , WCDMA B2/B1 , GSM1800/1900 DRx



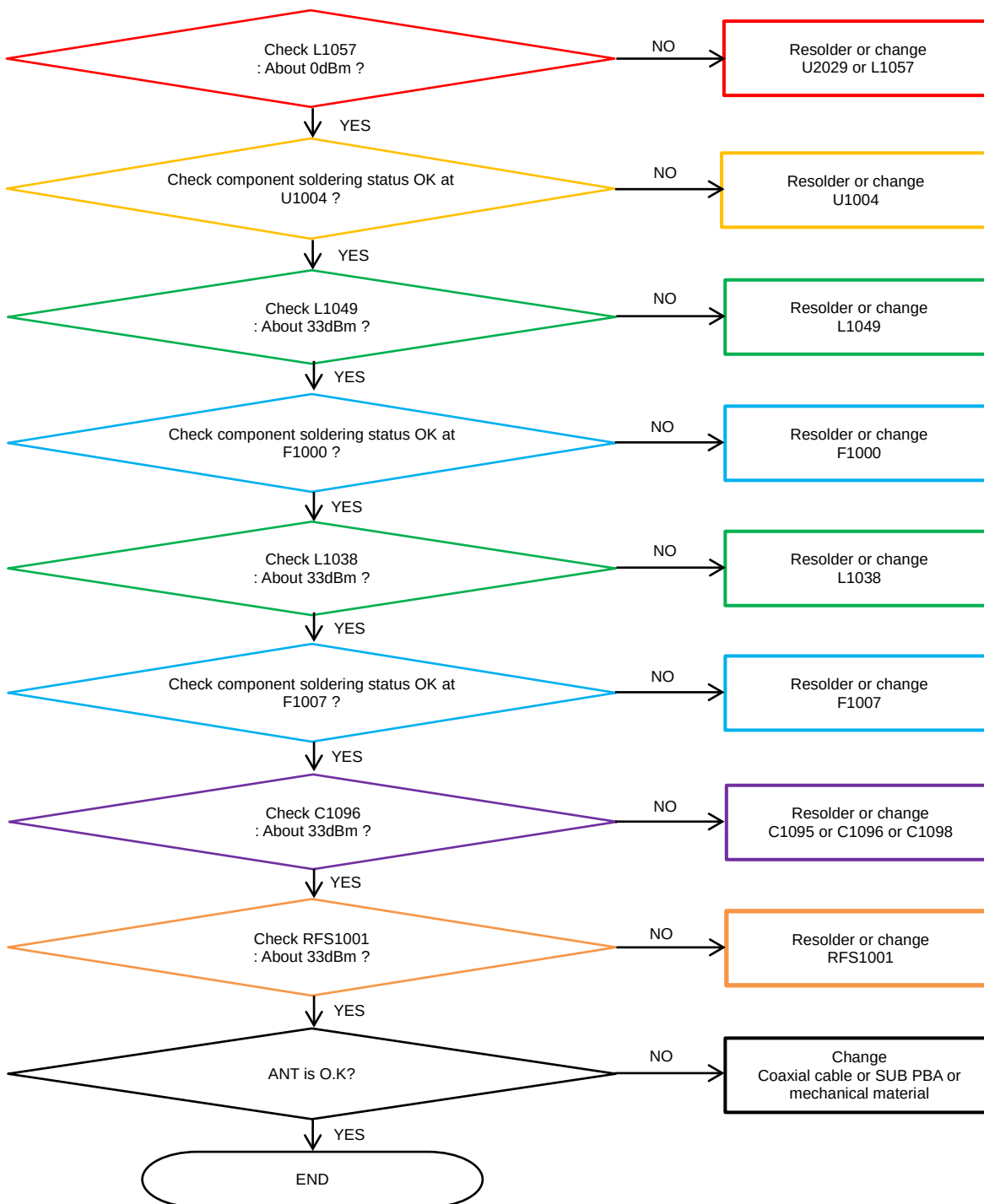
8. Level 3 Repair

8-4-29. WCDMA B5/8, LTE B5/8 Tx



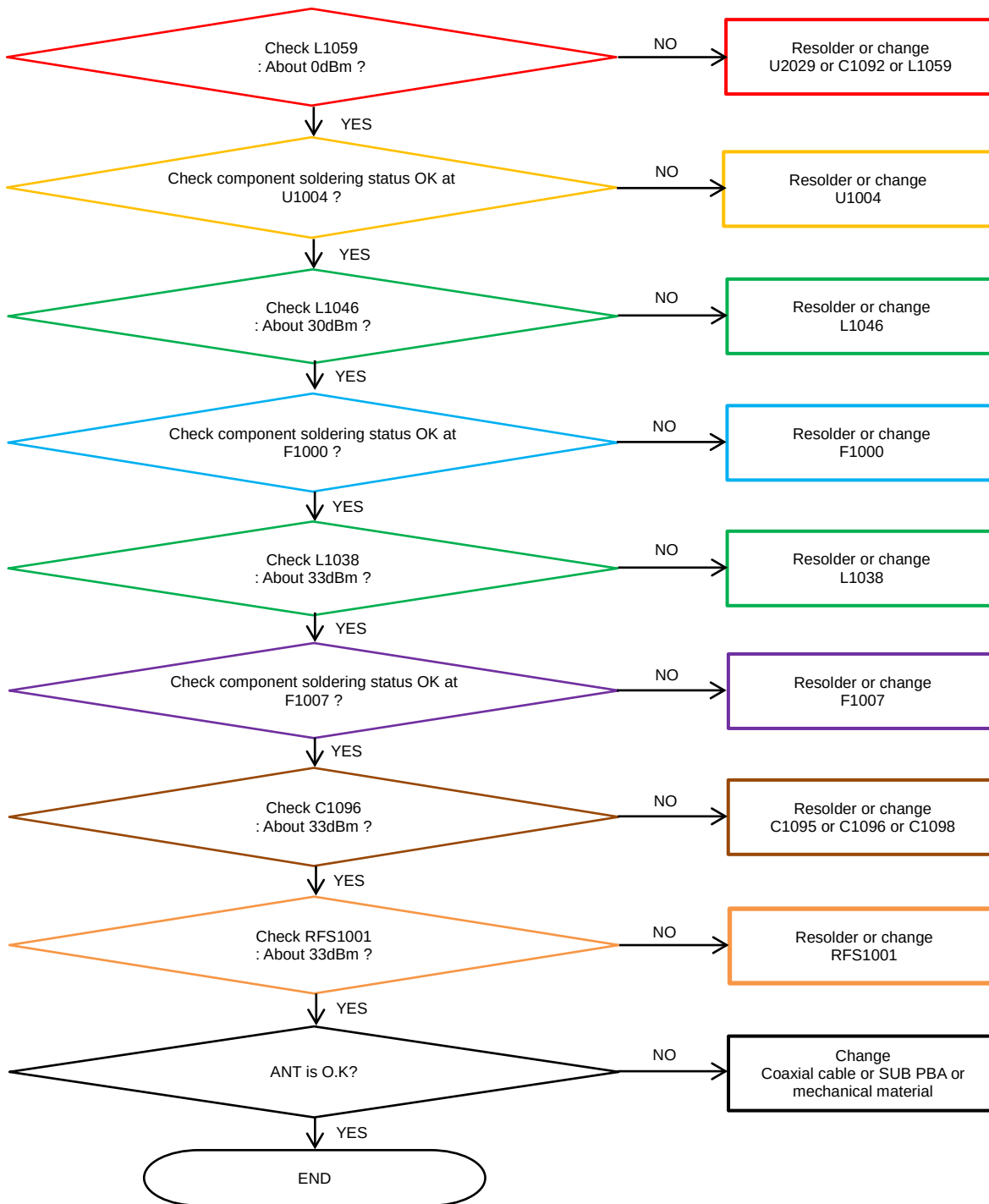
8. Level 3 Repair

8-4-30. GSM 850/900 Tx



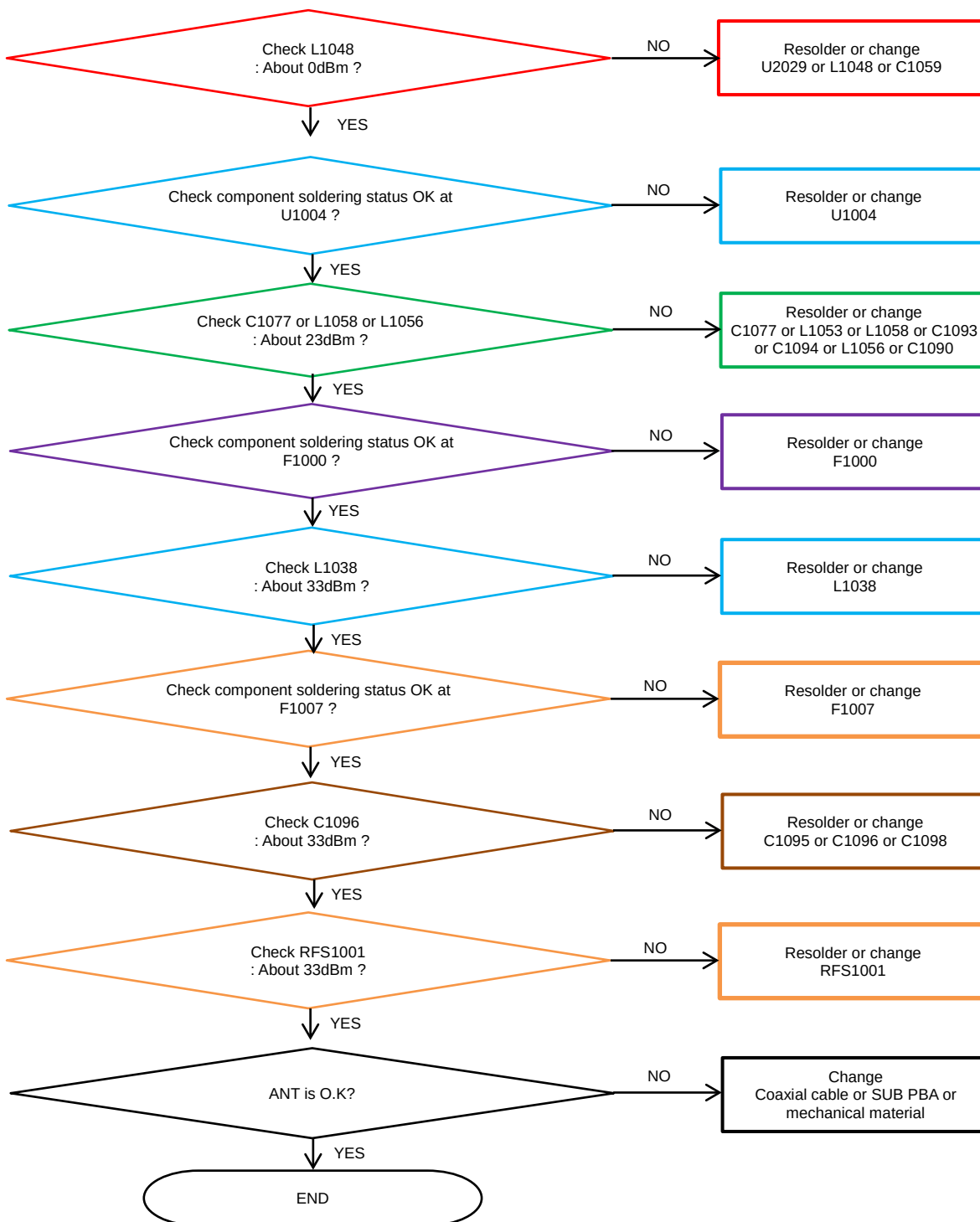
8. Level 3 Repair

8-4-31. GSM 1800/1900 Tx



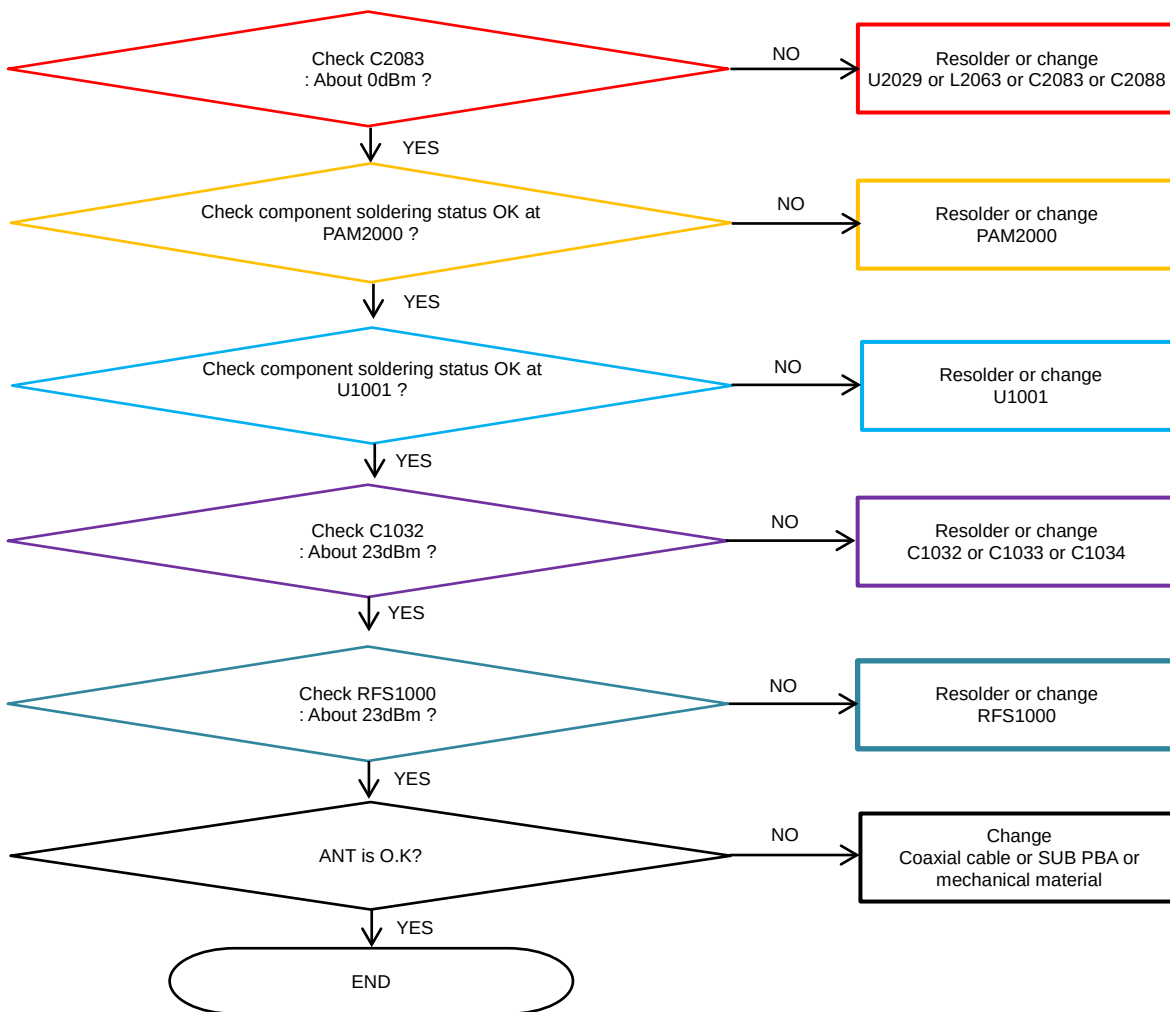
8. Level 3 Repair

8-4-32. WCDMA B1/2, TD-SCDMA B34/39, LTE B1/3/4/34/39 Tx



8. Level 3 Repair

8-4-33. LTE B7/38/40/41 Tx



8. Level 3 Repair

8-5. Service Schematics

- NC Point(Top View)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41
A	000	001	002	003	004	005	006	007	008	009	010	011	012	013	014	015	016	017	018	019	020	021	022	023	024	025	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040
B	001	002	003	004	005	006	007	008	009	010	011	012	013	014	015	016	017	018	019	020	021	022	023	024	025	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041
C	002	003	004	005	006	007	008	009	010	011	012	013	014	015	016	017	018	019	020	021	022	023	024	025	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041	
D	003	004	005	006	007	008	009	010	011	012	013	014	015	016	017	018	019	020	021	022	023	024	025	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041		
E	004	005	006	007	008	009	010	011	012	013	014	015	016	017	018	019	020	021	022	023	024	025	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041			
F	005	006	007	008	009	010	011	012	013	014	015	016	017	018	019	020	021	022	023	024	025	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041				
G	006	007	008	009	010	011	012	013	014	015	016	017	018	019	020	021	022	023	024	025	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041					
H	007	008	009	010	011	012	013	014	015	016	017	018	019	020	021	022	023	024	025	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041						
I	008	009	010	011	012	013	014	015	016	017	018	019	020	021	022	023	024	025	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041							
J	009	010	011	012	013	014	015	016	017	018	019	020	021	022	023	024	025	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041								
K	010	011	012	013	014	015	016	017	018	019	020	021	022	023	024	025	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041									
L	011	012	013	014	015	016	017	018	019	020	021	022	023	024	025	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041										
M	012	013	014	015	016	017	018	019	020	021	022	023	024	025	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041											
N	013	014	015	016	017	018	019	020	021	022	023	024	025	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041												
O	014	015	016	017	018	019	020	021	022	023	024	025	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041													
P	015	016	017	018	019	020	021	022	023	024	025	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041														
Q	016	017	018	019	020	021	022	023	024	025	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041															
R	017	018	019	020	021	022	023	024	025	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041																
S	018	019	020	021	022	023	024	025	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041																	
T	019	020	021	022	023	024	025	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041																		
U	020	021	022	023	024	025	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041																			
V	021	022	023	024	025	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041																				
W	022	023	024	025	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041																					
X	023	024	025	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041																						
Y	024	025	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041																							
Z	025	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041																								
AA	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041																									
AB	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041																										
AC	028	029	030	031	032	033	034	035	036	037	038	039	040	041																											
AD	029	030	031	032	033	034	035	036	037	038	039	040	041																												
AE	030	031	032	033	034	035	036	037	038	039	040	041																													
AF	031	032	033	034	035	036	037	038	039	040	041																														
AG	032	033	034	035	036	037	038	039	040	041																															
AH	033	034	035	036	037	038	039	040	041																																
AI	034	035	036	037	038	039	040	041																																	
AJ	035	036	037	038	039	040	041																																		
AK	036	037	038	039	040	041																																			
AL	037	038	039	040	041																																				
AM	038	039	040	041																																					
AN	039	040	041																																						
AO	040	041																																							
AP	041																																								

9. Reference Abbreviation

Reference Abbreviation

- **AAC**: Advanced Audio Coding.
- **AVC** : Advanced Video Coding.
- **BER** : Bit Error Rate
- **BPSK**: Binary Phase Shift Keying
- **CA** : Conditional Access
- **CDM** : Code Division Multiplexing
- **C/I** : Carrier to Interference
- **DMB** : Digital Multimedia Broadcasting
- **EN** : European Standard
- **ES** : Elementary Stream
- **ETSI**: European Telecommunications Standards Institute
- **MPEG**: Moving Picture Experts Group
- **PN** : Pseudo-random Noise
- **PS** : Pilot Symbol
- **QPSK**: Quadrature Phase Shift Keying
- **RS** : Reed-Solomon
- **SI** : Service Information
- **TDM** : Time Division Multiplexing
- **TS** : Transport Stream

SAMSUNG